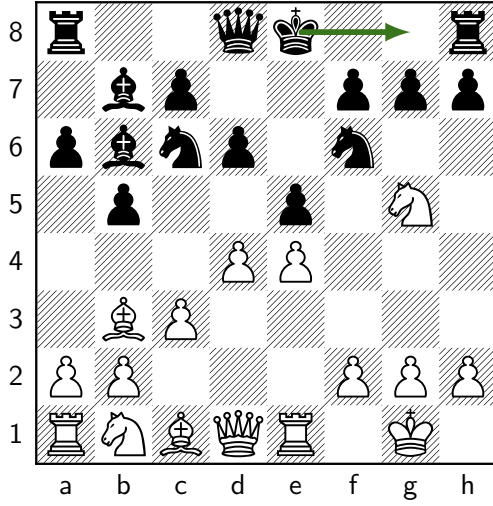


Melanie-BOTS (corpus: CCRL engines) — match game 3

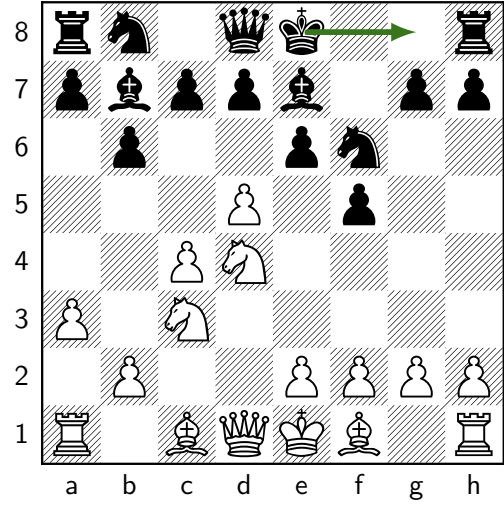
Melanie (Deep_Vision)

Abstract

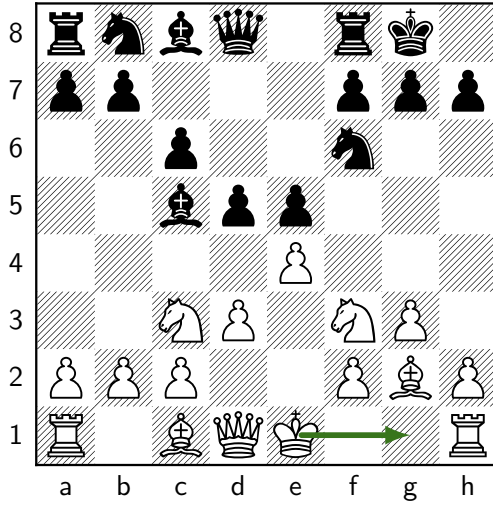
Every move below was produced by relational analogy to named master games (human and/or top-engine – the exact source is named per move). Each figure shows our position with Melanie’s move (a) and the top retrieved analog positions (b–d) with the master move played there; the caption names the exact source and the piece-role correspondence under μ – cross-type maps (a rook playing a queen’s role) are called out, as no material/eval engine explains a move that way.



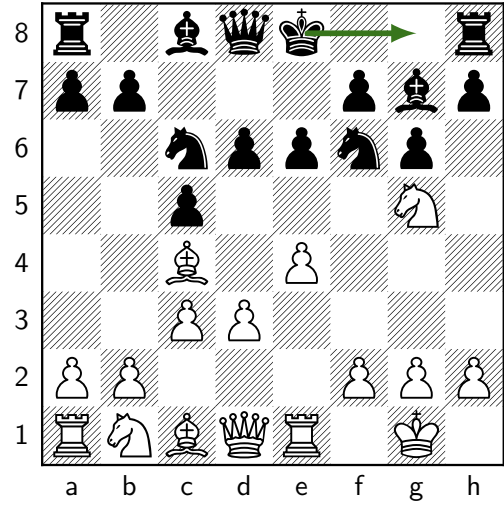
(a)



(b)

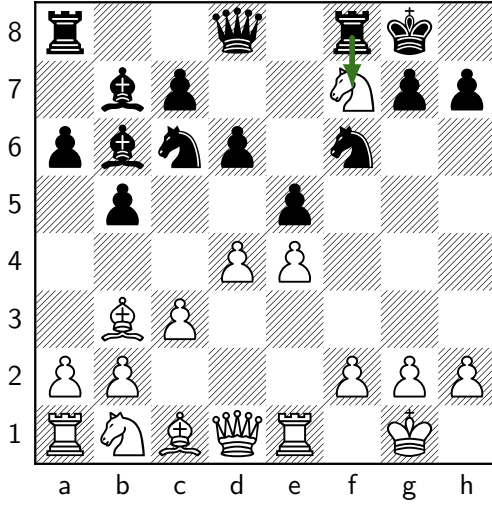


(c)

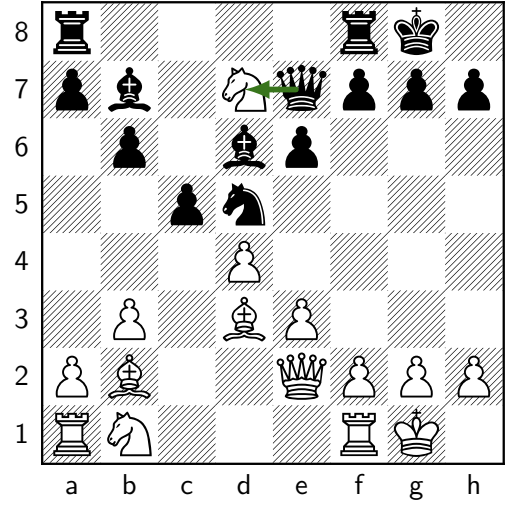


(d)

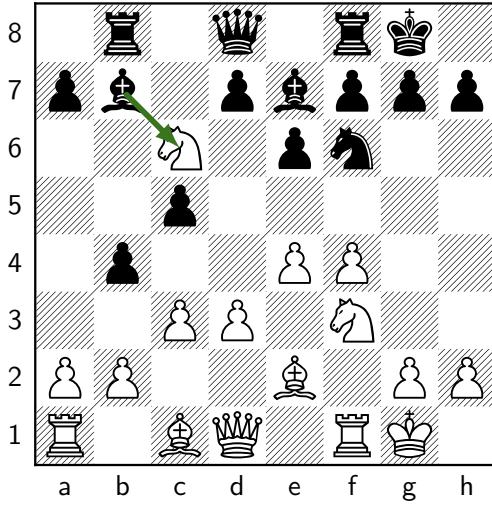
Figure 1: **Move 10, Black: 0-0.** (a) Our position; Melanie plays 0-0 (e8g8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1915` (ply 13) [structural distance $d=68$, QAP obj=105]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Pb5* as their *Ra8*, our *pg2* as their *Rh8*; transferred move e8g8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#311` (ply 12) [structural distance $d=61$, QAP obj=98]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Pb5* as their *Ra1*, our *Rh8* as their *Qd1*; transferred move e8g8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2807` (ply 15) [structural distance $d=48$, QAP obj=77]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Bb7* as their *Ra8*, our *Pe5* as their *Bc8*; transferred move e8g8 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to 0-0 (summed QAP objective 584, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



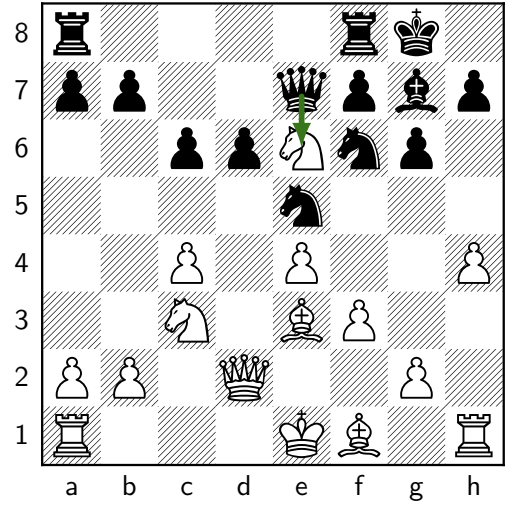
(a)



(b)

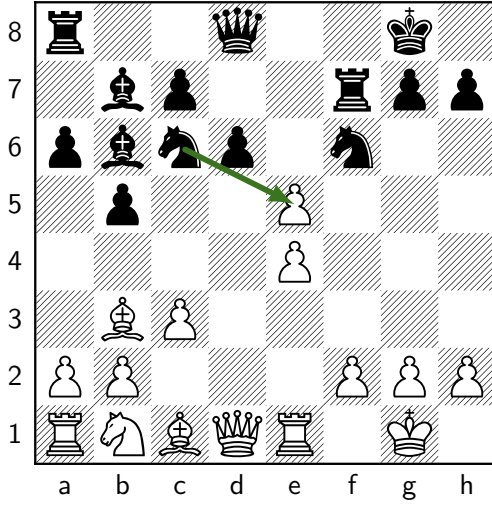


(c)

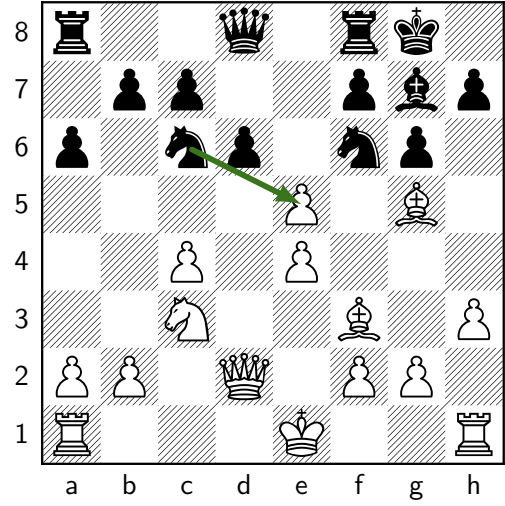


(d)

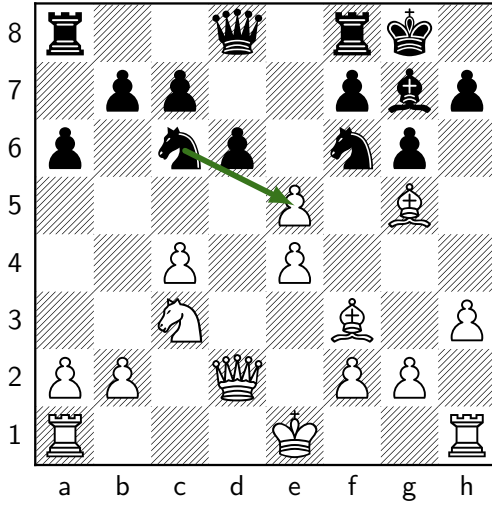
Figure 2: **Move 11, Black: Rxf7.** (a) Our position; Melanie plays Rxf7 (f8f7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#1584` (ply 23) [structural distance $d=82$, QAP obj=57]: the master played Qxd7; under μ our Rf1 plays the role of that game’s Qe2 (a **cross-type** correspondence); cross-type roles: our Qd8 as their Rf8, our Nc6 as their Pb6; transferred move f8f7 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#2250` (ply 21) [structural distance $d=84$, QAP obj=108]: the master played Bxc6; under μ our Rf1 plays the role of that game’s Bb2 (a **cross-type** correspondence); cross-type roles: our Qd8 as their Rb8, our Pg7 as their Rf8; transferred move f8e8. (d) Retrieved from `ccrl_top_p0000#664` (ply 23) [structural distance $d=89$, QAP obj=90]: the master played Qxe6; under μ our Rf1 plays the role of that game’s Qe2 (a **cross-type** correspondence); cross-type roles: our pg2 as their Ra8, our Qd8 as their Rf8; transferred move f8e8. Synthesis: 1 retrieved analog transferred to Rxf7 (summed QAP objective 57, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



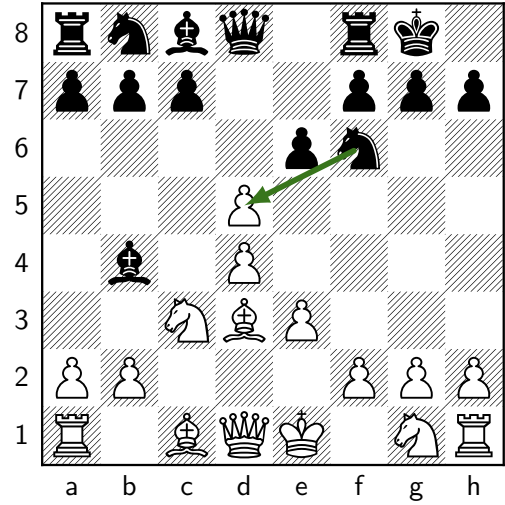
(a)



(b)

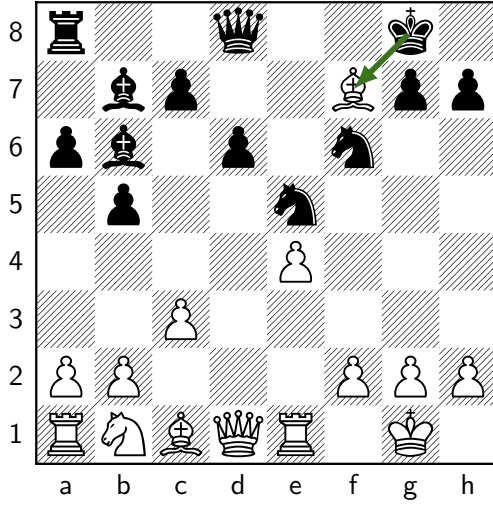


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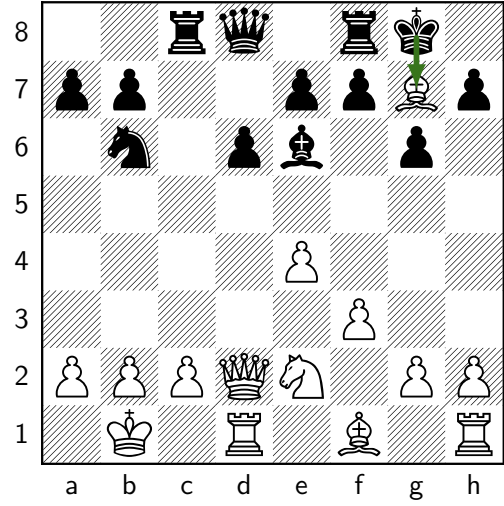


(d)

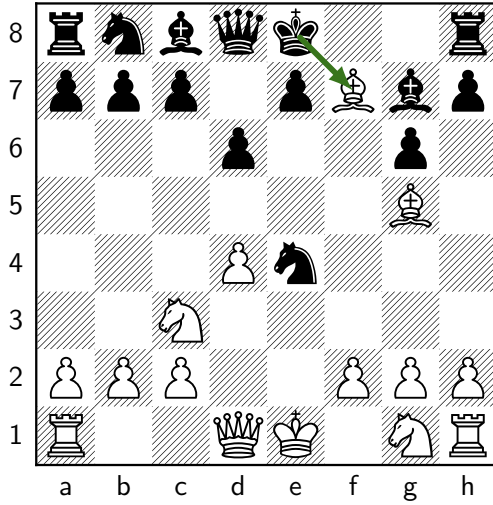
Figure 3: **Move 12, Black: Nxe5.** (a) Our position; Melanie plays Nxe5 (c6e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `cctl_top_p0003#2900` (ply 21) [structural distance $d=92$, QAP obj=142]: the master played Nxe5; under μ our *Nc3* plays the role of that game’s *Nc3*; cross-type roles: our *pc3* as their *Ra8*, our *Bb7* as their *Rf8*; transferred move c6e5 (**endorses this move**). (c) Retrieved from `cctl_top_p0003#2901` (ply 21) [structural distance $d=92$, QAP obj=142]: the master played Nxe5; under μ our *Nc3* plays the role of that game’s *Nc3*; cross-type roles: our *pc3* as their *Ra8*, our *Bb7* as their *Rf8*; transferred move c6e5 (**endorses this move**). (d) Retrieved from `cctl_top_p0001#2378` (ply 11) [structural distance $d=74$, QAP obj=51]: the master played Nxd5; under μ our *Nc3* plays the role of that game’s *Nf3*; cross-type roles: our *bc1* as their *Ra8*, our *Pb5* as their *Nb8*; transferred move c6e5 (**endorses this move**). Synthesis: 17 retrieved analogs transferred to Nxe5 (summed QAP objective 930, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



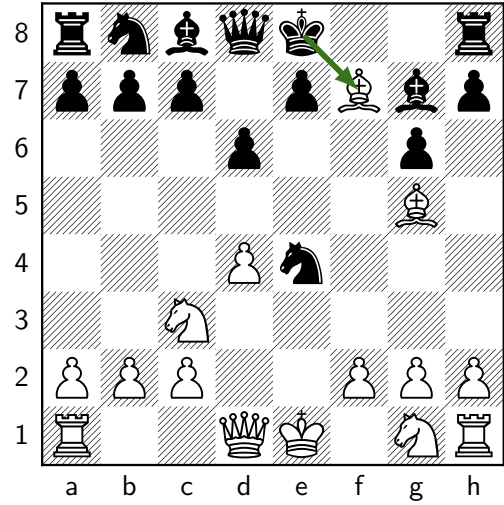
(a)



(b)

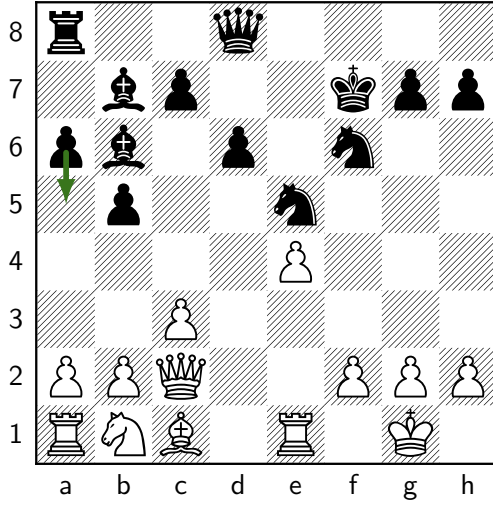


(c)

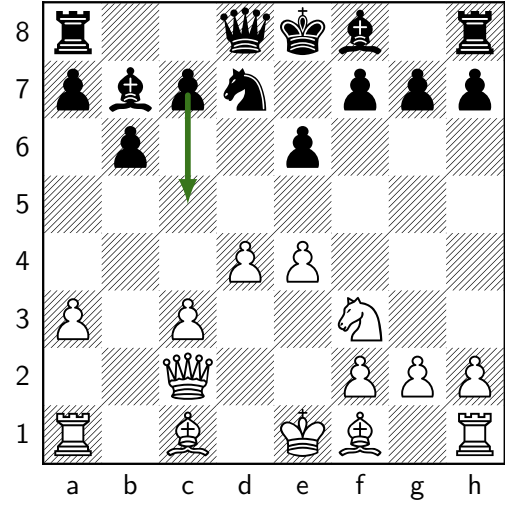


(d)

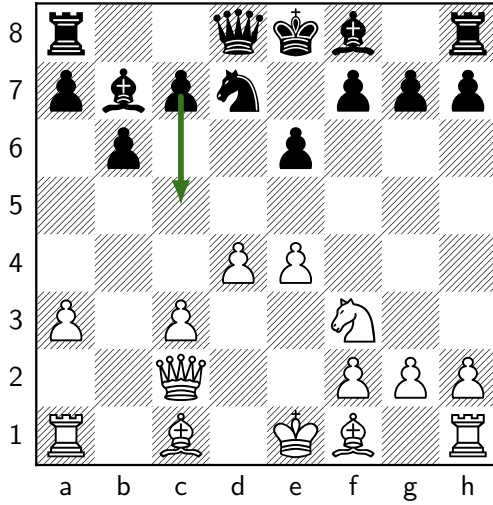
Figure 4: **Move 13, Black: Kxf7.** (a) Our position; Melanie plays Kxf7 (g8f7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#1498` (ply 27) [structural distance $d=104$, QAP obj=29]: the master played Kxg7; under μ our *Kg1* plays the role of that game’s *Kg1*; cross-type roles: our *Bb6* as their *Rc8*, our *Nf6* as their *Rf8*; transferred move g8f7 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#774` (ply 11) [structural distance $d=100$, QAP obj=92]: the master played Kxf7; under μ our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *Nf6* as their *Ra8*, our *Ra8* as their *Nb8*; transferred move g8h8. (d) Retrieved from `ccrl_top_p0000#775` (ply 11) [structural distance $d=100$, QAP obj=92]: the master played Kxf7; under μ our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *Nf6* as their *Ra8*, our *Ra8* as their *Nb8*; transferred move g8h8. Synthesis: 1 retrieved analog transferred to Kxf7 (summed QAP objective 29, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



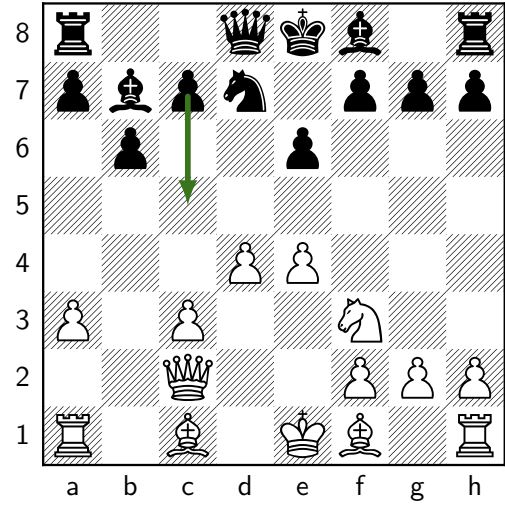
(a)



(b)

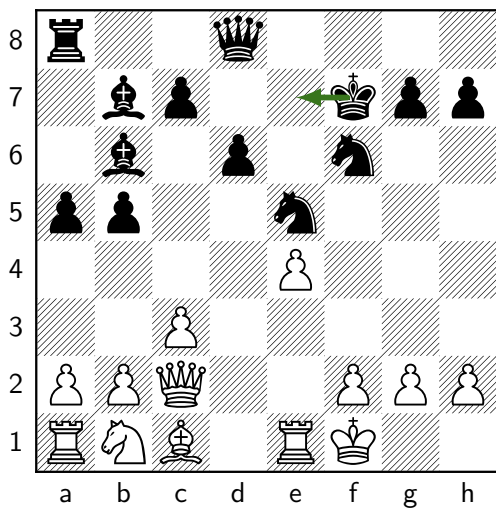


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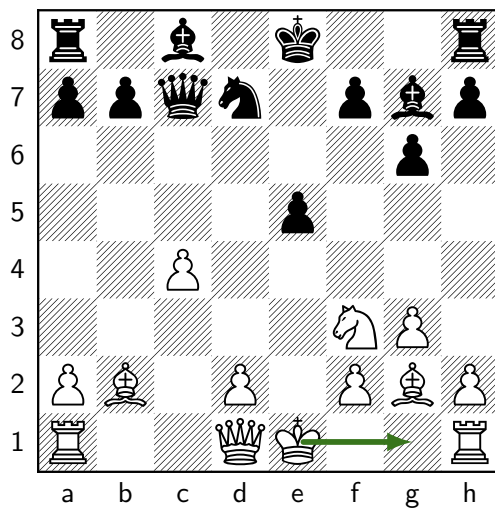


(d)

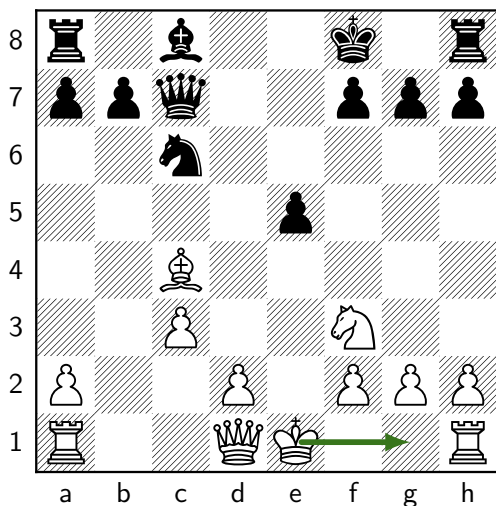
Figure 5: **Move 14, Black: a5.** (a) Our position; Melanie plays a5 (a6a5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1969` (ply 17) [structural distance $d=103$, QAP obj=38]: the master played c5; under μ our *Pa3* plays the role of that game’s *Pc2*; cross-type roles: our *Ph7* as their *Ra8*, our *Ne5* as their *Qd8*; transferred move a6a5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1970` (ply 17) [structural distance $d=103$, QAP obj=38]: the master played c5; under μ our *Pa3* plays the role of that game’s *Pc2*; cross-type roles: our *Ph7* as their *Ra8*, our *Ne5* as their *Qd8*; transferred move a6a5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1971` (ply 17) [structural distance $d=103$, QAP obj=38]: the master played c5; under μ our *Pa3* plays the role of that game’s *Pc2*; cross-type roles: our *Ph7* as their *Ra8*, our *Ne5* as their *Qd8*; transferred move a6a5 (**endorses this move**). Synthesis: 18 retrieved analogs transferred to a5 (summed QAP objective 691, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



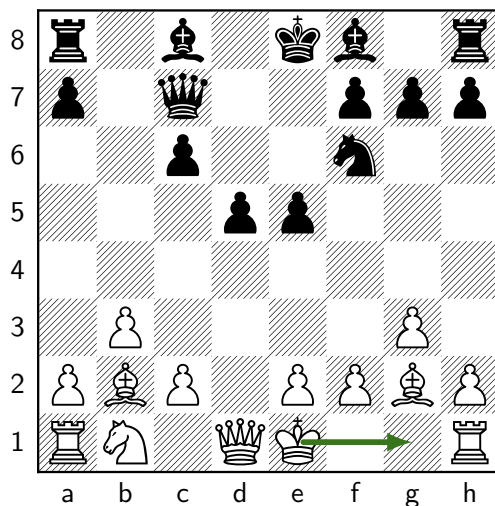
(a)



(b)

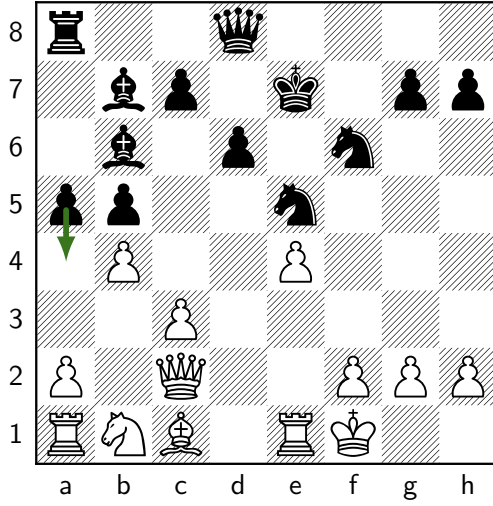


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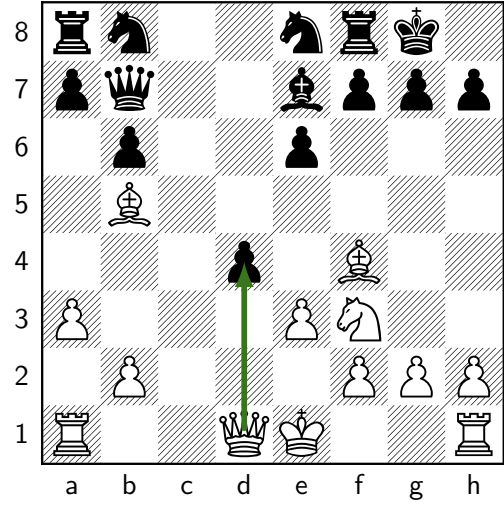


(d)

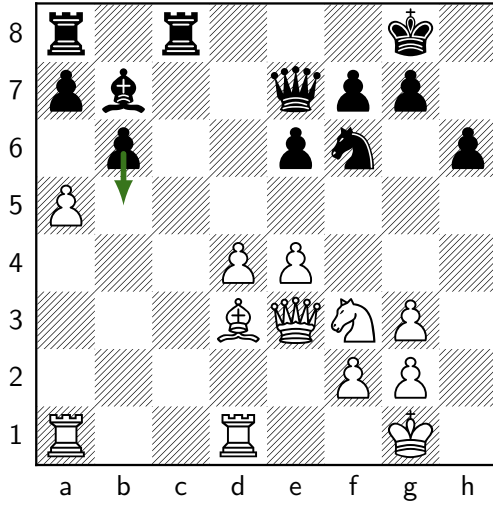
Figure 6: **Move 15, Black: Ke7.** (a) Our position; Melanie plays Ke7 (f7e7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#3130` (ply 22) [structural distance $d=95$, QAP obj=46]: the master played 0-0; under μ our *Kf2* plays the role of that game’s *Ke1*; cross-type roles: our *pg2* as their *Ra1*, our *nb1* as their *Qd1*; transferred move f7e7 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#1566` (ply 20) [structural distance $d=87$, QAP obj=36]: the master played 0-0; under μ our *Kf2* plays the role of that game’s *Ke1*; cross-type roles: our *Pg7* as their *Ra1*, our *Ne5* as their *Qd1*; transferred move f7e7 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#445` (ply 16) [structural distance $d=92$, QAP obj=36]: the master played 0-0; under μ our *Kf2* plays the role of that game’s *Ke1*; cross-type roles: our *pc3* as their *Ra1*, our *Ra8* as their *Nb1*; transferred move f7e7 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Ke7 (summed QAP objective 209, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



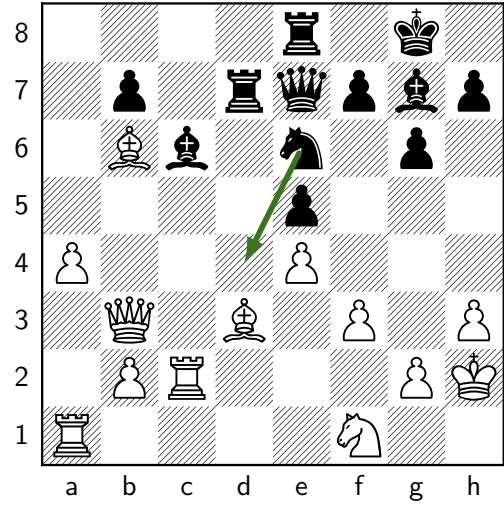
(a)



(b)

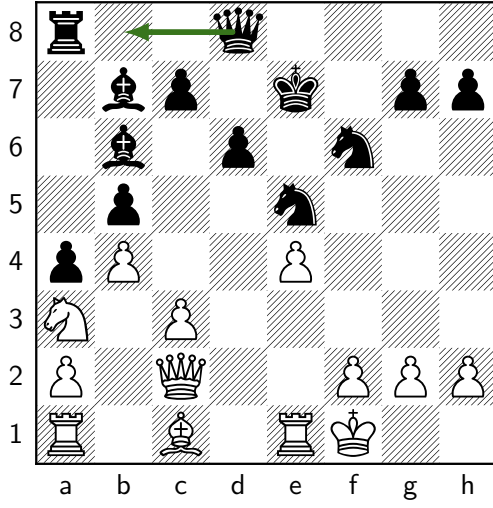


(c)

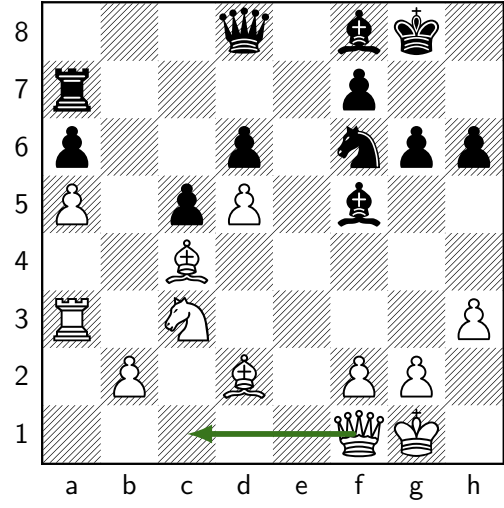


(d)

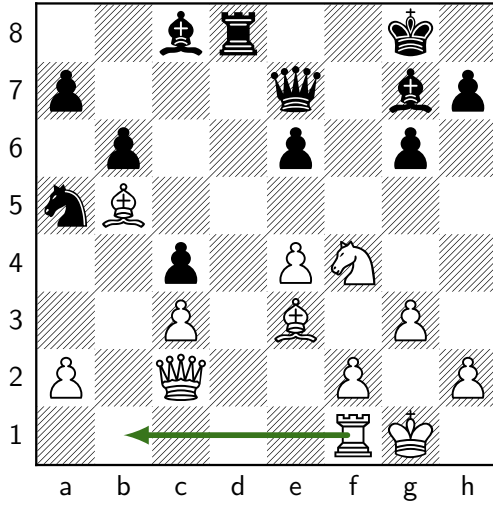
Figure 7: **Move 16, Black: a4.** (a) Our position; Melanie plays a4 (a5a4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#509` (ply 26) [structural distance $d=93$, QAP obj=70]: the master played $Qxd4$; under μ our $Pa4$ plays the role of that game’s $Qd1$ (a **cross-type** correspondence); cross-type roles: our $Bb6$ as their $Ra1$, our $Bb7$ as their $Pg2$; transferred move $a5a4$ (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#897` (ply 39) [structural distance $d=87$, QAP obj=60]: the master played $b5$; under μ our $Pa4$ plays the role of that game’s $Pb3$; cross-type roles: our $Ph7$ as their $Ra8$, our $Qd8$ as their $Rc8$; transferred move $a5a4$ (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3479` (ply 45) [structural distance $d=90$, QAP obj=47]: the master played $Nd4$; under μ our $Pa4$ plays the role of that game’s $Ne3$ (a **cross-type** correspondence); cross-type roles: our $Qd8$ as their $Re8$, our $Nf6$ as their $Rd7$; transferred move $a5a4$ (**endorses this move**). Synthesis: 7 retrieved analogs transferred to a4 (summed QAP objective 258, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



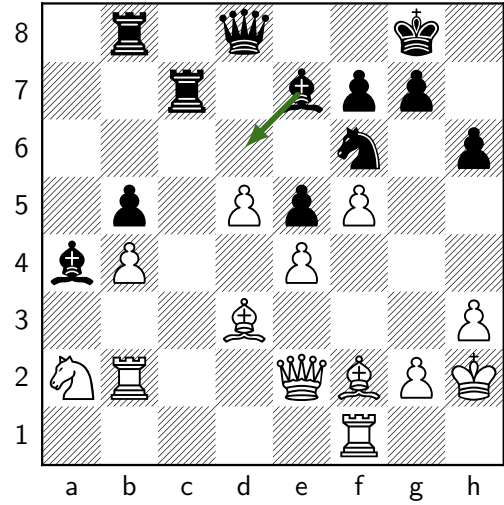
(a)



(b)

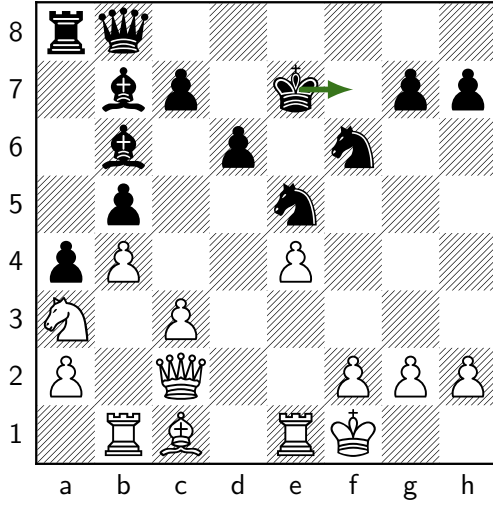


(c)

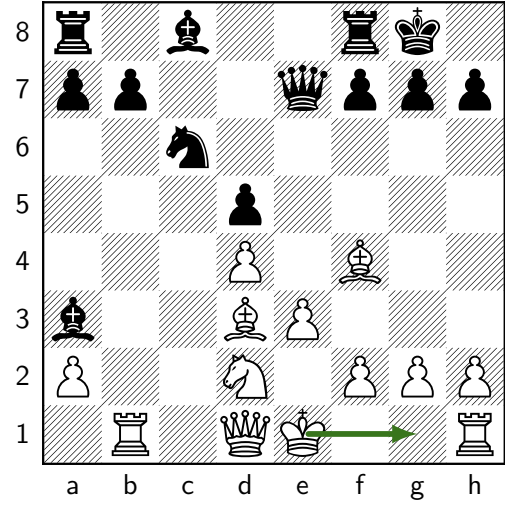


(d)

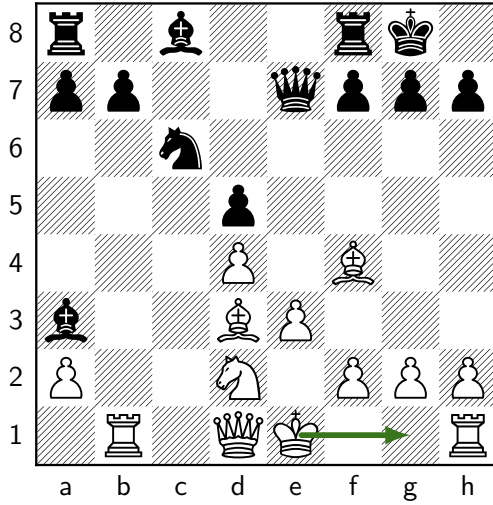
Figure 8: **Move 17, Black: Qb8.** (a) Our position; Melanie plays Qb8 (d8b8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#838` (ply 46) [structural distance $d=80$, QAP obj=78]: the master played Qc1; under μ our Qd1 plays the role of that game’s Qf1; cross-type roles: our bc1 as their Ra3, our Nf6 as their Bc4; transferred move d8b8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2103` (ply 38) [structural distance $d=82$, QAP obj=38]: the master played Rb1; under μ our Qd1 plays the role of that game’s Rf1 (a **cross-type** correspondence); cross-type roles: our Bb7 as their Pa2, our Ne5 as their Qc2; transferred move d8b8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2420` (ply 49) [structural distance $d=84$, QAP obj=79]: the master played Bd6; under μ our Bb3 plays the role of that game’s Be2; cross-type roles: our Ph7 as their Rb8, our Bb7 as their Qd8; transferred move b6c5. Synthesis: 2 retrieved analogs transferred to Qb8 (summed QAP objective 117, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



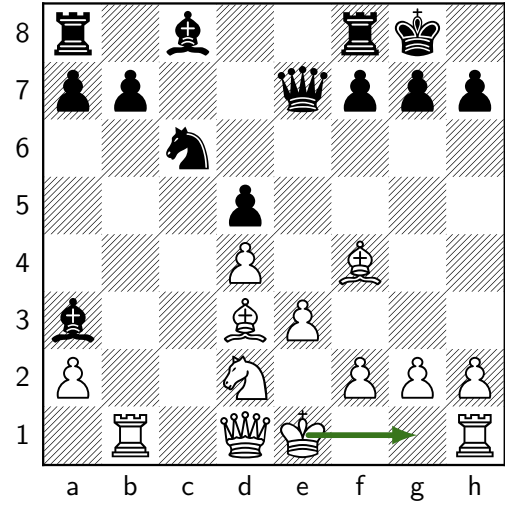
(a)



(b)

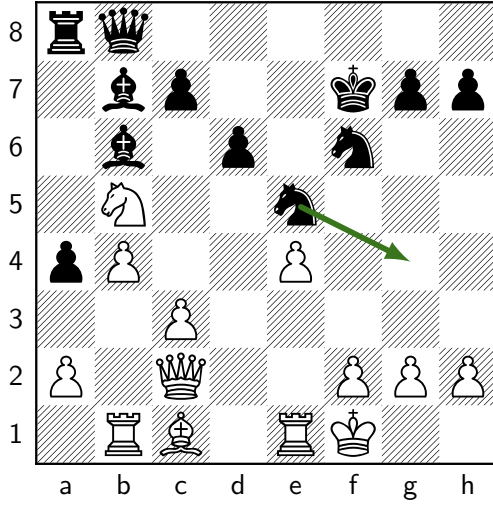


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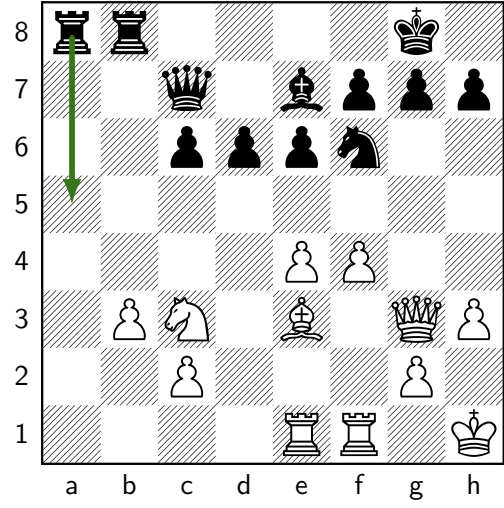


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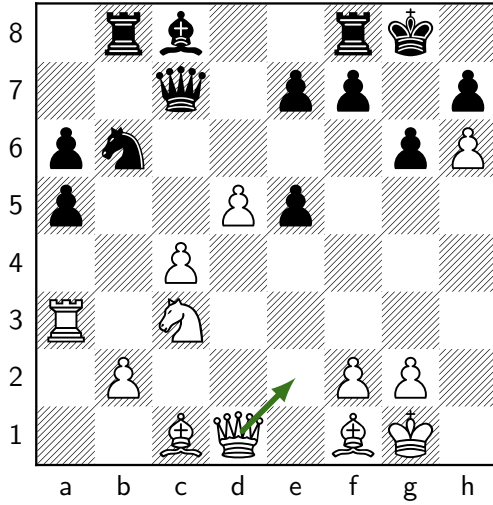
Figure 9: **Move 18, Black: Kf7.** (a) Our position; Melanie plays Kf7 (e7f7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#932` (ply 26) [structural distance $d=91$, QAP obj=49]: the master played 0-0; under μ our *Ke2* plays the role of that game’s *Ke1*; cross-type roles: our *Bb6* as their *Rb1*, our *Pd6* as their *Qd1*; transferred move e7f7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#933` (ply 26) [structural distance $d=91$, QAP obj=49]: the master played 0-0; under μ our *Ke2* plays the role of that game’s *Ke1*; cross-type roles: our *Bb6* as their *Rb1*, our *Pd6* as their *Qd1*; transferred move e7f7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1669` (ply 26) [structural distance $d=91$, QAP obj=49]: the master played 0-0; under μ our *Ke2* plays the role of that game’s *Ke1*; cross-type roles: our *Bb6* as their *Rb1*, our *Pd6* as their *Qd1*; transferred move e7f7 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Kf7 (summed QAP objective 296, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



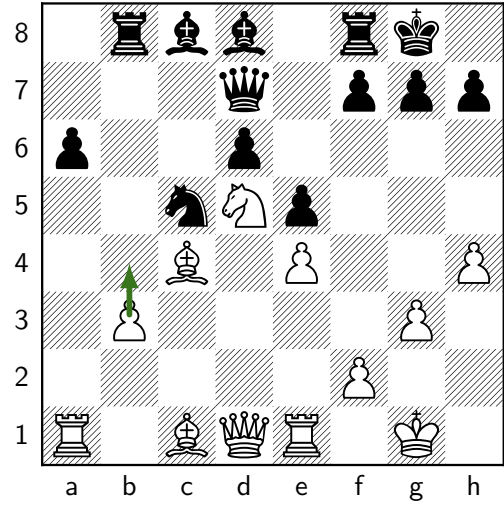
(a)



(b)

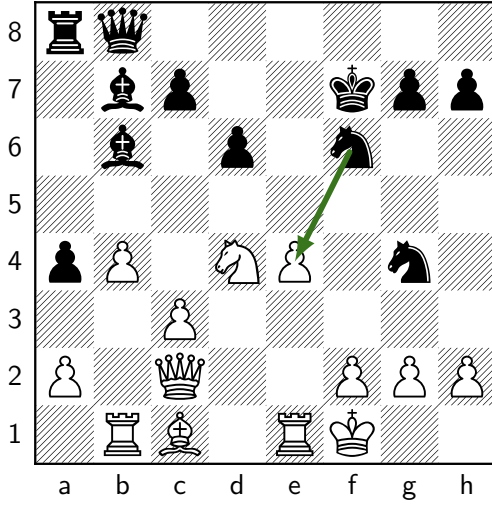


(c)

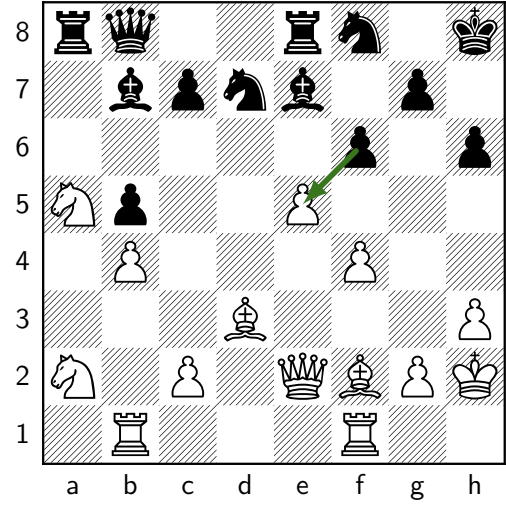


(d)

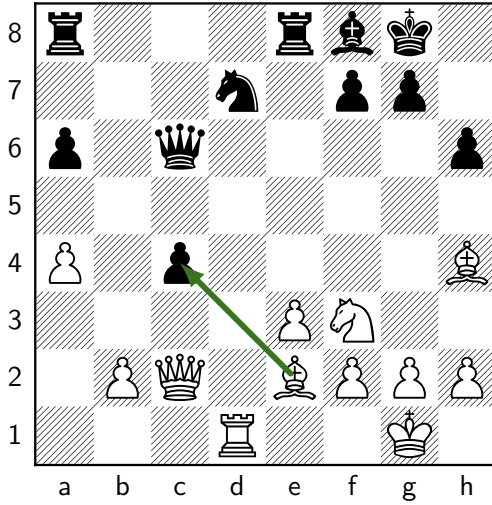
Figure 10: **Move 19, Black: Neg4.** (a) Our position; Melanie plays **Neg4** (e5g4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#245` (ply 41) [structural distance $d=95$, QAP obj=40]: the master played **Ra5**; under μ our **Ne4** plays the role of that game’s **Ra1** (a **cross-type** correspondence); cross-type roles: our **Nf6** as their **Rb8**, our **Ph7** as their **Qc7**; transferred move **e5g4** (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#505` (ply 40) [structural distance $d=96$, QAP obj=84]: the master played **Qe2**; under μ our **Qb1** plays the role of that game’s **Qd1**; cross-type roles: our **pb4** as their **Bf1**, our **Bb7** as their **Pb2**; transferred move **b8c8**. (d) Retrieved from `ccr1_top_p0002#3842` (ply 34) [structural distance $d=91$, QAP obj=83]: the master played **b4**; under μ our **Pd3** plays the role of that game’s **Pb3**; cross-type roles: our **qc2** as their **Ra1**, our **Bb6** as their **Qd1**; transferred move **d6d5**. Synthesis: 1 retrieved analog transferred to **Neg4** (summed QAP objective 40, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



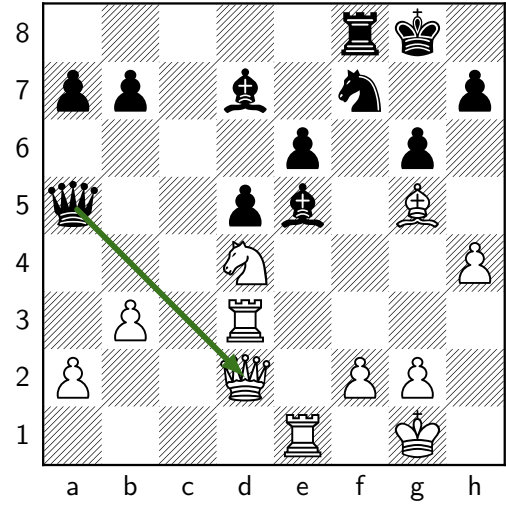
(a)



(b)

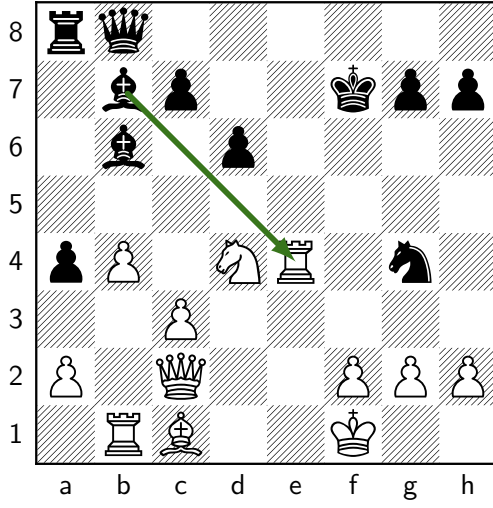


(c)

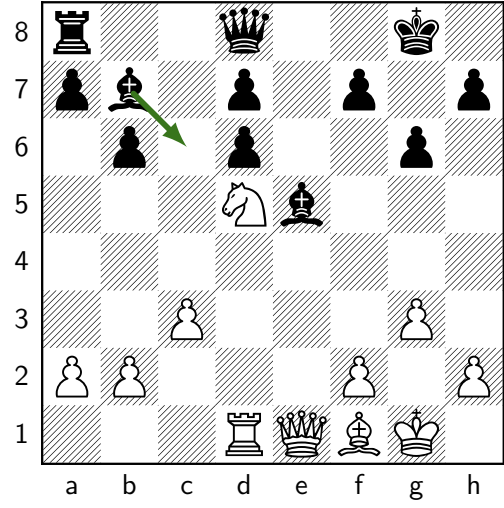


(d)

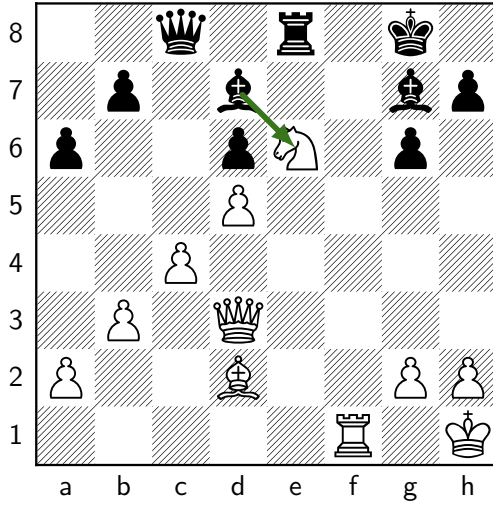
Figure 11: **Move 20, Black: Nxe4**. (a) Our position; Melanie plays Nxe4 (f6e4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0003#310 (ply 53) [structural distance $d=101$, QAP obj=37]: the master played fxe5; under μ our Nf3 plays the role of that game’s Pf3 (a **cross-type** correspondence); cross-type roles: our qc2 as their Ra8, our pb4 as their Qb8; transferred move f6e4 (**endorses this move**). (c) Retrieved from ccr1_top_p0002#2158 (ply 42) [structural distance $d=105$, QAP obj=34]: the master played Bxc4; under μ our Nf3 plays the role of that game’s Be2 (a **cross-type** correspondence); cross-type roles: our Ng4 as their Rd1, our Bb7 as their Qc2; transferred move f6e4 (**endorses this move**). (d) Retrieved from ccr1_top_p0000#3625 (ply 53) [structural distance $d=101$, QAP obj=15]: the master played Qxd2; under μ our Nf3 plays the role of that game’s Qa4 (a **cross-type** correspondence); cross-type roles: our Pc7 as their Rf8, our Ra8 as their Pa7; transferred move f6e4 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Nxe4 (summed QAP objective 86, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



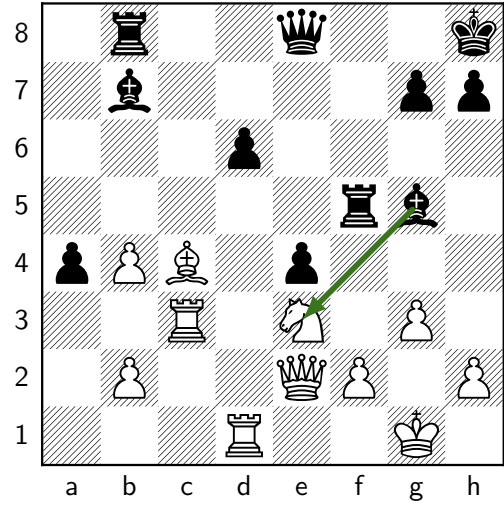
(a)



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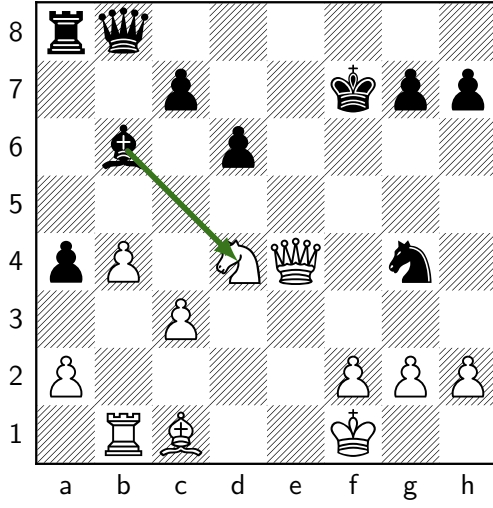


(c)

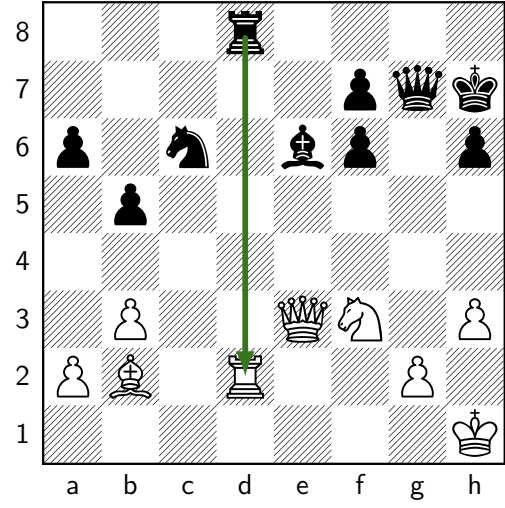


(d)

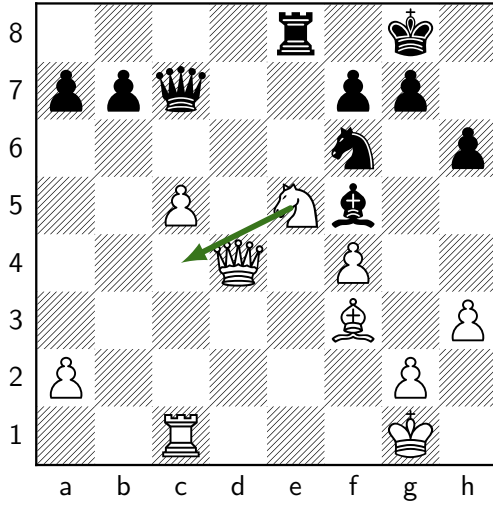
Figure 12: **Move 21, Black: Bxe4**. (a) Our position; Melanie plays **Bxe4** (b7e4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#702` (ply 35) [structural distance $d=86$, QAP obj=60]: the master played **Bc6**; under μ our *Bb2* plays the role of that game’s *Bb2*; cross-type roles: our *Pa4* as their *Qd8*, our *Qb8* as their *Ph7*; transferred move **b7e4** (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2225` (ply 45) [structural distance $d=76$, QAP obj=43]: the master played **Bxe6**; under μ our *Bb2* plays the role of that game’s *Bd2*; cross-type roles: our *Ra8* as their *Qc8*, our *Bb7* as their *Re8*; transferred move **b7e4** (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1377` (ply 49) [structural distance $d=85$, QAP obj=28]: the master played **Bxe3**; under μ our *Bb2* plays the role of that game’s *Bg4*; cross-type roles: our *Pc7* as their *Rb8*, our *pc3* as their *Qe8*; transferred move **b7e4** (**endorses this move**). Synthesis: 5 retrieved analogs transferred to **Bxe4** (summed QAP objective 183, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



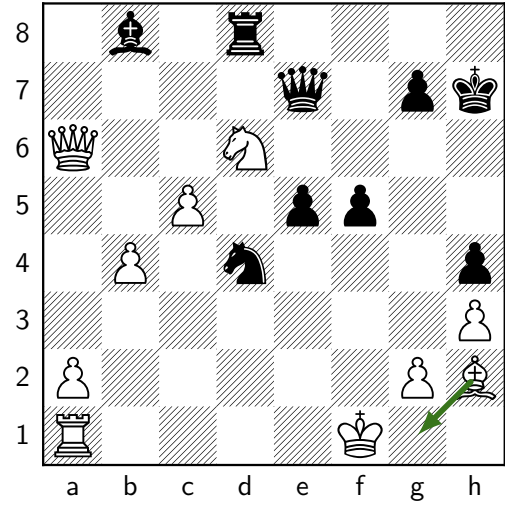
(a)



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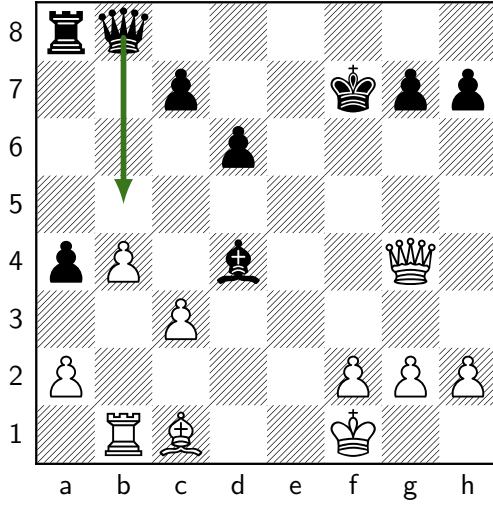


(c)

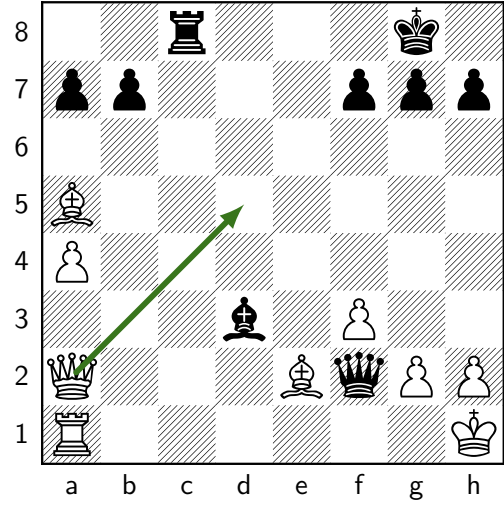


(d)

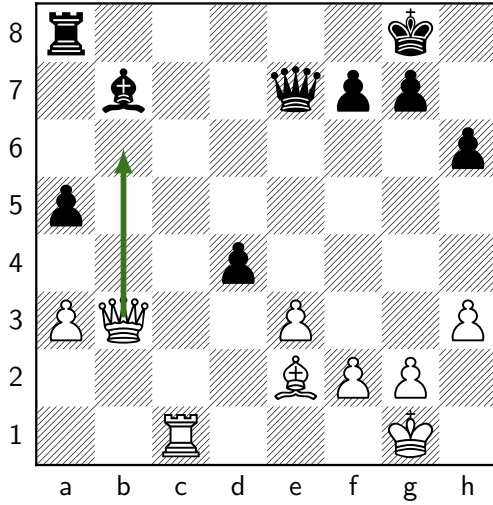
Figure 13: **Move 22, Black: Bxd4.** (a) Our position; Melanie plays Bxd4 (b6d4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2914` (ply 57) [structural distance $d=69$, QAP obj=41]: the master played Rxd2; under μ our Bb3 plays the role of that game’s Rd1 (a **cross-type** correspondence); cross-type roles: our Qb8 as their Pf7, our Pa4 as their Qg7; transferred move b6d4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#740` (ply 50) [structural distance $d=70$, QAP obj=62]: the master played Nc4; under μ our Ng5 plays the role of that game’s Ne5; cross-type roles: our Pc7 as their Rc1, our Qb8 as their Bf3; transferred move g4e5. (d) Retrieved from `ccr1_top_p0002#1872` (ply 56) [structural distance $d=73$, QAP obj=62]: the master played Bg1; under μ our Pd3 plays the role of that game’s Bh2 (a **cross-type** correspondence); cross-type roles: our Bb6 as their Ph3, our bc1 as their nd4; transferred move d6d5. Synthesis: 1 retrieved analog transferred to Bxd4 (summed QAP objective 41, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



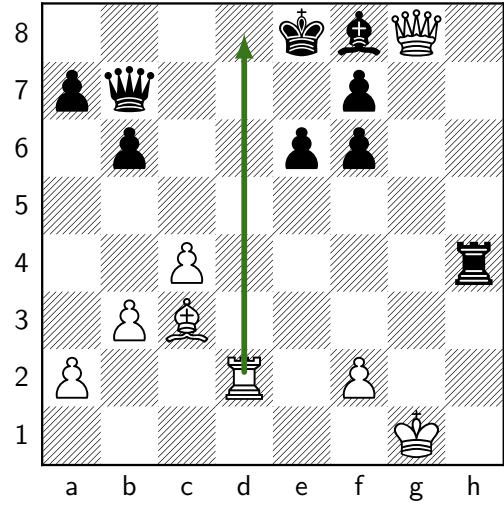
(a)



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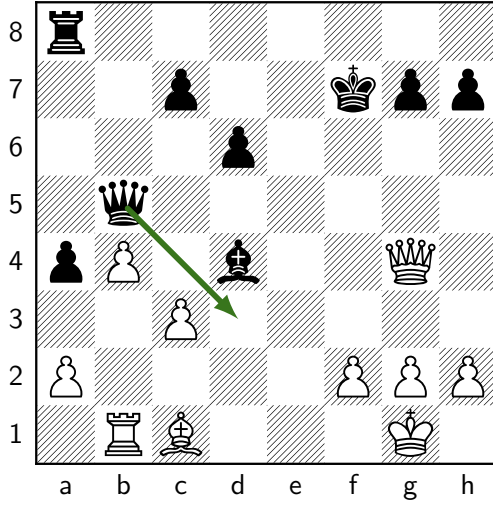


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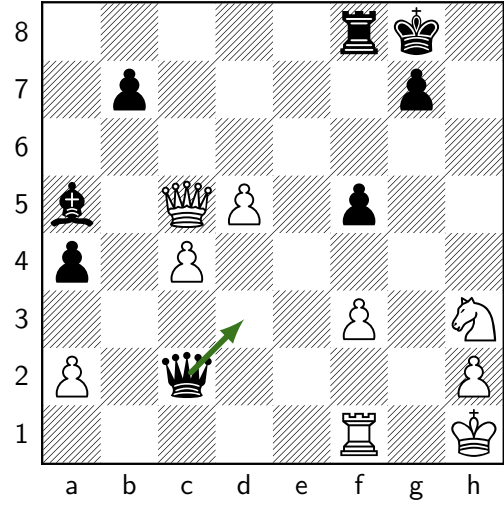


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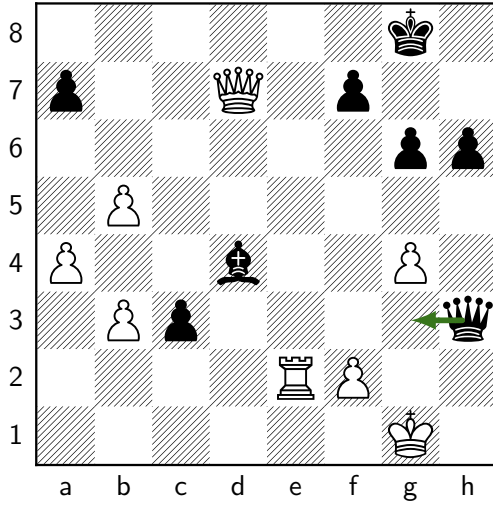
Figure 14: **Move 23, Black: Qb5+.** (a) Our position; Melanie plays Qb5+ (b8b5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2208` (ply 54) [structural distance $d=53$, QAP obj=51]: the master played Qd5; under μ our Qb1 plays the role of that game’s Qa2; cross-type roles: our pc3 as their qf2, our Bd4 as their Pg2; transferred move b8b5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#662` (ply 44) [structural distance $d=61$, QAP obj=36]: the master played Qb6; under μ our Qb1 plays the role of that game’s Qb3; cross-type roles: our pf2 as their Rc1, our Ra8 as their Pa3; transferred move b8b5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3762` (ply 54) [structural distance $d=59$, QAP obj=25]: the master played Rd8+; under μ our Qb1 plays the role of that game’s Rd2 (a **cross-type** correspondence); cross-type roles: our bc1 as their Pb3, our Ra8 as their Bc3; transferred move b8b5 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Qb5+ (summed QAP objective 149, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



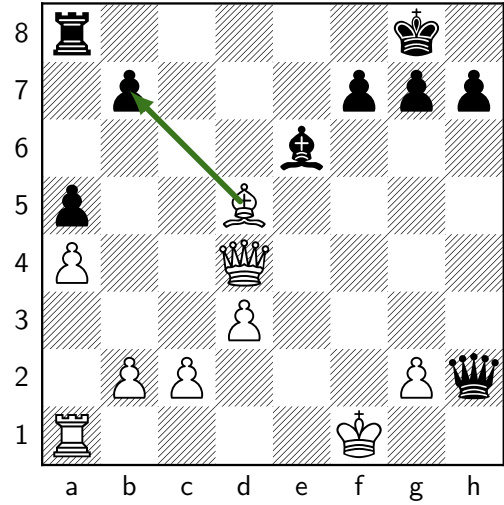
(a)



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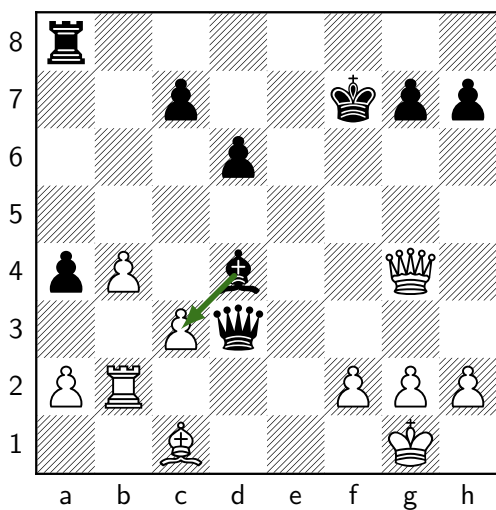


(c)

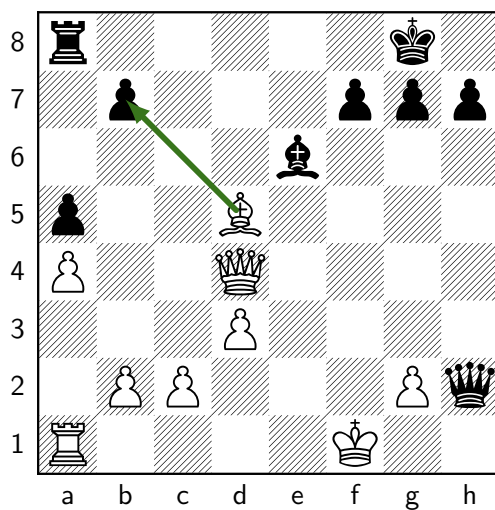


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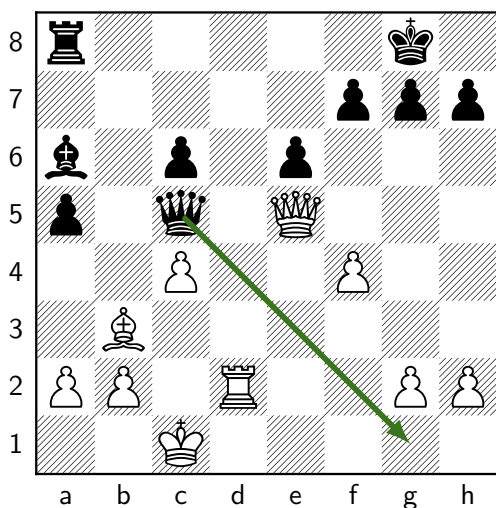
Figure 15: **Move 24, Black: Qd3.** (a) Our position; Melanie plays Qd3 (b5d3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#303` (ply 63) [structural distance $d=64$, QAP obj=30]: the master played Qd3; under μ our Qb4 plays the role of that game’s Qc7; cross-type roles: our Bd4 as their Rf8, our Ra8 as their Pg7; transferred move b5d3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3016` (ply 61) [structural distance $d=51$, QAP obj=61]: the master played Qg3+; under μ our Qb4 plays the role of that game’s Qh6; cross-type roles: our bc1 as their pa4, our rb1 as their pb3; transferred move b5h5. (d) Retrieved from `ccr1_top_p0003#1051` (ply 38) [structural distance $d=57$, QAP obj=60]: the master played Bxb7; under μ our Bd5 plays the role of that game’s Bd5; cross-type roles: our pa2 as their Ra1, our Ra8 as their Pb2; transferred move d4c3. Synthesis: 1 retrieved analog transferred to Qd3 (summed QAP objective 30, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



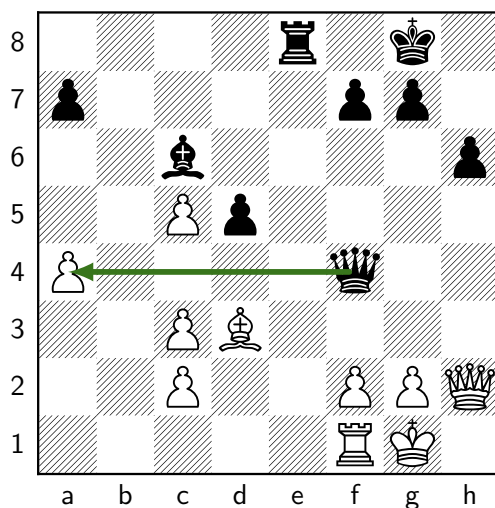
(a)



(b)



(c)



(d)

Figure 16: **Move 25, Black: Bxc3.** (a) Our position; Melanie plays Bxc3 (d4c3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#1051` (ply 38) [structural distance $d=60$, QAP obj=57]: the master played Bxb7; under μ our Bd5 plays the role of that game’s Bd5; cross-type roles: our pc3 as their be6; transferred move d4c3 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#3814` (ply 45) [structural distance $d=74$, QAP obj=45]: the master played Qg1+; under μ our Bd5 plays the role of that game’s Qc4 (a **cross-type** correspondence); cross-type roles: our Qd3 as their Ba6, our pc3 as their ge5; transferred move d4c3 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#1776` (ply 51) [structural distance $d=72$, QAP obj=42]: the master played Qxa4; under μ our Bd5 plays the role of that game’s Qf5 (a **cross-type** correspondence); cross-type roles: our pb4 as their Re8, our Qd3 as their Bc6; transferred move d4c3 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Bxc3 (summed QAP objective 248, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

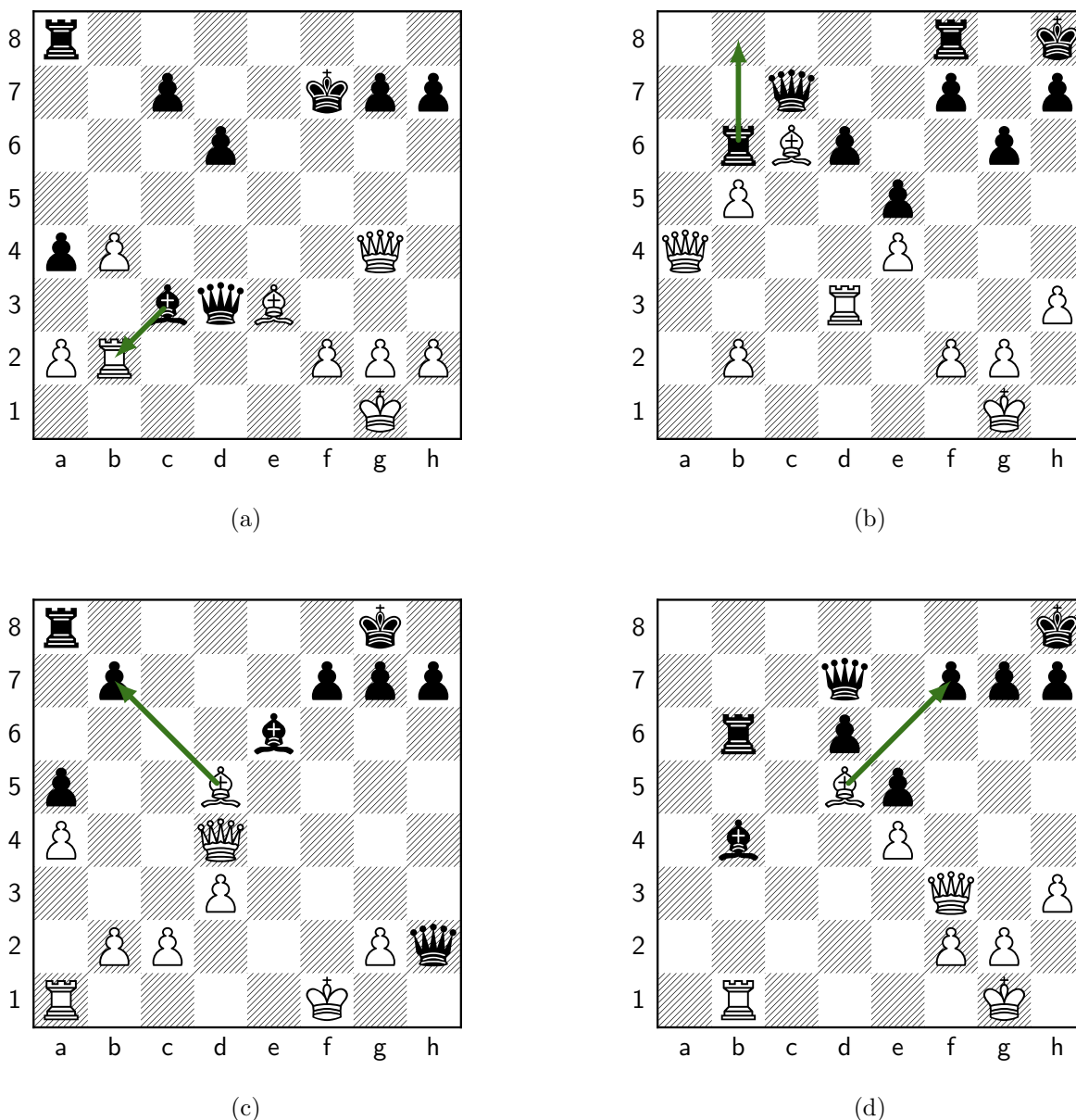
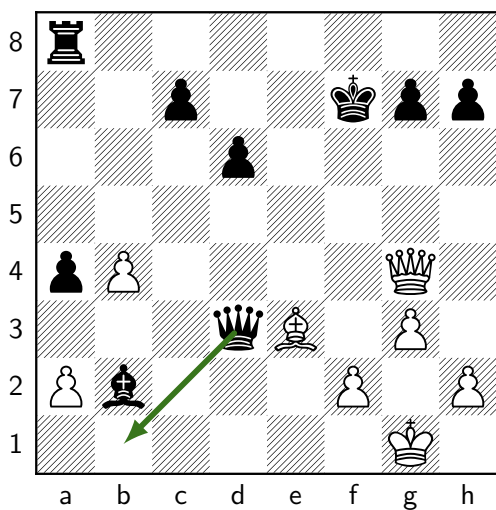
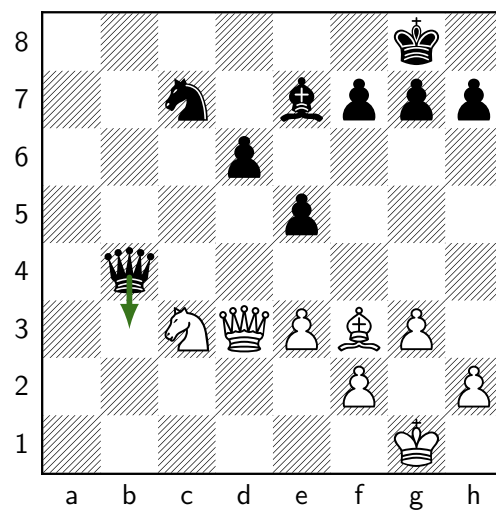


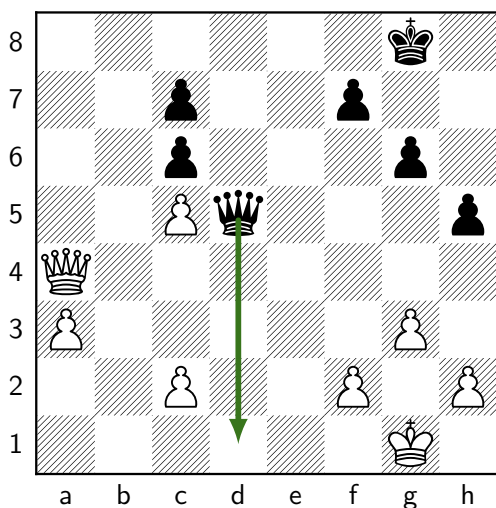
Figure 17: **Move 26, Black: Bxb2.** (a) Our position; Melanie plays Bxb2 (c3b2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2871` (ply 51) [structural distance $d=61$, QAP obj=25]: the master played Rbb8; under μ our Bc6 plays the role of that game’s Rb3 (a **cross-type** correspondence); cross-type roles: our Pg7 as their Rf8, our Bc3 as their Qc7; transferred move c3b2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1051` (ply 38) [structural distance $d=62$, QAP obj=16]: the master played Bxb7; under μ our Bc6 plays the role of that game’s Bd5; cross-type roles: our rb2 as their be6, our be3 as their pb7; transferred move c3b2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3501` (ply 52) [structural distance $d=68$, QAP obj=4]: the master played Bxf7; under μ our Bc6 plays the role of that game’s Bd5; cross-type roles: our Pg7 as their Rb1, our qg4 as their Pf2; transferred move c3b2 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Bxb2 (summed QAP objective 48, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



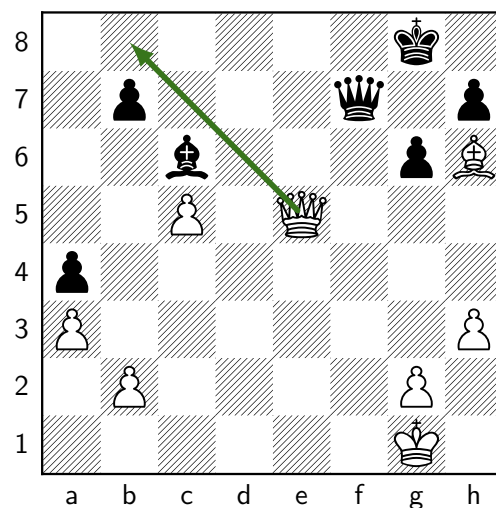
(a)



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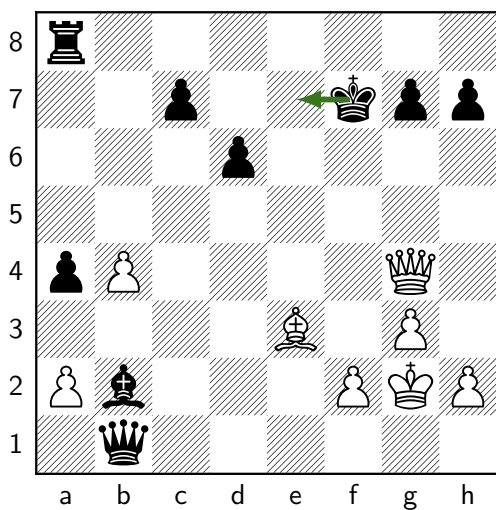


(c)

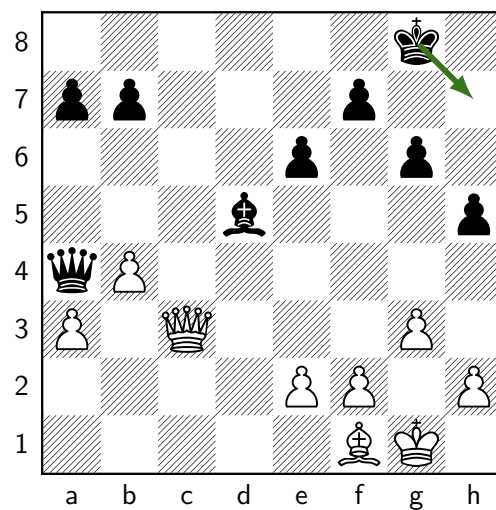


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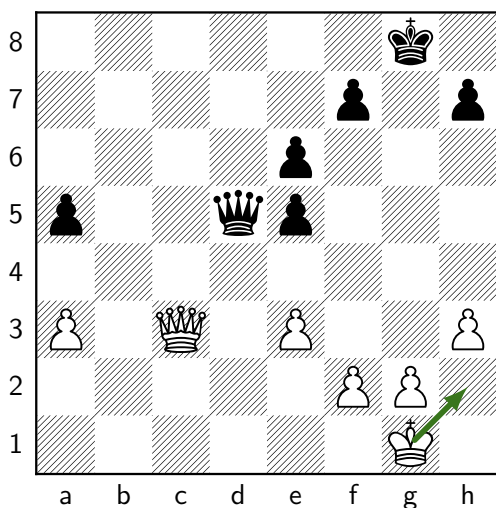
Figure 18: **Move 27, Black: Qb1+.** (a) Our position; Melanie plays Qb1+ (d3b1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1502` (ply 57) [structural distance $d=42$, QAP obj=23]: the master played Qb3; under μ our Qd6 plays the role of that game’s Qb5; cross-type roles: our Bb2 as their Nc7, our Pc7 as their Be7; transferred move d3b1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#998` (ply 59) [structural distance $d=40$, QAP obj=11]: the master played Qd1+; under μ our Qd6 plays the role of that game’s Qd4; cross-type roles: our Bb2 as their Pc7, our be3 as their pc5; transferred move d3b1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2709` (ply 68) [structural distance $d=40$, QAP obj=4]: the master played Qb8+; under μ our Qd6 plays the role of that game’s Qe5; cross-type roles: our Ra8 as their Ph3, our qg4 as their bc6; transferred move d3b1 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Qb1+ (summed QAP objective 62, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



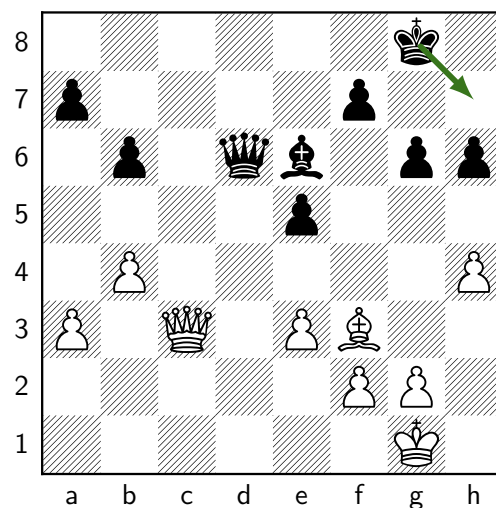
(a)



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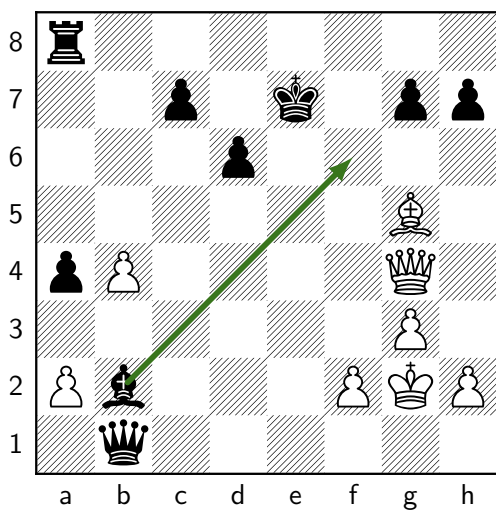


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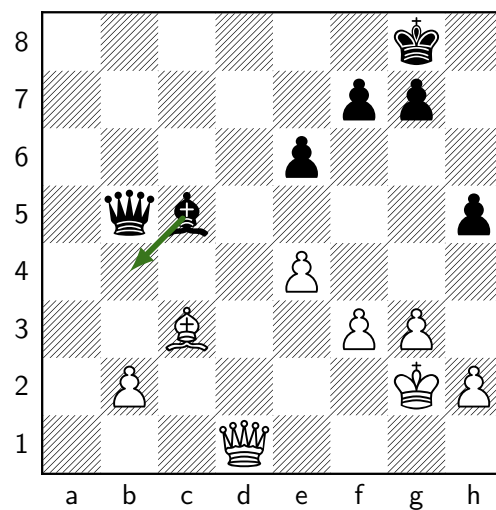


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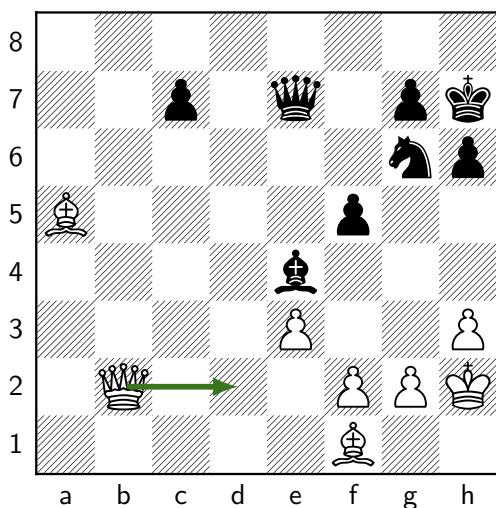
Figure 19: **Move 28, Black: Ke7**. (a) Our position; Melanie plays Ke7 (f7e7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0002#626 (ply 65) [structural distance $d=45$, QAP obj=34]: the master played Kh7; under μ our Kf2 plays the role of that game's Kg1; cross-type roles: our Bb2 as their Pa7, our be3 as their Pe6; transferred move f7e7 (**endorses this move**). (c) Retrieved from ccr1_top_p0000#700 (ply 54) [structural distance $d=47$, QAP obj=24]: the master played Kh2; under μ our Kf2 plays the role of that game's Kg1; cross-type roles: our Bb2 as their Pe3, our be3 as their Ph3; transferred move f7e7 (**endorses this move**). (d) Retrieved from ccr1_top_p0000#1961 (ply 65) [structural distance $d=38$, QAP obj=20]: the master played Kh7; under μ our Kf2 plays the role of that game's Kg1; cross-type roles: our Bb2 as their Pg6, our Ra8 as their ph4; transferred move f7e7 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Ke7 (summed QAP objective 80, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



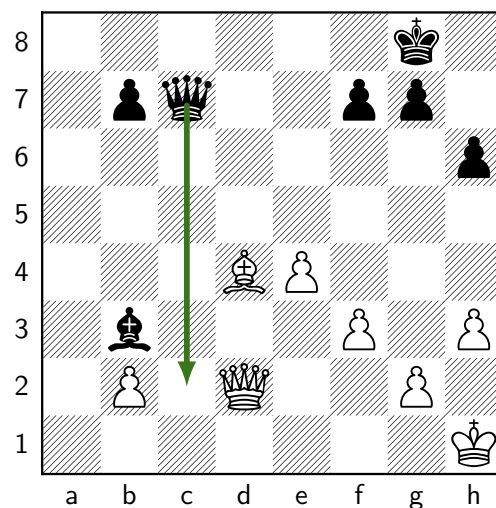
(a)



(b)



(c)



(d)

Figure 20: **Move 29, Black: Bf6.** (a) Our position; Melanie plays Bf6 (b2f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#5` (ply 63) [structural distance $d=45$, QAP obj=6]: the master played Bb4; under μ our Bb7 plays the role of that game’s Bc4; cross-type roles: our qg4 as their pb2, our pa2 as their qd1; transferred move b2f6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1685` (ply 60) [structural distance $d=44$, QAP obj=4]: the master played Qd2; under μ our Bb7 plays the role of that game’s Qb2 (a **cross-type** correspondence); cross-type roles: our Pg7 as their Bf1, our qg4 as their Pe3; transferred move b2f6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#876` (ply 57) [structural distance $d=38$, QAP obj=4]: the master played Qc2; under μ our Bb7 plays the role of that game’s Qc2 (a **cross-type** correspondence); cross-type roles: our Qb1 as their Bb3, our Ra8 as their pb2; transferred move b2f6 (**endorses this move**). Synthesis: 10 retrieved analogs transferred to Bf6 (summed QAP objective 38, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

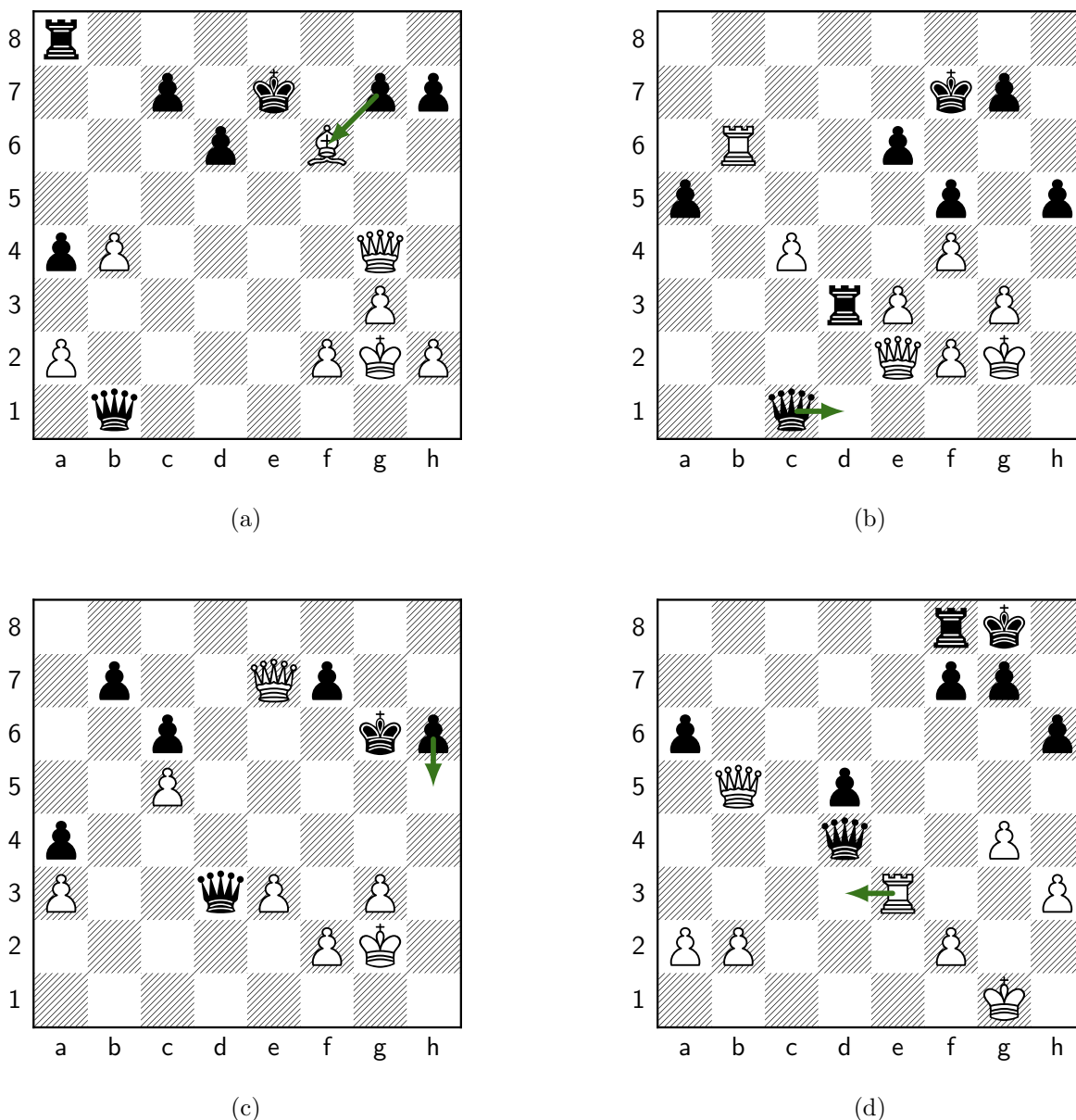
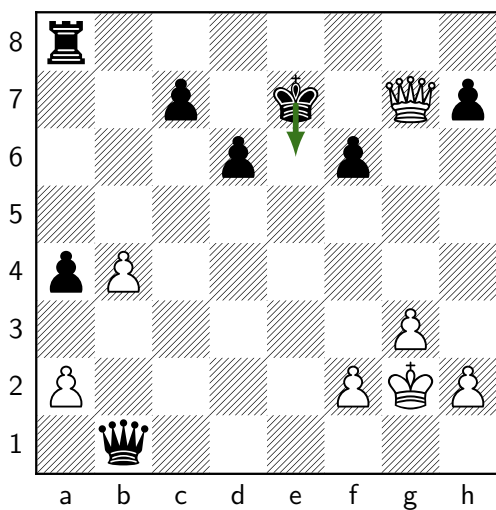
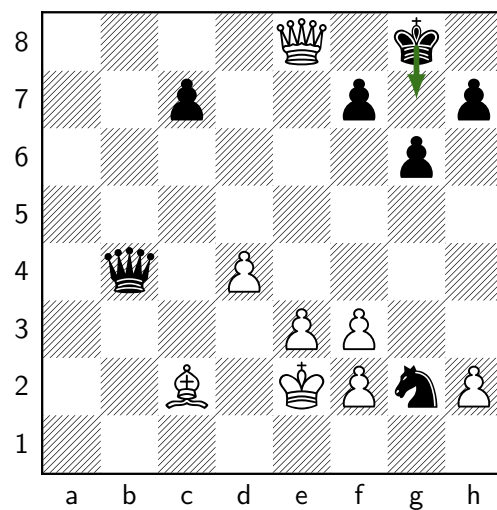


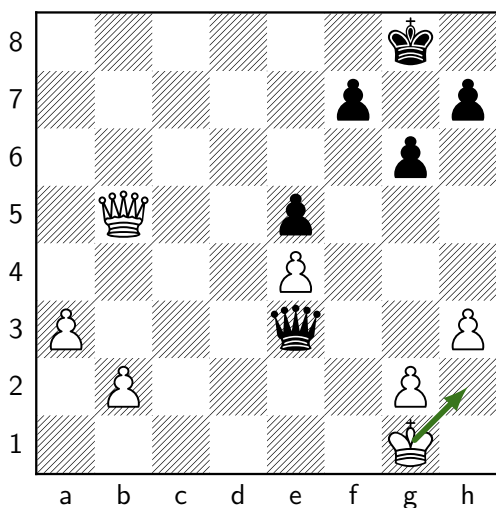
Figure 21: **Move 30, Black: gxf6.** (a) Our position; Melanie plays **gxf6** (**g7f6**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3253` (ply 71) [structural distance $d=53$, QAP obj=6]: the master played **Qd1**; under μ our *Pg2* plays the role of that game’s *Qc8* (a **cross-type** correspondence); cross-type roles: our *Ra8* as their *Pg7*, our *bf6* as their *rb6*; transferred move **g7f6** (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1494` (ply 69) [structural distance $d=50$, QAP obj=4]: the master played **h5**; under μ our *Pg2* plays the role of that game’s *Ph3*; cross-type roles: our *bf6* as their *Pb7*, our *Ra8* as their *pg3*; transferred move **g7f6** (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2120` (ply 54) [structural distance $d=41$, QAP obj=2]: the master played **Rd3**; under μ our *Pg2* plays the role of that game’s *Re3* (a **cross-type** correspondence); cross-type roles: our *Ra8* as their *Pa2*, our *bf6* as their *Pb2*; transferred move **g7f6** (**endorses this move**). Synthesis: 3 retrieved analogs transferred to **gxf6** (summed QAP objective 13, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



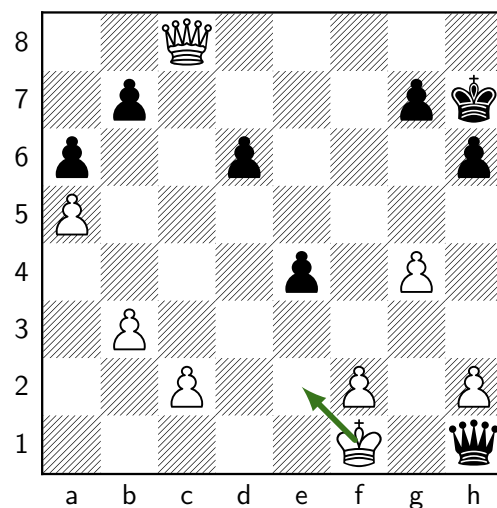
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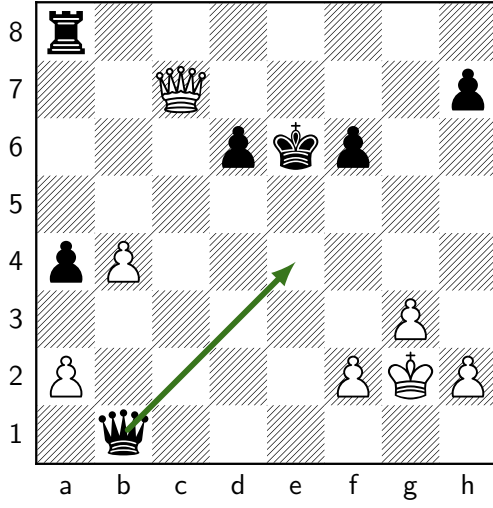


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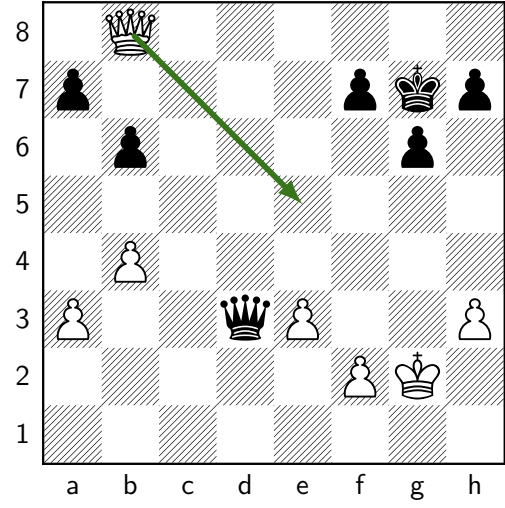


(d)

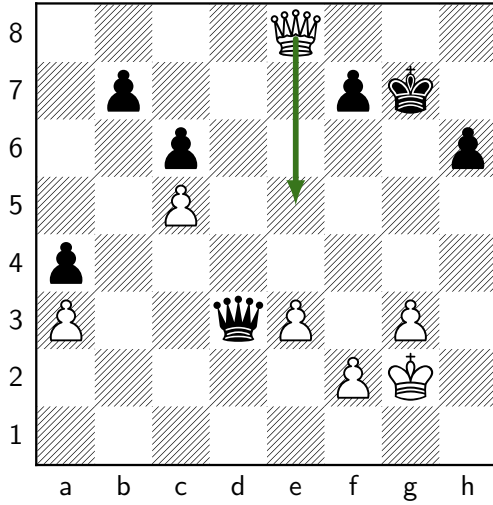
Figure 22: **Move 31, Black: Ke6**. (a) Our position; Melanie plays Ke6 (e7e6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3913` (ply 61) [structural distance $d=28$, QAP obj=82]: the master played Kg7; under μ our Ke2 plays the role of that game’s Kg1; cross-type roles: our pb4 as their bc2, our Ra8 as their Ng2; transferred move e7e6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1645` (ply 76) [structural distance $d=31$, QAP obj=77]: the master played Kh2; under μ our Ke2 plays the role of that game’s Kg1; transferred move e7e6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#103` (ply 74) [structural distance $d=27$, QAP obj=76]: the master played Ke2; under μ our Ke2 plays the role of that game’s Kf1; cross-type roles: our Ra8 as their Ph2, our Qb1 as their Pa5; transferred move e7e6 (**endorses this move**). Synthesis: 14 retrieved analogs transferred to Ke6 (summed QAP objective 795, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



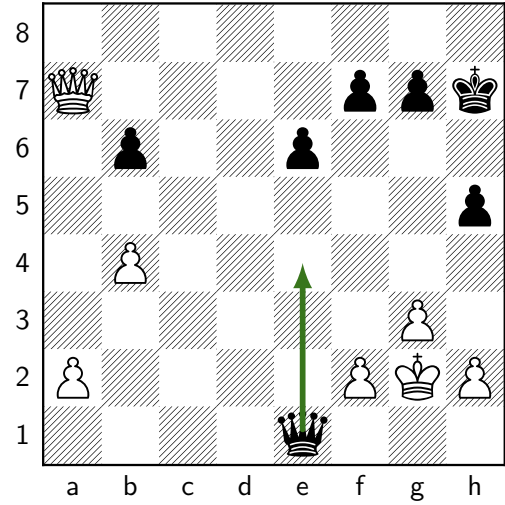
(a)



(b)

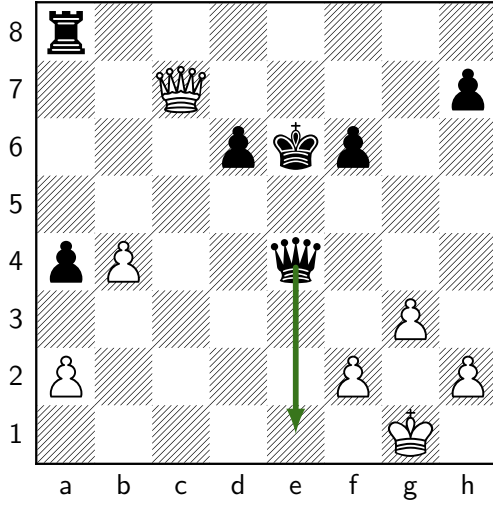


(c)

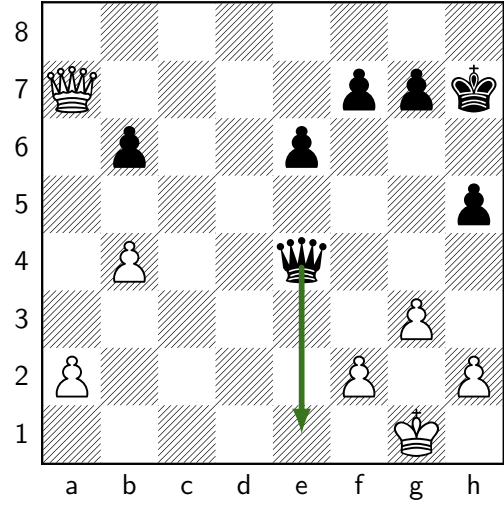


(d)

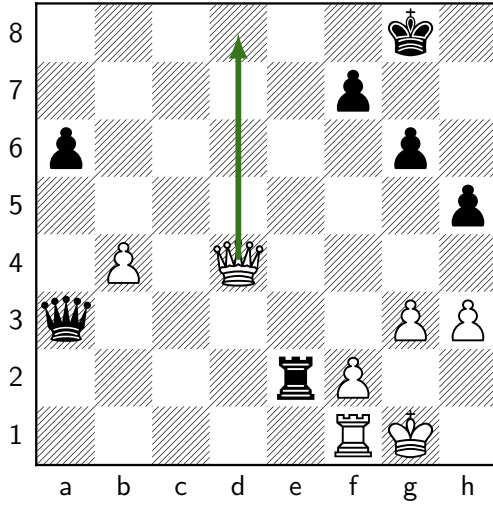
Figure 23: **Move 32, Black: Qe4+.** (a) Our position; Melanie plays Qe4+ (b1e4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2721` (ply 64) [structural distance $d=24$, QAP obj=66]: the master played Qe5+; under μ our Qb8 plays the role of that game’s Qb8; cross-type roles: our Ra8 as their Ph3; transferred move b1e4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1494` (ply 66) [structural distance $d=30$, QAP obj=58]: the master played Qe5+; under μ our Qb8 plays the role of that game’s Qe8; cross-type roles: our Ra8 as their Pc5; transferred move b1e4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3750` (ply 65) [structural distance $d=15$, QAP obj=57]: the master played Qe4+; under μ our Qb8 plays the role of that game’s Qe8; cross-type roles: our Ra8 as their Pe6; transferred move b1e4 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Qe4+ (summed QAP objective 268, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



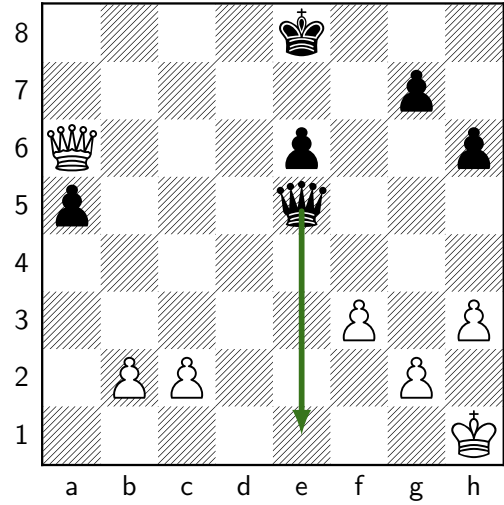
(a)



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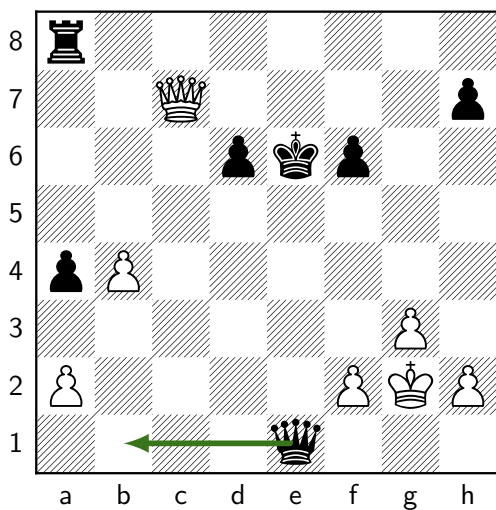


(c)

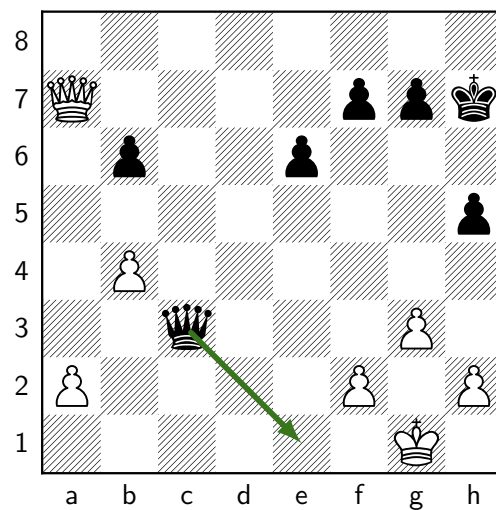


(d)

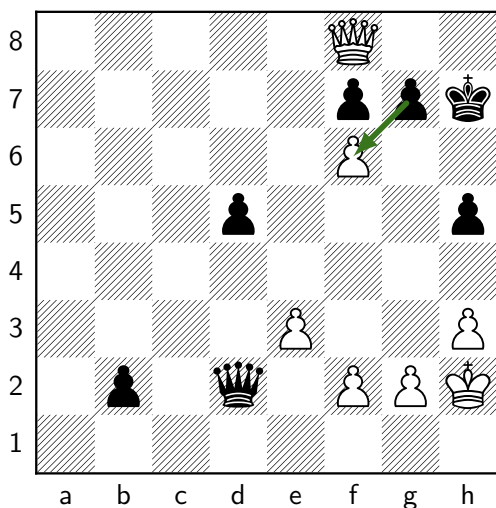
Figure 24: **Move 33, Black: Qe1+ (e4e1).** (a) Our position; Melanie plays Qe1+ (e4e1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3750` (ply 67) [structural distance $d=19$, QAP obj=34]: the master played Qe1+; under μ our Qe5 plays the role of that game’s Qe5; cross-type roles: our Ra8 as their Pe6; transferred move e4e1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2245` (ply 78) [structural distance $d=24$, QAP obj=32]: the master played Qd8+; under μ our Qe5 plays the role of that game’s Qd4; cross-type roles: our qc7 as their re2; transferred move e4e1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#665` (ply 67) [structural distance $d=28$, QAP obj=31]: the master played Qe1+; under μ our Qe5 plays the role of that game’s Qe4; cross-type roles: our Ra8 as their Pg7; transferred move e4e1 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Qe1+ (summed QAP objective 128, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



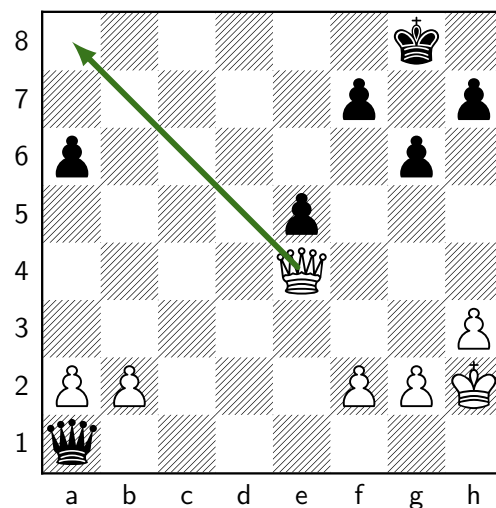
(a)



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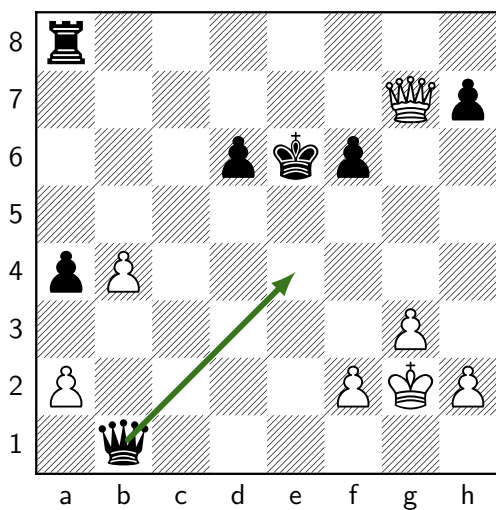


(c)

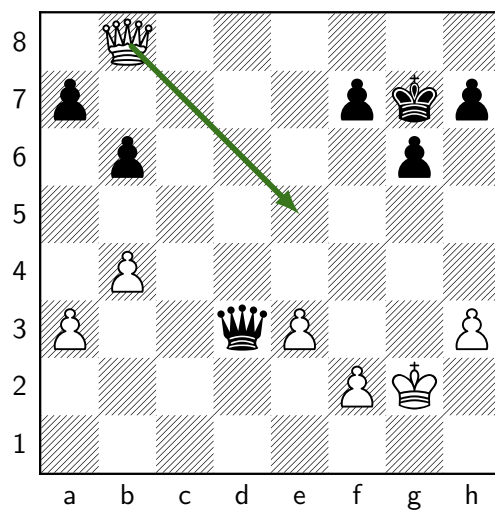


(d)

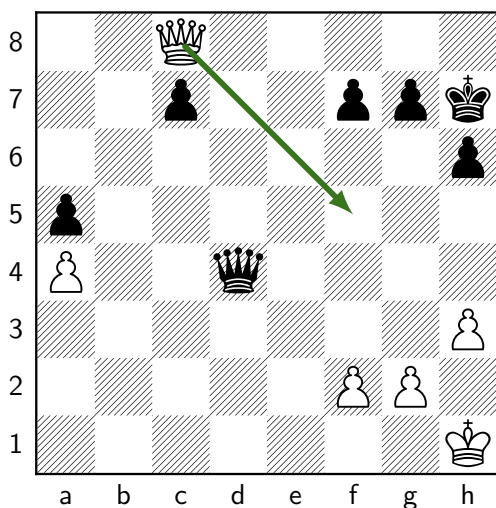
Figure 25: **Move 34, Black: Qb1**. (a) Our position; Melanie plays Qb1 (e1b1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3750` (ply 63) [structural distance $d=25$, QAP obj=45]: the master played Qe1+; under μ our Qe8 plays the role of that game’s Qc6; cross-type roles: our Ra8 as their ph2; transferred move e1b1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2764` (ply 71) [structural distance $d=23$, QAP obj=78]: the master played gxf6; under μ our Pd3 plays the role of that game’s Pg2; cross-type roles: our qc7 as their pf6, our Ra8 as their Pb2; transferred move d6d5. (d) Retrieved from `ccr1_top_p0000#98` (ply 50) [structural distance $d=18$, QAP obj=60]: the master played Qa8+; under μ our Qe8 plays the role of that game’s Qe4; cross-type roles: our Ra8 as their Pf2; transferred move e1a1. Synthesis: 2 retrieved analogs transferred to Qb1 (summed QAP objective 76, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



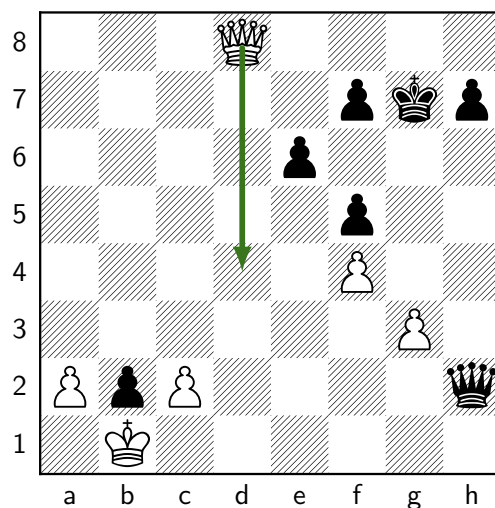
(a)



(b)

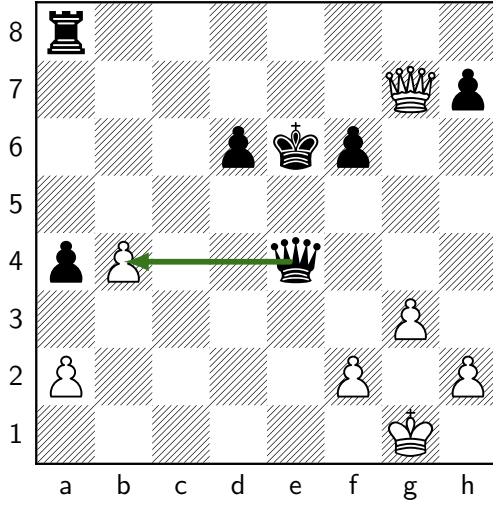


(c)

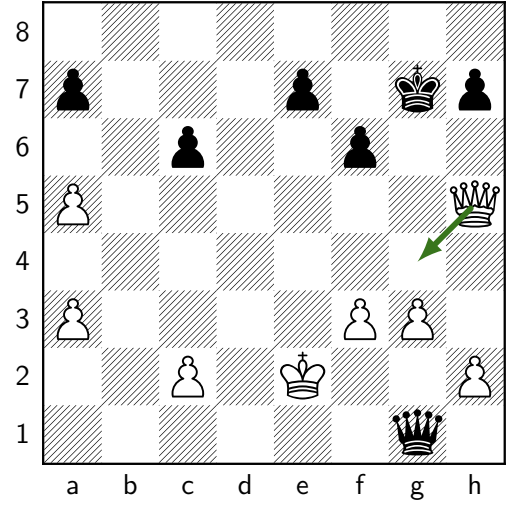


(d)

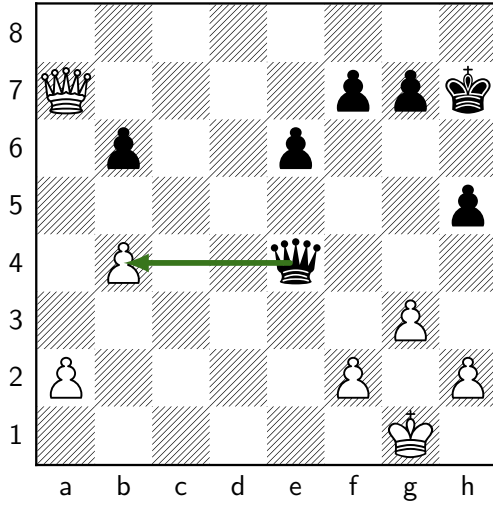
Figure 26: **Move 35, Black: Qe4+.** (a) Our position; Melanie plays Qe4+ (b1e4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2721` (ply 64) [structural distance $d=16$, QAP obj=74]: the master played Qe5+; under μ our Qb8 plays the role of that game’s Qb8; cross-type roles: our Ra8 as their Pb4; transferred move b1e4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#911` (ply 66) [structural distance $d=23$, QAP obj=41]: the master played Qf5+; under μ our Qb8 plays the role of that game’s Qc8; cross-type roles: our Ra8 as their pa5; transferred move b1e4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2301` (ply 70) [structural distance $d=29$, QAP obj=31]: the master played Qd4+; under μ our Qb8 plays the role of that game’s Qd8; cross-type roles: our Ra8 as their pb2; transferred move b1e4 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Qe4+ (summed QAP objective 205, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



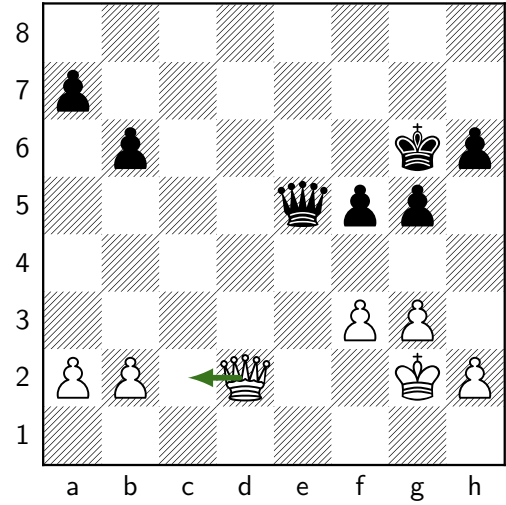
(a)



(b)

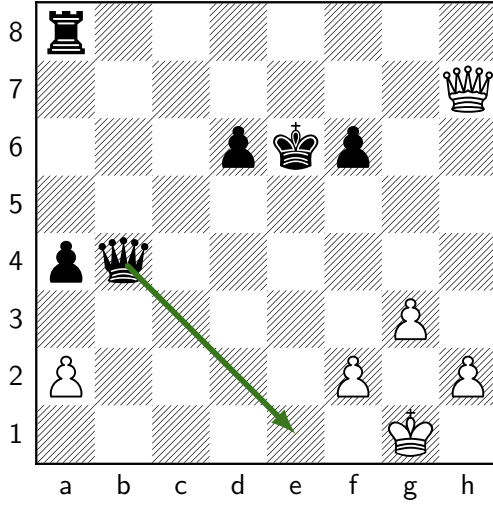


(c)

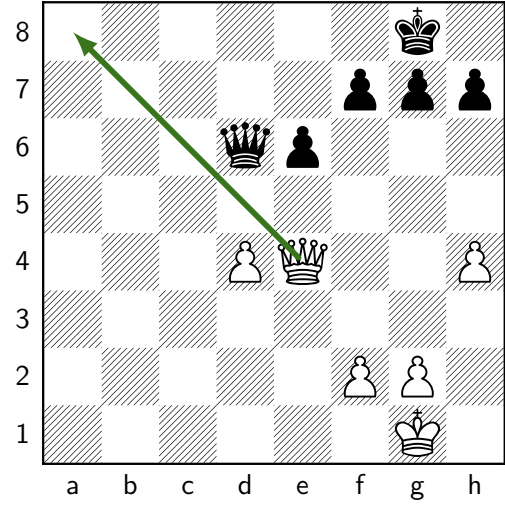


(d)

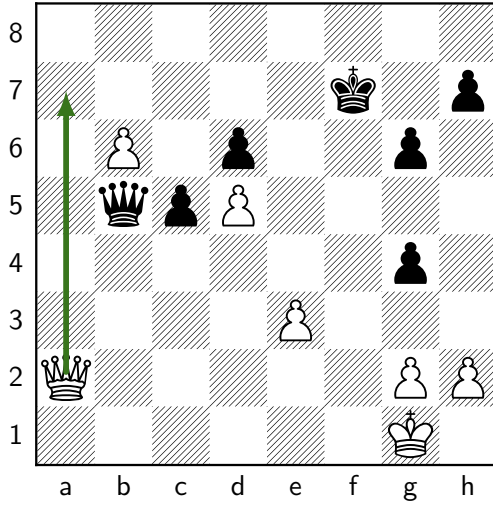
Figure 27: **Move 36, Black: Qxb4**. (a) Our position; Melanie plays Qxb4 (e4b4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1968` (ply 58) [structural distance $d=24$, QAP obj=45]: the master played Qg4+; under μ our Qe5 plays the role of that game’s Qh5; cross-type roles: our Ra8 as their Pa3; transferred move e4b4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3750` (ply 71) [structural distance $d=17$, QAP obj=33]: the master played Qxb4; under μ our Qe5 plays the role of that game’s Qe5; cross-type roles: our Ra8 as their Pe6; transferred move e4b4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1867` (ply 64) [structural distance $d=25$, QAP obj=29]: the master played Qc2; under μ our Qe5 plays the role of that game’s Qd2; cross-type roles: our Ra8 as their Pa2; transferred move e4b4 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Qxb4 (summed QAP objective 156, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



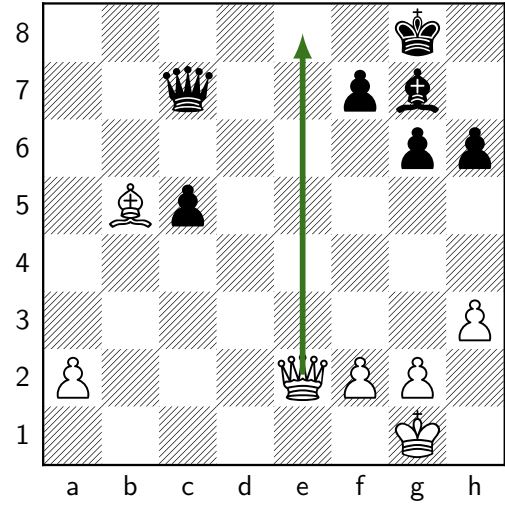
(a)



(b)

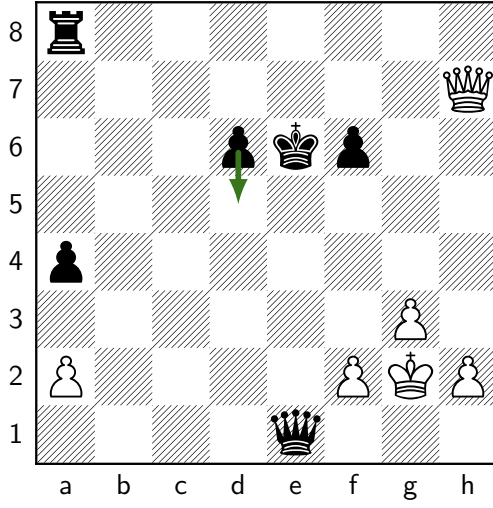


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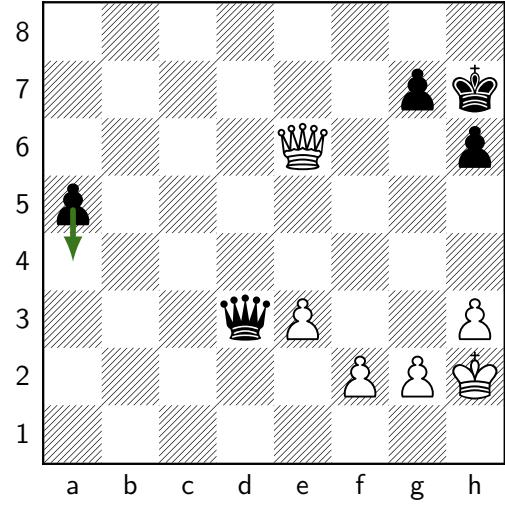


(d)

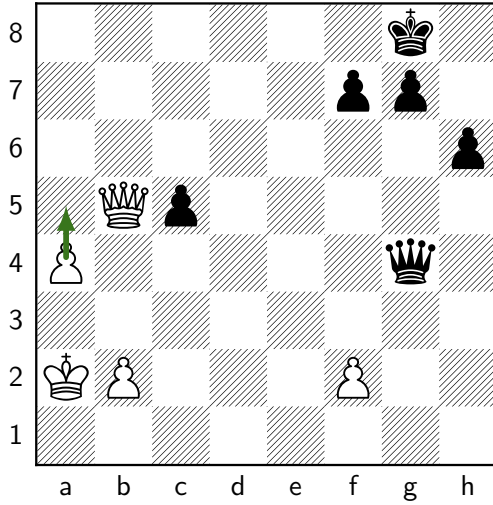
Figure 28: **Move 37, Black: Qe1+.** (a) Our position; Melanie plays Qe1+ (b4e1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `cctl_top_p0000#2185` (ply 58) [structural distance $d=31$, QAP obj=11]: the master played Qa8+; under μ our Qb5 plays the role of that game’s Qe4; cross-type roles: our Ra8 as their Ph4, our pg3 as their qd6; transferred move b4e1 (**endorses this move**). (c) Retrieved from `cctl_top_p0001#1457` (ply 80) [structural distance $d=29$, QAP obj=7]: the master played Qa7+; under μ our Qb5 plays the role of that game’s Qa2; cross-type roles: our qh7 as their Pe3, our pa2 as their qb5; transferred move b4e1 (**endorses this move**). (d) Retrieved from `cctl_top_p0000#34` (ply 82) [structural distance $d=32$, QAP obj=3]: the master played Qe8+; under μ our Qb5 plays the role of that game’s Qe2; cross-type roles: our Ra8 as their Pa2, our qh7 as their Ph3; transferred move b4e1 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Qe1+ (summed QAP objective 25, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



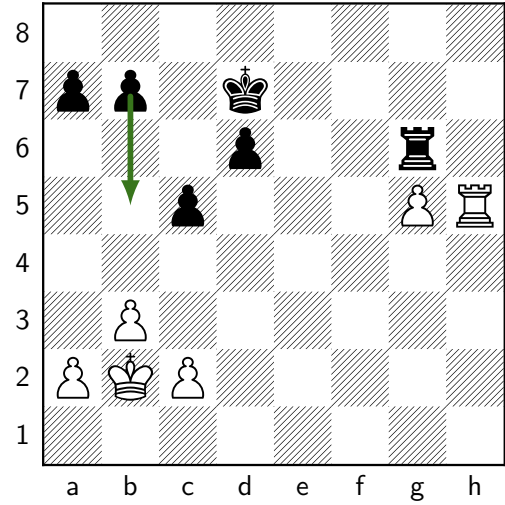
(a)



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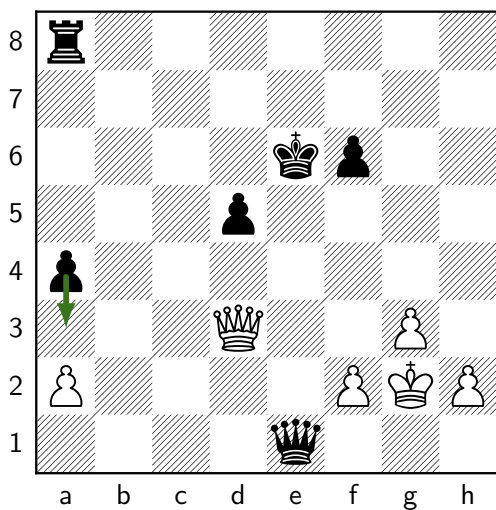


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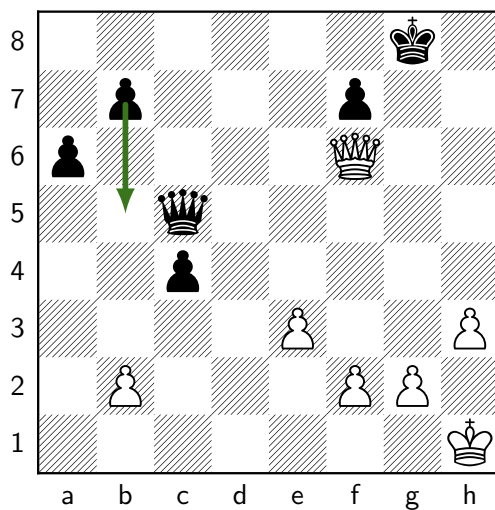


(d)

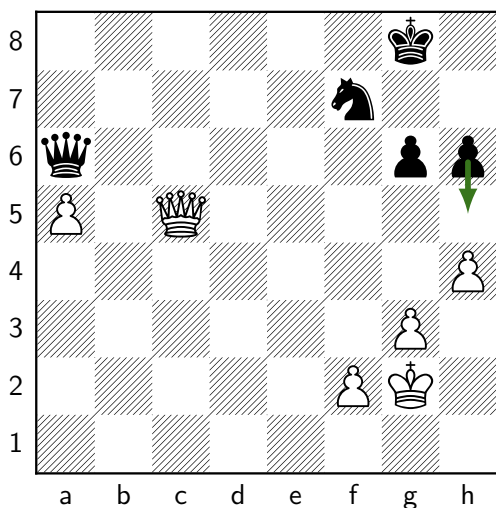
Figure 29: **Move 38, Black: d5.** (a) Our position; Melanie plays d5 (d6d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1509` (ply 89) [structural distance $d=19$, QAP obj=22]: the master played a4; under μ our $Pd3$ plays the role of that game’s $Pa4$; cross-type roles: our $Ra8$ as their $Ph6$; transferred move d6d5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1168` (ply 66) [structural distance $d=24$, QAP obj=19]: the master played a5; under μ our $Pd3$ plays the role of that game’s $Pa4$; cross-type roles: our $qh7$ as their $Pb2$, our $pg3$ as their $qg4$; transferred move d6d5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#433` (ply 75) [structural distance $d=26$, QAP obj=19]: the master played b5; under μ our $Pd3$ plays the role of that game’s $Pb2$; cross-type roles: our $Ra8$ as their $Pa7$, our $Qe1$ as their $Rg6$; transferred move d6d5 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to d5 (summed QAP objective 76, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



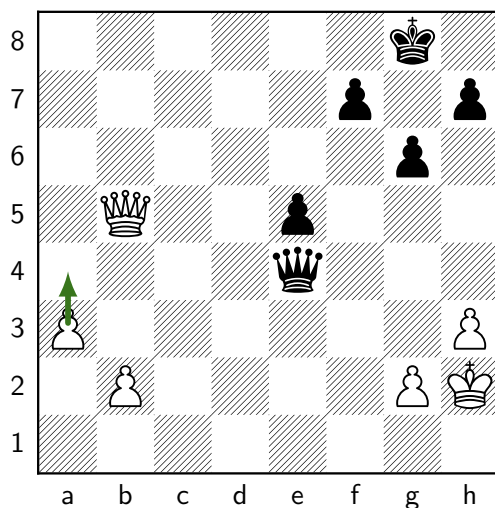
(a)



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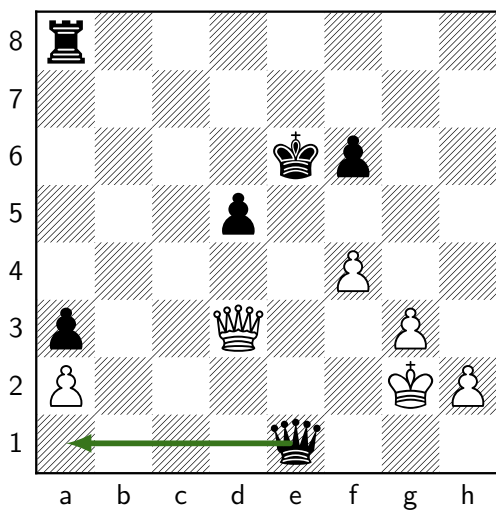


(c)

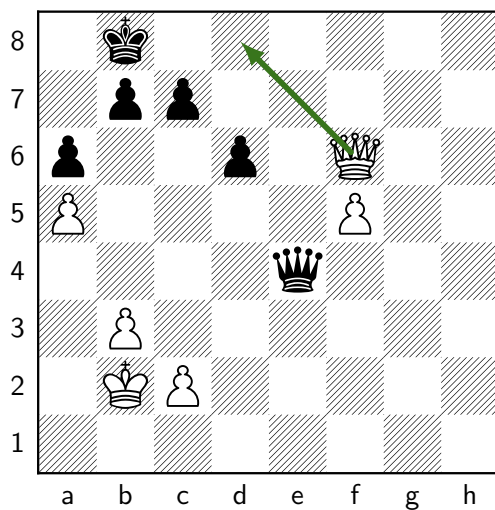


(d)

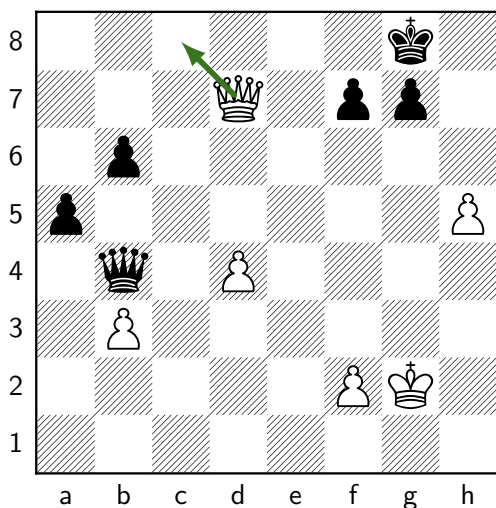
Figure 30: **Move 39, Black: a3.** (a) Our position; Melanie plays **a3** (**a4a3**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3316` (ply 87) [structural distance $d=24$, QAP obj=29]: the master played **b5**; under μ our $Pa5$ plays the role of that game’s $Pb2$; cross-type roles: our $Ra8$ as their $pg2$; transferred move **a4a3** (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2039` (ply 69) [structural distance $d=25$, QAP obj=21]: the master played **h5**; under μ our $Pa5$ plays the role of that game’s $Ph3$; cross-type roles: our $Pf6$ as their $Nf7$; transferred move **a4a3** (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1645` (ply 78) [structural distance $d=22$, QAP obj=18]: the master played **a4**; under μ our $Pa5$ plays the role of that game’s $Pa3$; cross-type roles: our $Ra8$ as their $pf7$; transferred move **a4a3** (**endorses this move**). Synthesis: 3 retrieved analogs transferred to **a3** (summed QAP objective 67, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



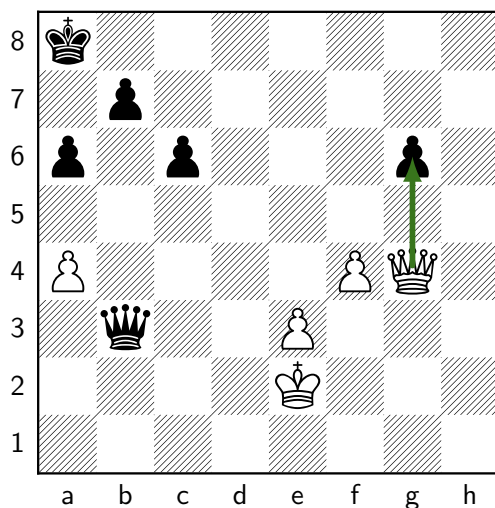
(a)



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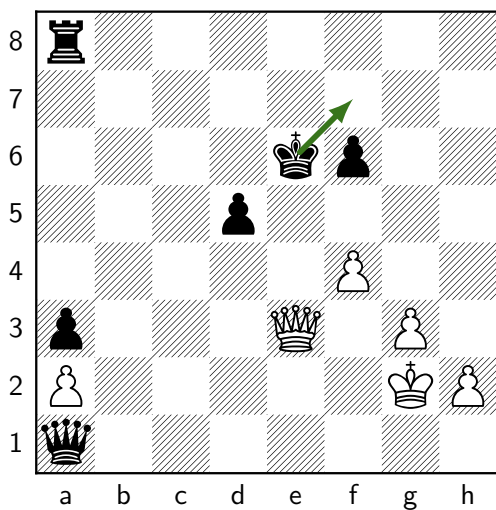


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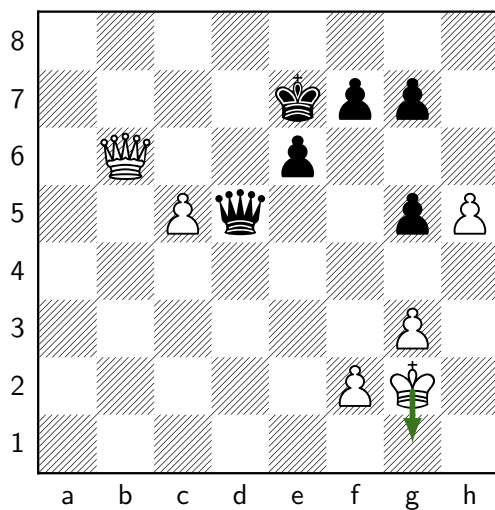


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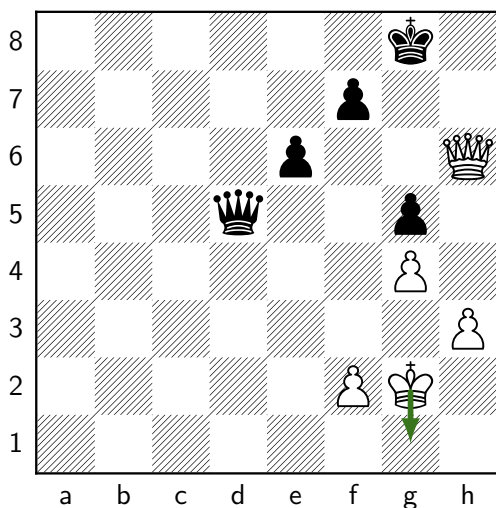
Figure 31: **Move 40, Black: Qa1**. (a) Our position; Melanie plays **Qa1 (e1a1)**. Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#787` (ply 88) [structural distance $d=18$, QAP obj=35]: the master played **Qd8+**; under μ our *Qe8* plays the role of that game's *Qf6*; cross-type roles: our *Ra8* as their *pb7*; transferred move **e1a1 (endorses this move)**. (c) Retrieved from `ccr1_top_p0000#434` (ply 94) [structural distance $d=14$, QAP obj=32]: the master played **Qc8+**; under μ our *Qe8* plays the role of that game's *Qd7*; cross-type roles: our *Ra8* as their *Pf2*; transferred move **e1a1 (endorses this move)**. (d) Retrieved from `ccr1_top_p0002#1938` (ply 76) [structural distance $d=17$, QAP obj=31]: the master played **Qxg6**; under μ our *Qe8* plays the role of that game's *Qg4*; transferred move **e1a1 (endorses this move)**. Synthesis: 13 retrieved analogs transferred to **Qa1** (summed QAP objective 319, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



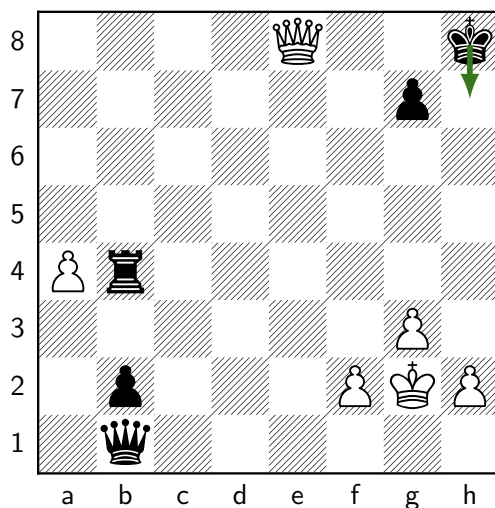
(a)



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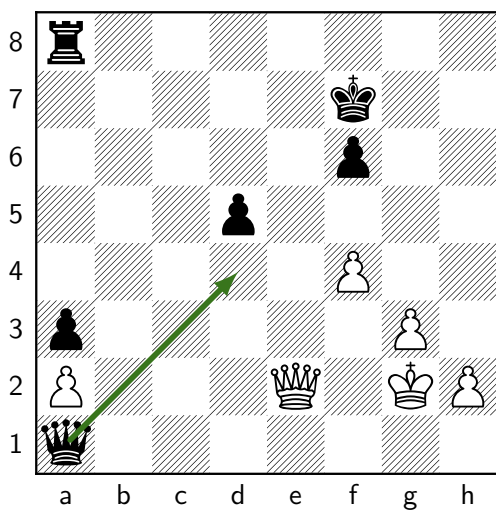


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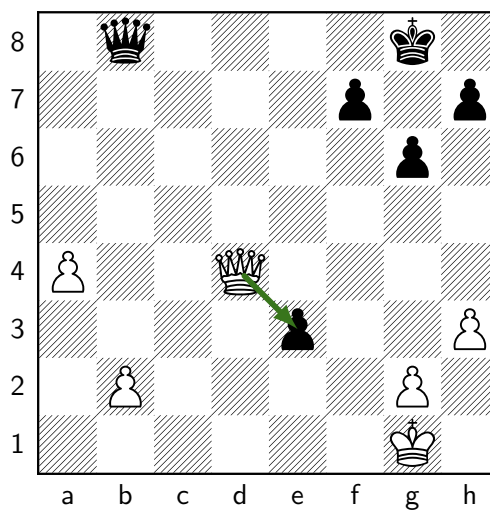


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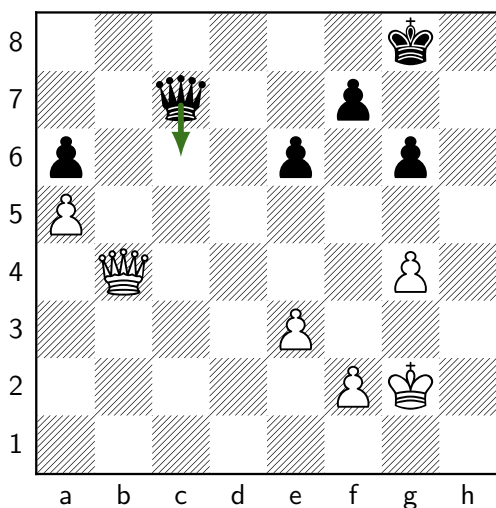
Figure 32: **Move 41, Black: Kf7.** (a) Our position; Melanie plays Kf7 (e6f7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#3321` (ply 88) [structural distance $d=19$, QAP obj=79]: the master played Kg1; under μ our *Ke3* plays the role of that game’s *Kg2*; cross-type roles: our *Ra8* as their *pg7*; transferred move e6f7 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#623` (ply 82) [structural distance $d=26$, QAP obj=77]: the master played Kg1; under μ our *Ke3* plays the role of that game’s *Kg2*; transferred move e6f7 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#2373` (ply 83) [structural distance $d=25$, QAP obj=71]: the master played Kh7; under μ our *Ke3* plays the role of that game’s *Kh1*; transferred move e6f7 (**endorses this move**). Synthesis: 29 retrieved analogs transferred to Kf7 (summed QAP objective 1613, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



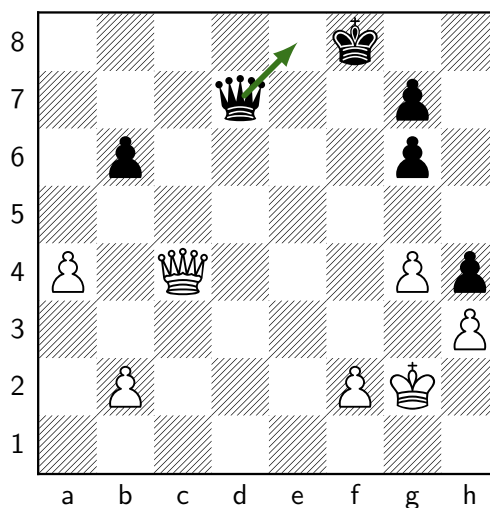
(a)



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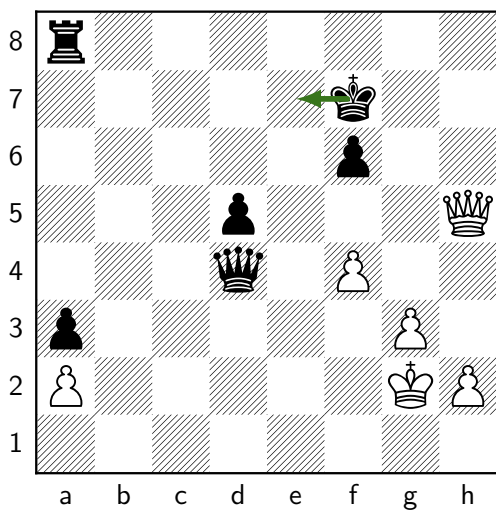


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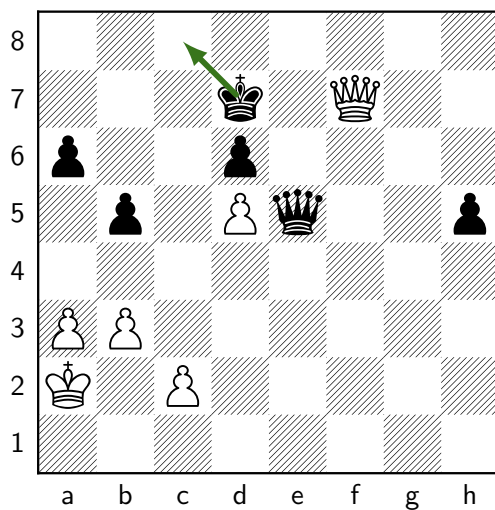


(d)

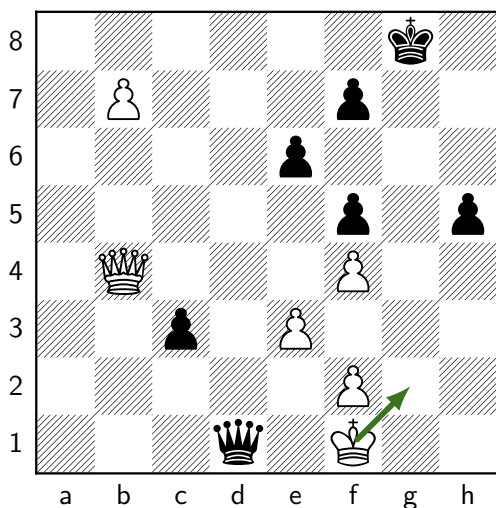
Figure 33: **Move 42, Black: Qd4.** (a) Our position; Melanie plays Qd4 (a1d4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1645` (ply 86) [structural distance $d=23$, QAP obj=21]: the master played Qxe3; under μ our Qa8 plays the role of that game’s Qd4; cross-type roles: our Ra8 as their Ph3; transferred move a1d4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2113` (ply 89) [structural distance $d=20$, QAP obj=14]: the master played Qc6+; under μ our Qa8 plays the role of that game’s Qc2; cross-type roles: our Ra8 as their pe3; transferred move a1d4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2619` (ply 97) [structural distance $d=22$, QAP obj=13]: the master played Qe8; under μ our Qa8 plays the role of that game’s Qd2; cross-type roles: our Ra8 as their Pg6; transferred move a1d4 (**endorses this move**). Synthesis: 10 retrieved analogs transferred to Qd4 (summed QAP objective 124, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



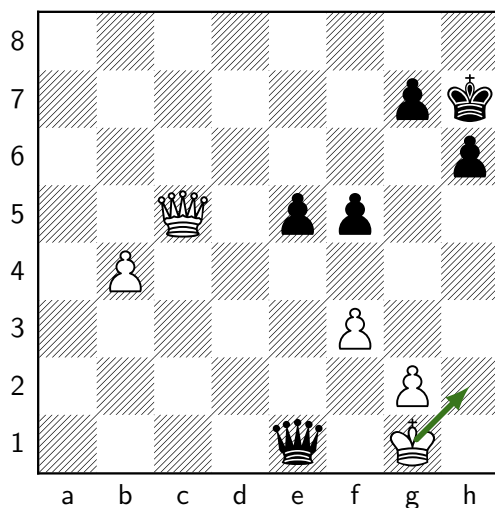
(a)



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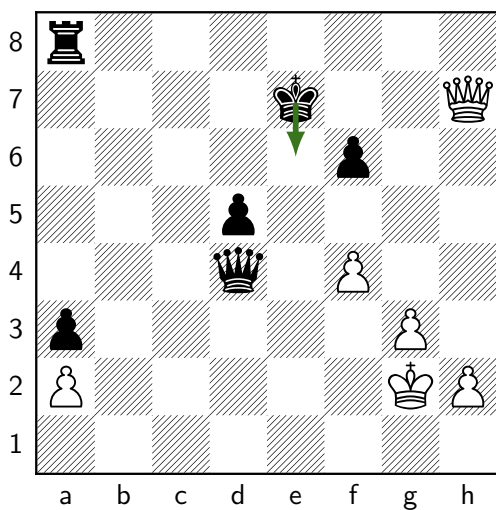


(c)

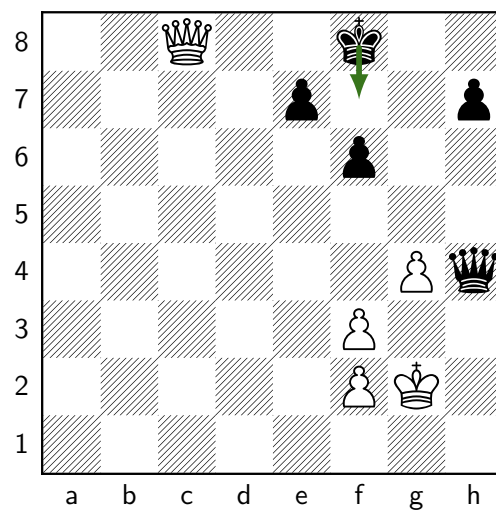


(d)

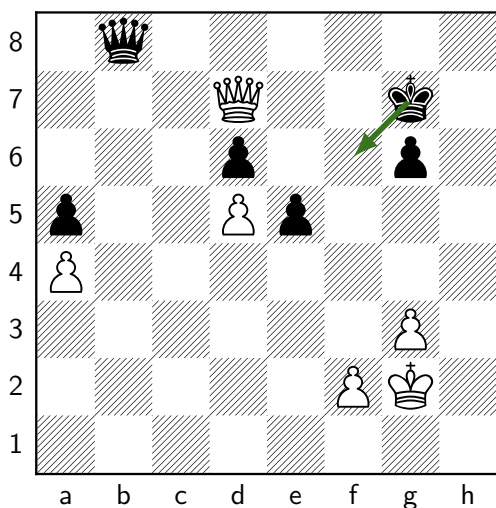
Figure 34: **Move 43, Black: Ke7**. (a) Our position; Melanie plays **Ke7** (**f7e7**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2312` (ply 99) [structural distance $d=18$, QAP obj=70]: the master played **Kc8**; under μ our **Kf2** plays the role of that game’s **Kd2**; cross-type roles: our **Ra8** as their **Pa6**; transferred move **f7e7** (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1769` (ply 84) [structural distance $d=20$, QAP obj=69]: the master played **Kg2**; under μ our **Kf2** plays the role of that game’s **Kf1**; cross-type roles: our **Qd4** as their **Pe3**, our **Ra8** as their **Qb4**; transferred move **f7e7** (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#36` (ply 96) [structural distance $d=23$, QAP obj=68]: the master played **Kh2**; under μ our **Kf2** plays the role of that game’s **Kg1**; transferred move **f7e7** (**endorses this move**). Synthesis: 15 retrieved analogs transferred to **Ke7** (summed QAP objective 770, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



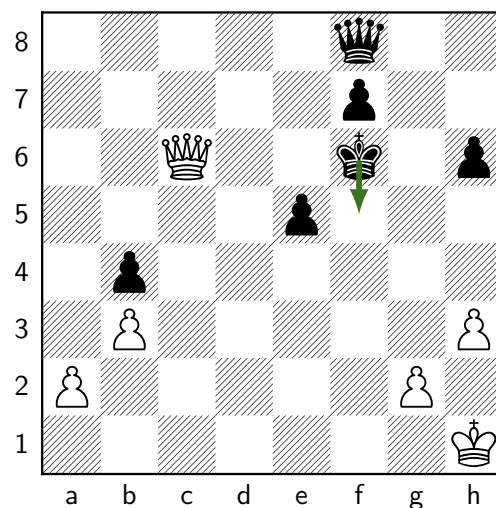
(a)



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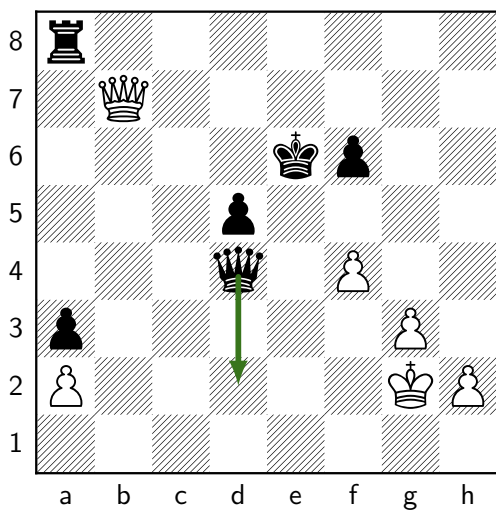


(c)

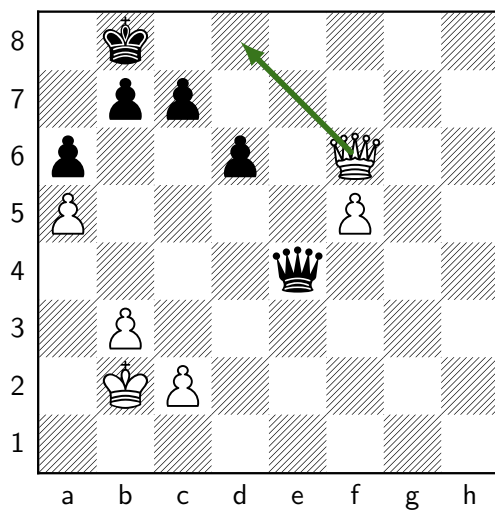


(d)

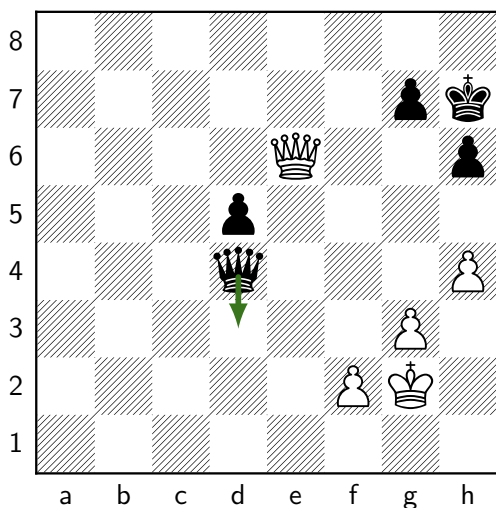
Figure 35: **Move 44, Black: Ke6 (e7e6)**. (a) Our position; Melanie plays Ke6 (e7e6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#3881` (ply 87) [structural distance $d=23$, QAP obj=74]: the master played Kf7; under μ our Ke2 plays the role of that game’s Kf1; cross-type roles: our Ra8 as their Ph7; transferred move e7e6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#1512` (ply 87) [structural distance $d=22$, QAP obj=73]: the master played Kf6; under μ our Ke2 plays the role of that game’s Kg2; cross-type roles: our Ra8 as their Qb8, our Qd4 as their Pd6; transferred move e7e6 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#2914` (ply 77) [structural distance $d=23$, QAP obj=71]: the master played Kf5; under μ our Ke2 plays the role of that game’s Kf3; cross-type roles: our Ra8 as their pb3; transferred move e7e6 (**endorses this move**). Synthesis: 16 retrieved analogs transferred to Ke6 (summed QAP objective 909, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



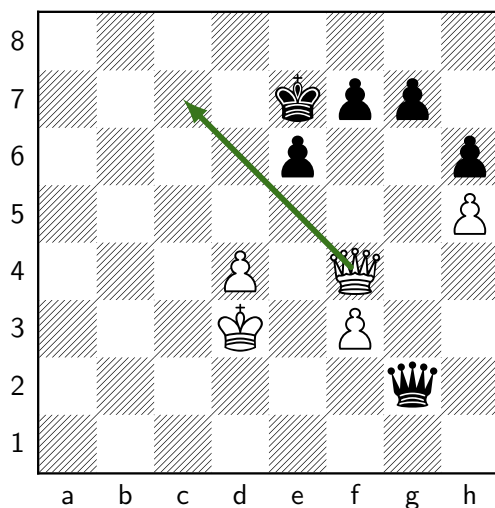
(a)



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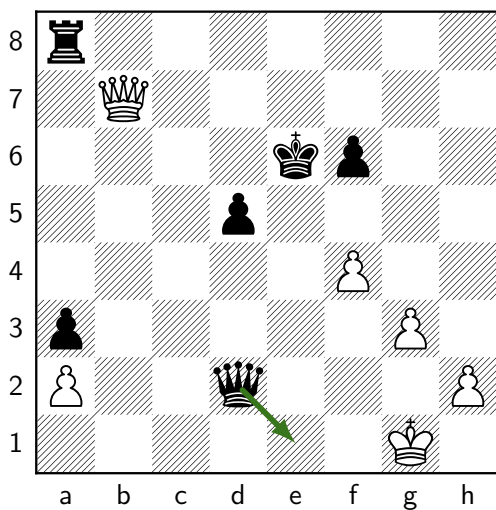


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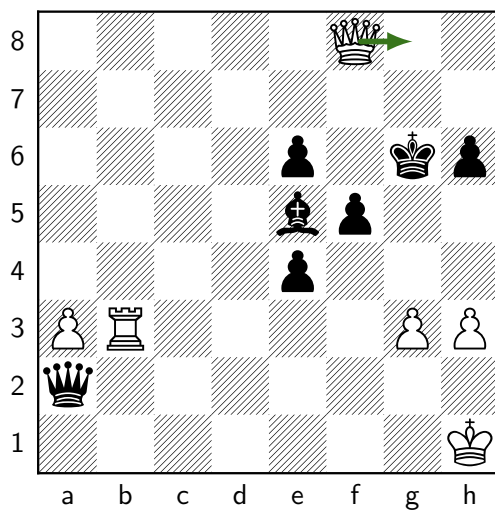


(d)

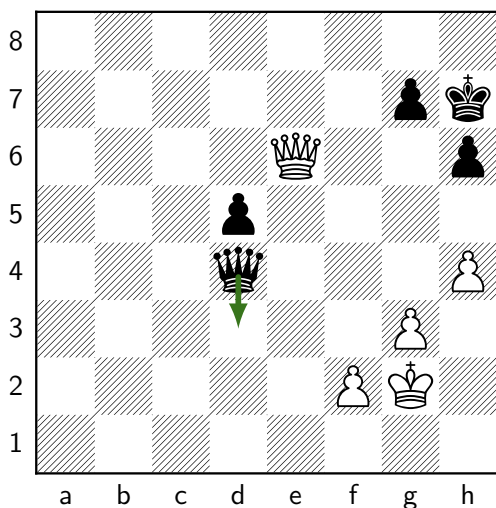
Figure 36: **Move 45, Black: Qd2+.** (a) Our position; Melanie plays Qd2+ (d4d2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#787` (ply 88) [structural distance $d=25$, QAP obj=5]: the master played Qd8+; under μ our Qd5 plays the role of that game's Qf6; cross-type roles: our Ra8 as their Pc2; transferred move d4d2 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#1110` (ply 85) [structural distance $d=22$, QAP obj=13]: the master played Qd3; under μ our Qd5 plays the role of that game's Qd5; cross-type roles: our Ra8 as their Ph6; transferred move d4c3. (d) Retrieved from `ccrl_top_p0000#2185` (ply 94) [structural distance $d=24$, QAP obj=13]: the master played Qc7+; under μ our Qd5 plays the role of that game's Qf4; transferred move d4a1. Synthesis: 2 retrieved analogs transferred to Qd2+ (summed QAP objective 10, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



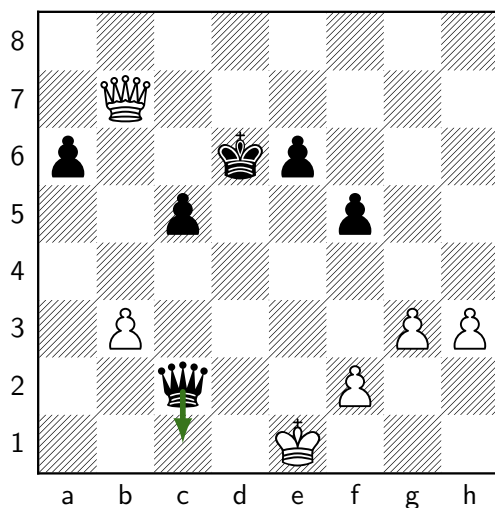
(a)



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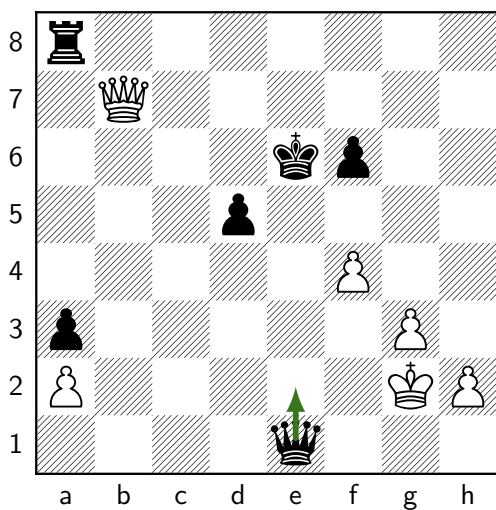


(c)

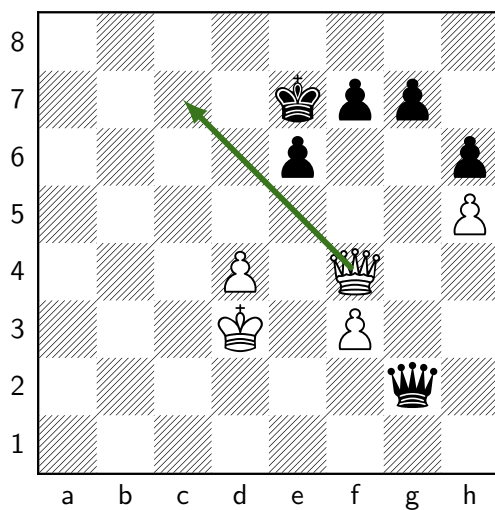


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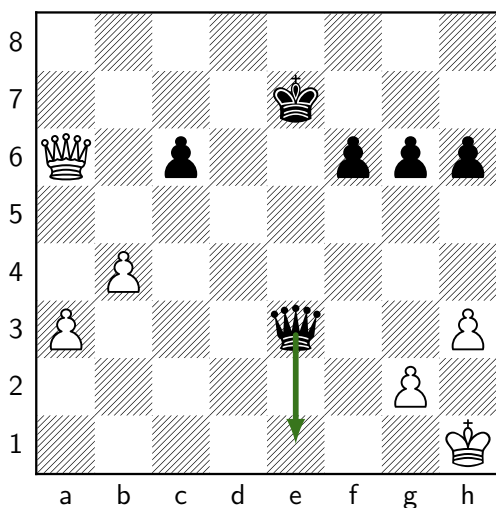
Figure 37: **Move 46, Black: Qe1+.** (a) Our position; Melanie plays Qe1+ (d2e1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2536` (ply 104) [structural distance $d=23$, QAP obj=25]: the master played Qg8+; under μ our Qd7 plays the role of that game’s Qf8; cross-type roles: our Pa3 as their be5; transferred move d2e1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1110` (ply 85) [structural distance $d=29$, QAP obj=11]: the master played Qd3; under μ our Qd7 plays the role of that game’s Qd5; transferred move d2c3. (d) Retrieved from `ccr1_top_p0000#662` (ply 105) [structural distance $d=29$, QAP obj=11]: the master played Qc1+; under μ our Qd7 plays the role of that game’s Qc7; cross-type roles: our Ra8 as their ph3; transferred move d2d1. Synthesis: 1 retrieved analog transferred to Qe1+ (summed QAP objective 25, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



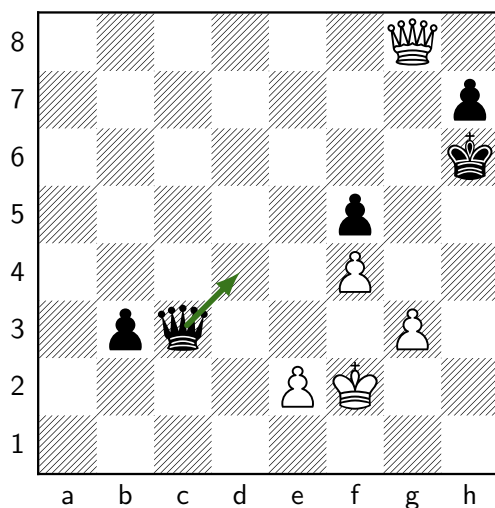
(a)



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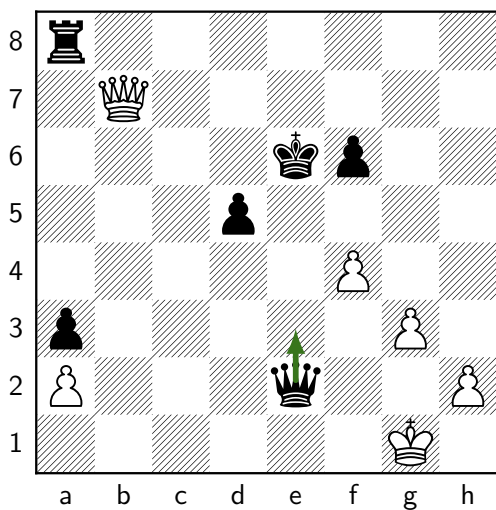


(c)

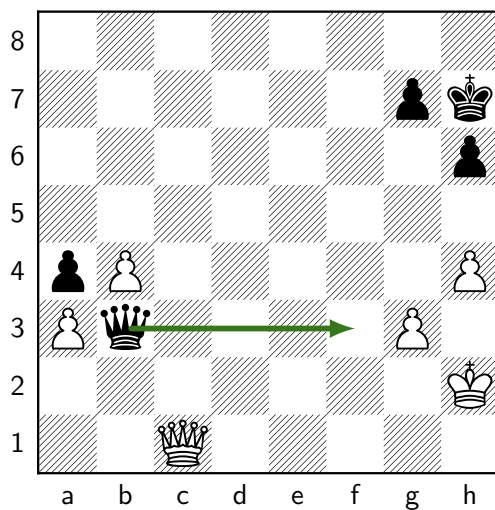


(d)

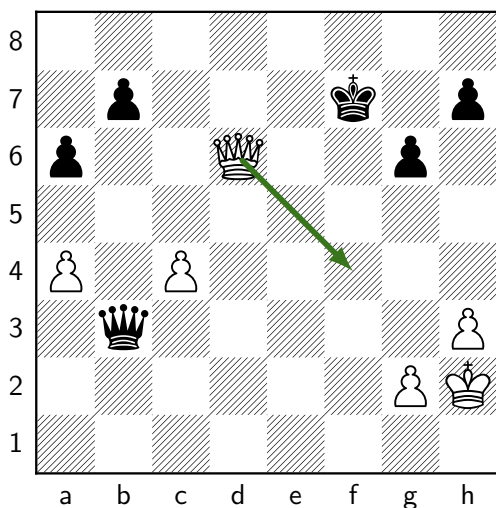
Figure 38: **Move 47, Black: Qe2+.** (a) Our position; Melanie plays Qe2+ (e1e2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2185` (ply 94) [structural distance $d=26$, QAP obj=6]: the master played Qc7+; under μ our Qe8 plays the role of that game’s Qf4; cross-type roles: our Ra8 as their Pf3; transferred move e1e2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#968` (ply 105) [structural distance $d=24$, QAP obj=5]: the master played Qe1+; under μ our Qe8 plays the role of that game’s Qe6; cross-type roles: our ph2 as their qa6, our Ra8 as their Pg6; transferred move e1e2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2645` (ply 83) [structural distance $d=24$, QAP obj=4]: the master played Qd4+; under μ our Qe8 plays the role of that game’s Qc6; cross-type roles: our Ra8 as their Ph7; transferred move e1e2 (**endorses this move**). Synthesis: 11 retrieved analogs transferred to Qe2+ (summed QAP objective 45, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



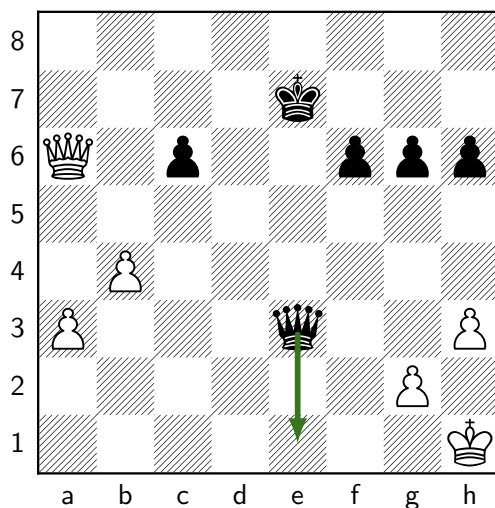
(a)



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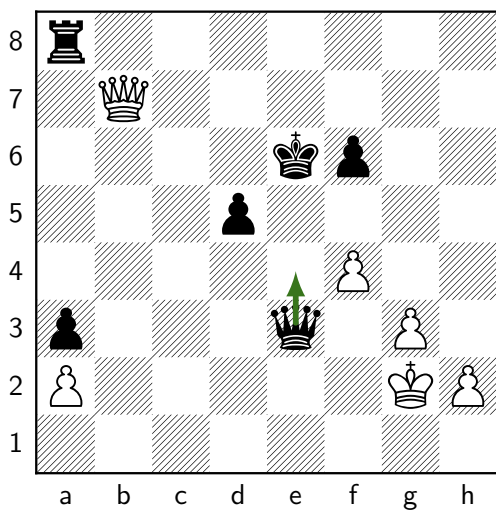


(c)

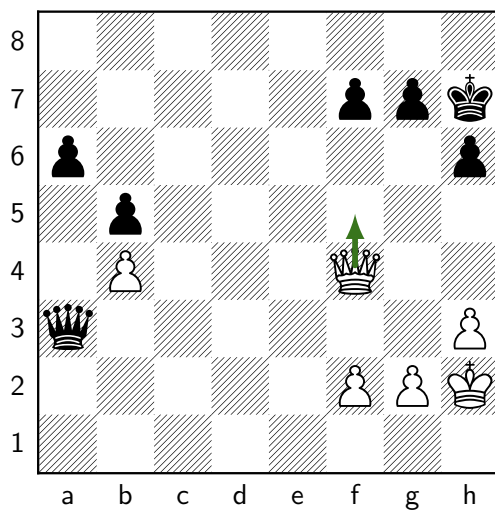


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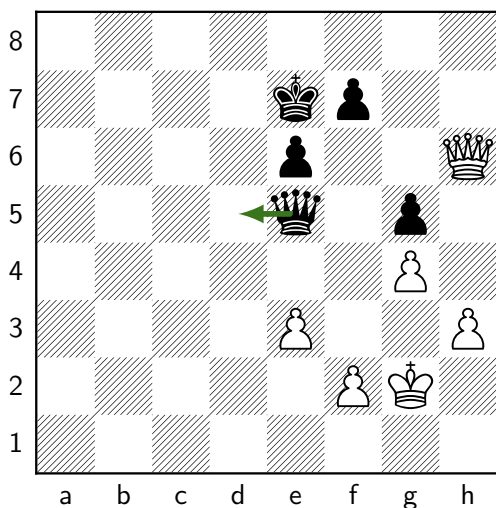
Figure 39: **Move 48, Black: Qe3+.** (a) Our position; Melanie plays Qe3+ (e2e3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top.p0000#3076` (ply 103) [structural distance $d=26$, QAP obj=2]: the master played Qf3; under μ our Qe7 plays the role of that game’s Qb6; cross-type roles: our Ra8 as their pb4; transferred move e2e3 (**endorses this move**). (c) Retrieved from `ccr1_top.p0000#2225` (ply 86) [structural distance $d=23$, QAP obj=9]: the master played Qf4+; under μ our Qe7 plays the role of that game’s Qd6; cross-type roles: our Ra8 as their Pc4; transferred move e2b5. (d) Retrieved from `ccr1_top.p0002#968` (ply 105) [structural distance $d=26$, QAP obj=7]: the master played Qe1+; under μ our Qe7 plays the role of that game’s Qe6; cross-type roles: our pf4 as their qa6, our Ra8 as their Pg6; transferred move e2e1. Synthesis: 1 retrieved analog transferred to Qe3+ (summed QAP objective 2, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



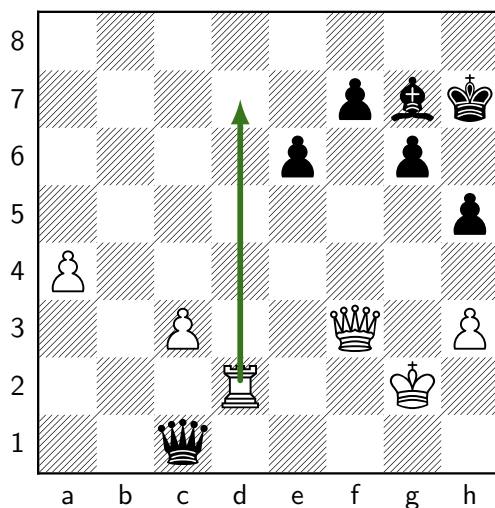
(a)



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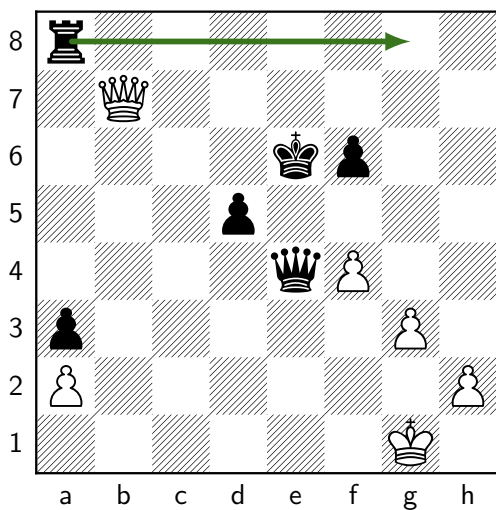


(c)

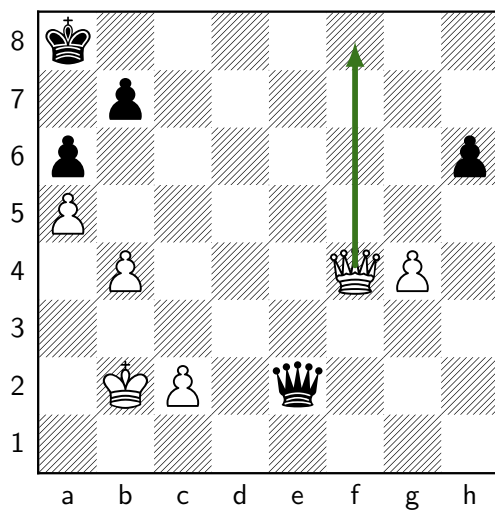


(d)

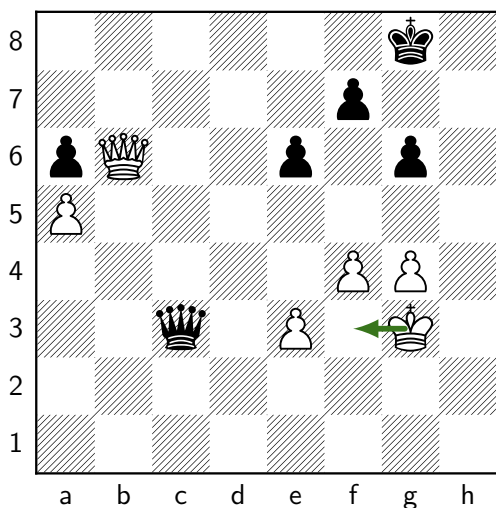
Figure 40: **Move 49, Black: Qe4+.** (a) Our position; Melanie plays **Qe4+** (**e3e4**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2507` (ply 86) [structural distance $d=27$, QAP obj=4]: the master played **Qf5+**; under μ our *Qe6* plays the role of that game’s *Qf4*; cross-type roles: our *Ra8* as their *pb5*; transferred move **e3e4** (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2735` (ply 109) [structural distance $d=29$, QAP obj=4]: the master played **Qd5+**; under μ our *Qe6* plays the role of that game’s *Qe4*; cross-type roles: our *Ra8* as their *Pf7*; transferred move **e3e4** (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#913` (ply 88) [structural distance $d=22$, QAP obj=25]: the master played **Rd7**; under μ our *Ra1* plays the role of that game’s *Rd2*; transferred move **a8a4**. Synthesis: 2 retrieved analogs transferred to **Qe4+** (summed QAP objective 8, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



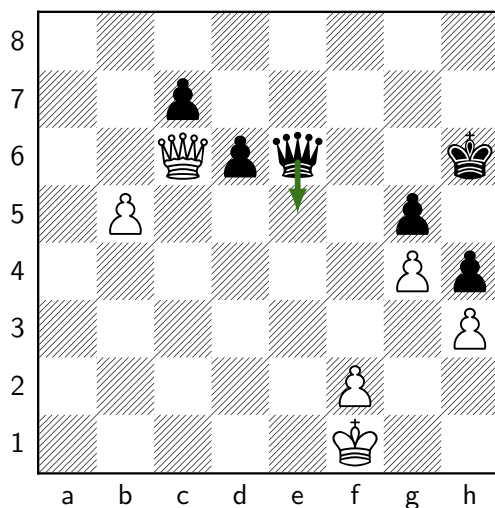
(a)



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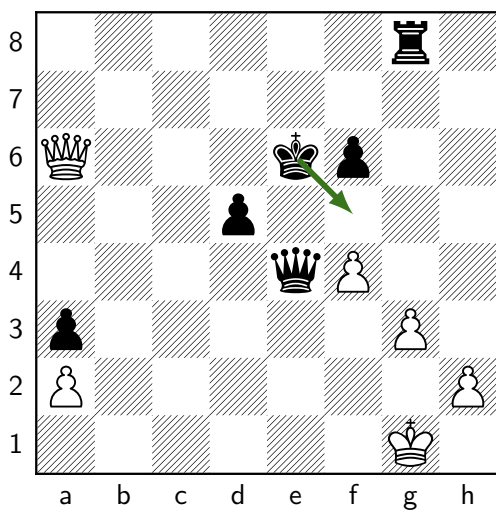


(c)

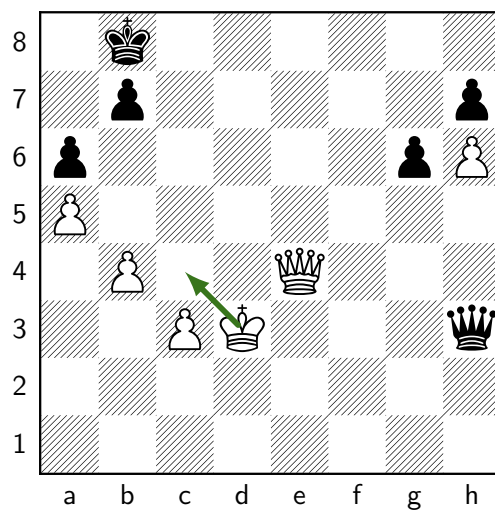


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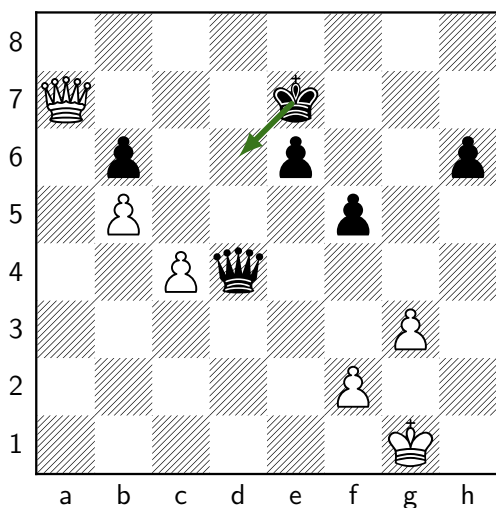
Figure 41: **Move 50, Black: Rg8.** (a) Our position; Melanie plays Rg8 (a8g8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#541` (ply 106) [structural distance $d=25$, QAP obj=28]: the master played Qf8+; under μ our Qe5 plays the role of that game's Qf4; cross-type roles: our Ra8 as their Pg4; transferred move e4e1. (c) Retrieved from `ccrl_top_p0002#2113` (ply 108) [structural distance $d=22$, QAP obj=27]: the master played Kf3; under μ our Ke3 plays the role of that game's Kg3; cross-type roles: our Ra8 as their Pa5; transferred move e6f5. (d) Retrieved from `ccrl_top_p0001#1096` (ply 107) [structural distance $d=24$, QAP obj=27]: the master played Qe5; under μ our Qe5 plays the role of that game's Qe3; cross-type roles: our Ra8 as their Pc7; transferred move e4c4. Synthesis: no retrieved analog transferred to Rg8 – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



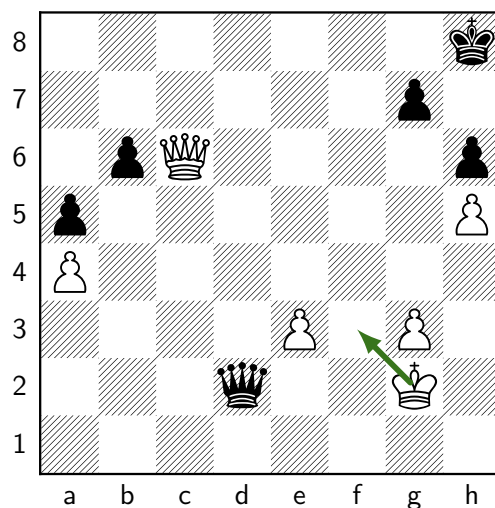
(a)



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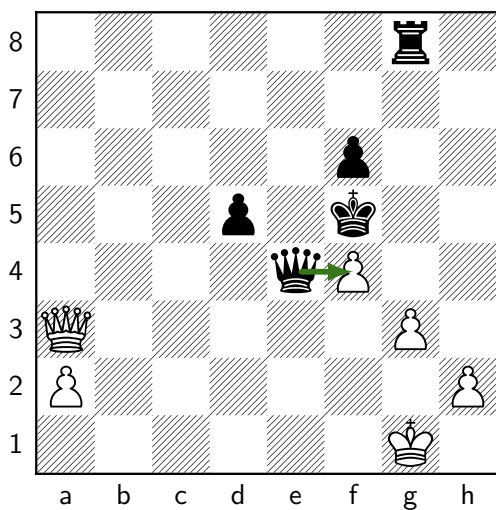


(c)

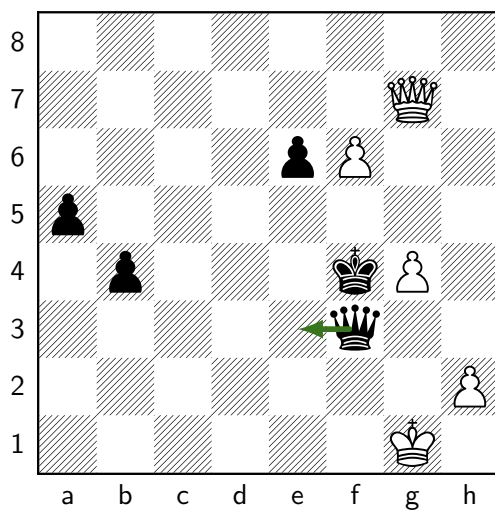


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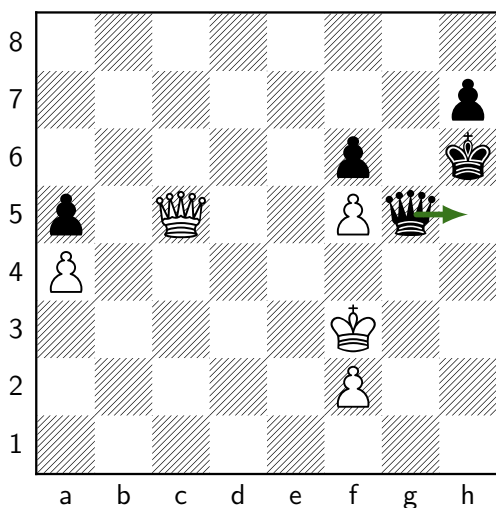
Figure 42: **Move 51, Black: Kf5**. (a) Our position; Melanie plays Kf5 (e6f5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3786` (ply 90) [structural distance $d=18$, QAP obj=71]: the master played Kc4; under μ our Ke3 plays the role of that game’s Kd3; cross-type roles: our Rg8 as their pa6; transferred move e6f5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#323` (ply 111) [structural distance $d=22$, QAP obj=70]: the master played Kd6; under μ our Ke3 plays the role of that game’s Ke2; cross-type roles: our Rg8 as their Ph6; transferred move e6f5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#2748` (ply 100) [structural distance $d=21$, QAP obj=55]: the master played Kf3; under μ our Ke3 plays the role of that game’s Kg2; cross-type roles: our Rg8 as their pb6; transferred move e6f5 (**endorses this move**). Synthesis: 13 retrieved analogs transferred to Kf5 (summed QAP objective 650, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



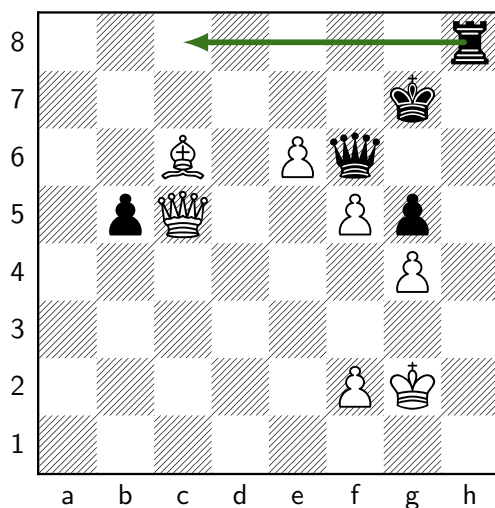
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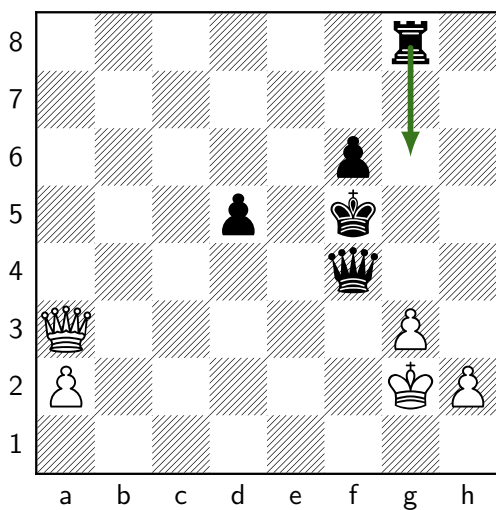


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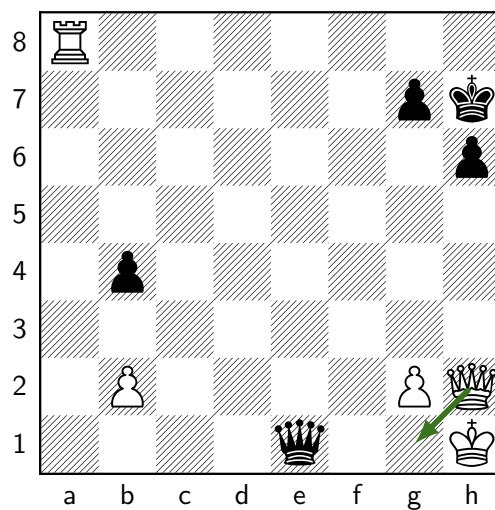


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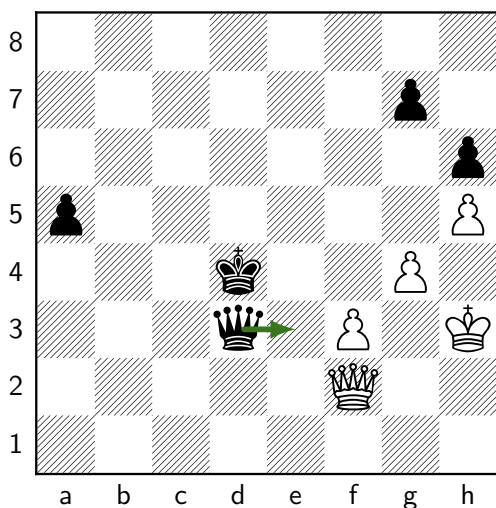
Figure 43: **Move 52, Black: Qxf4.** (a) Our position; Melanie plays Qxf4 (e4f4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#78` (ply 95) [structural distance $d=43$, QAP obj=54]: the master played Qe3+; under μ our Qe5 plays the role of that game’s Qf6; cross-type roles: our Rg8 as their Pb4; transferred move e4f4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3243` (ply 111) [structural distance $d=41$, QAP obj=26]: the master played Qh5+; under μ our Qe5 plays the role of that game’s Qg4; cross-type roles: our pg3 as their qc5, our Rg8 as their pa4; transferred move e4f4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2043` (ply 101) [structural distance $d=37$, QAP obj=65]: the master played Rc8; under μ our Rg1 plays the role of that game’s Rh1; cross-type roles: our Kf5 as their bc6; transferred move g8c8. Synthesis: 2 retrieved analogs transferred to Qxf4 (summed QAP objective 80, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



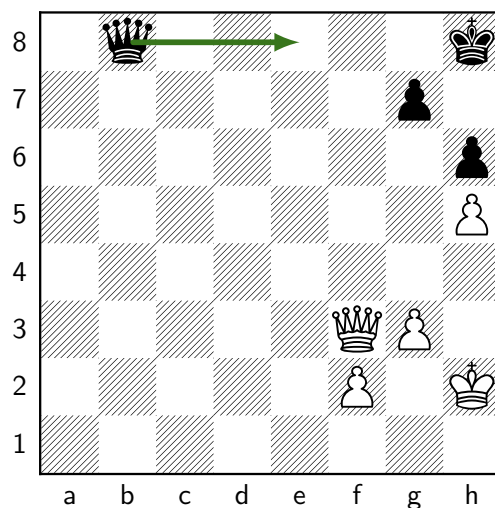
(a)



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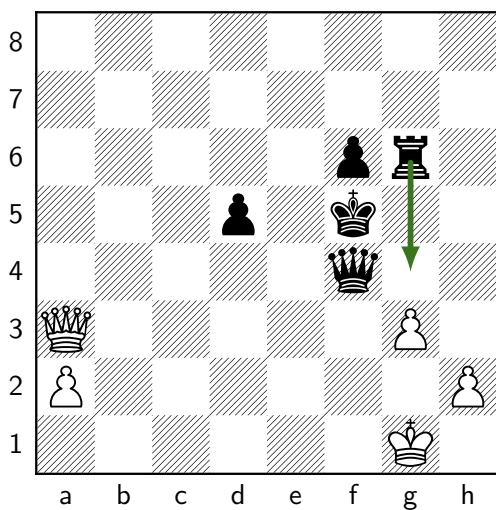


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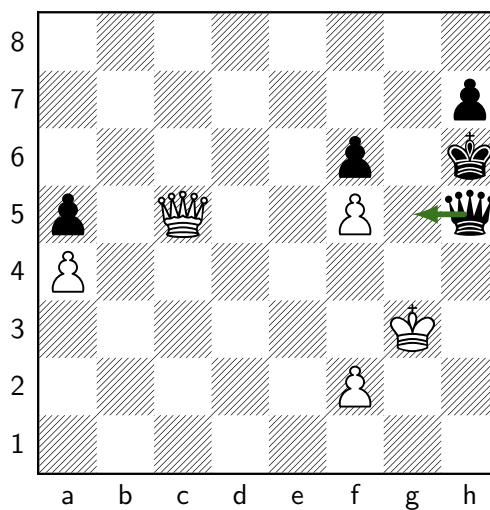


(d)

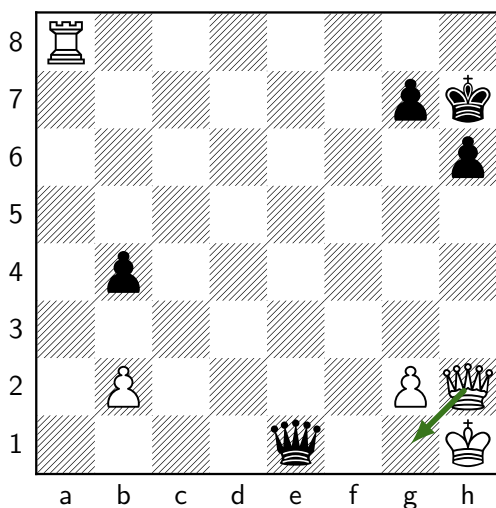
Figure 44: **Move 53, Black: Rg6**. (a) Our position; Melanie plays Rg6 (g8g6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3047` (ply 102) [structural distance $d=31$, QAP obj=19]: the master played Qg1; under μ our Rg1 plays the role of that game’s Qh2 (a **cross-type** correspondence); cross-type roles: our Pd5 as their qe1, our qa3 as their pb4; transferred move g8g6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2150` (ply 111) [structural distance $d=34$, QAP obj=10]: the master played Qe3; under μ our Rg1 plays the role of that game’s Qd6 (a **cross-type** correspondence); cross-type roles: our qa3 as their Ph6, our Qf4 as their pg4; transferred move g8g6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2626` (ply 95) [structural distance $d=35$, QAP obj=24]: the master played Qe8; under μ our Rg1 plays the role of that game’s Qb1 (a **cross-type** correspondence); cross-type roles: our Qf4 as their Pg7; transferred move g8h8. Synthesis: 3 retrieved analogs transferred to Rg6 (summed QAP objective 47, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



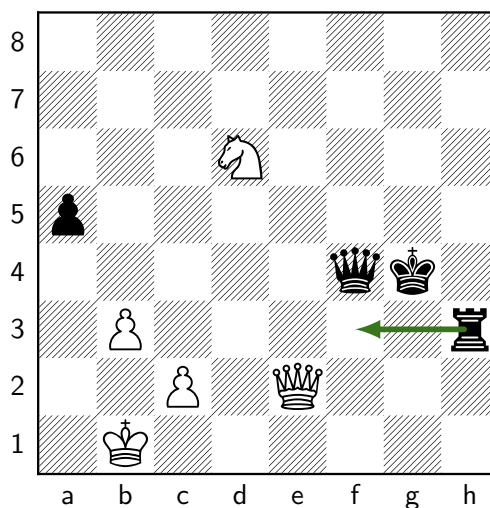
(a)



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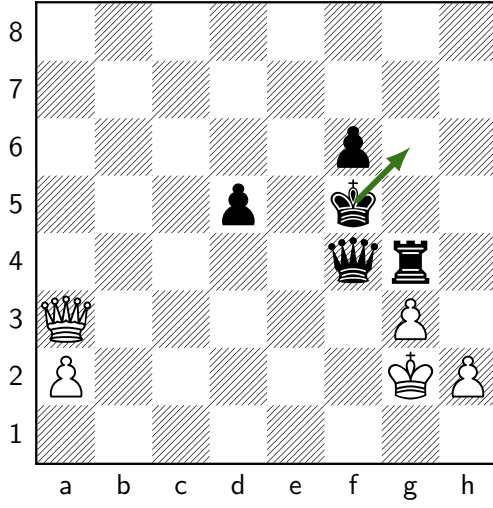


(c)

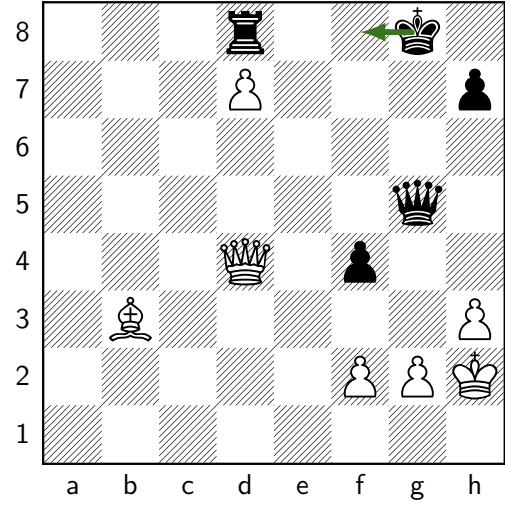


(d)

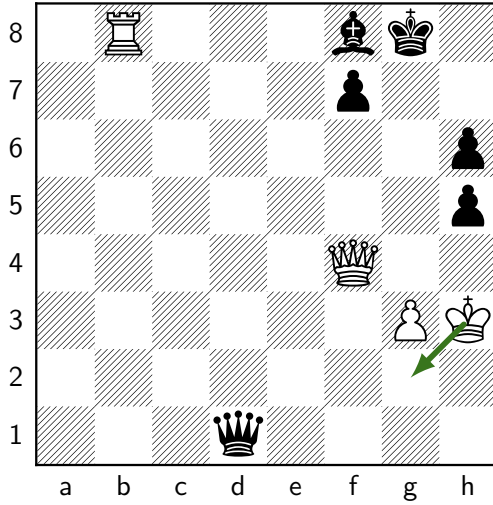
Figure 45: **Move 54, Black: Rg4**. (a) Our position; Melanie plays Rg4 (g6g4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0001#3243 (ply 113) [structural distance $d=37$, QAP obj=22]: the master played Qg5+; under μ our Rg3 plays the role of that game’s Qh4 (a **cross-type** correspondence); cross-type roles: our Qf4 as their Pf6; transferred move g6g4 (**endorses this move**). (c) Retrieved from ccr1_top_p0002#3047 (ply 106) [structural distance $d=32$, QAP obj=18]: the master played Qg1; under μ our Rg3 plays the role of that game’s Qh2 (a **cross-type** correspondence); cross-type roles: our Qf4 as their Ra8; transferred move g6g4 (**endorses this move**). (d) Retrieved from ccr1_top_p0001#2890 (ply 105) [structural distance $d=35$, QAP obj=3]: the master played Rf3; under μ our Rg3 plays the role of that game’s Rh6; cross-type roles: our Pf6 as their nd6, our qa3 as their pb3; transferred move g6g4 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Rg4 (summed QAP objective 75, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



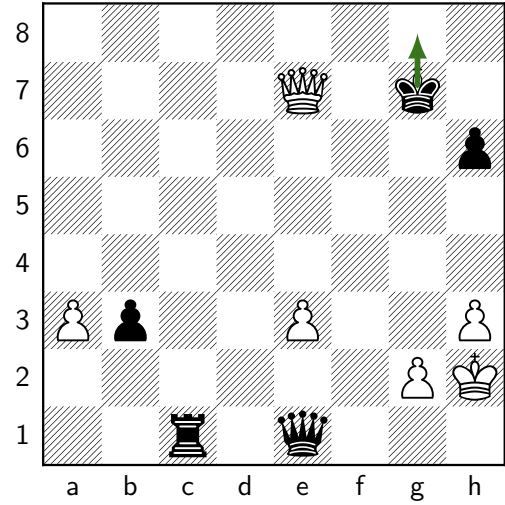
(a)



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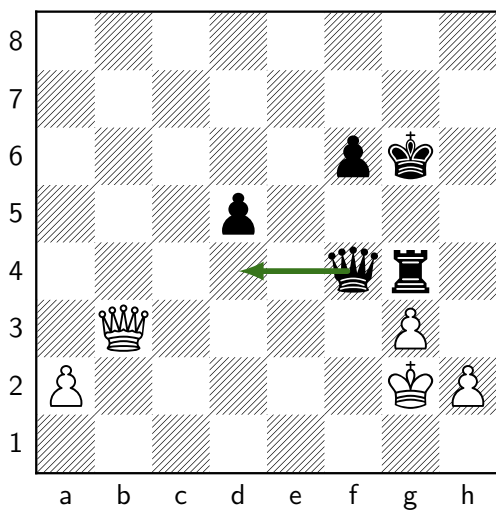


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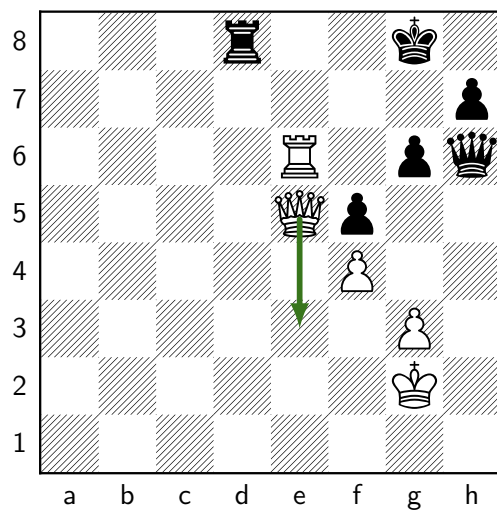


(d)

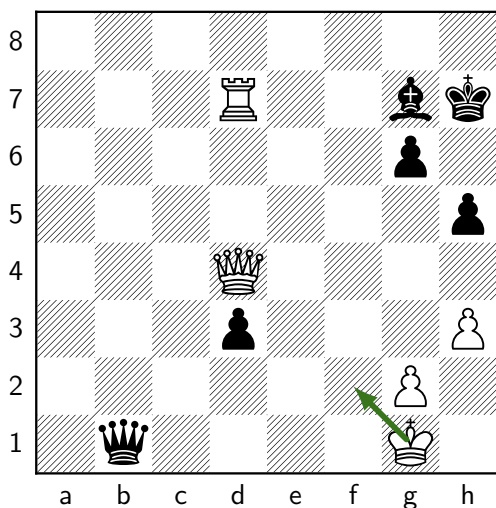
Figure 46: **Move 55, Black: Kg6.** (a) Our position; Melanie plays Kg6 (f5g6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2819` (ply 109) [structural distance $d=40$, QAP obj=25]: the master played Kf8; under μ our Kf4 plays the role of that game’s Kg1; cross-type roles: our Pf6 as their bb3; transferred move f5g6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1644` (ply 94) [structural distance $d=45$, QAP obj=19]: the master played Kg2; under μ our Kf4 plays the role of that game’s Kh3; cross-type roles: our pg3 as their bf8; transferred move f5g6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1828` (ply 109) [structural distance $d=42$, QAP obj=14]: the master played Kg8; under μ our Kf4 plays the role of that game’s Kg2; cross-type roles: our Qf4 as their Rc1, our Rg4 as their Qe1; transferred move f5g6 (**endorses this move**). Synthesis: 19 retrieved analogs transferred to Kg6 (summed QAP objective 151, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



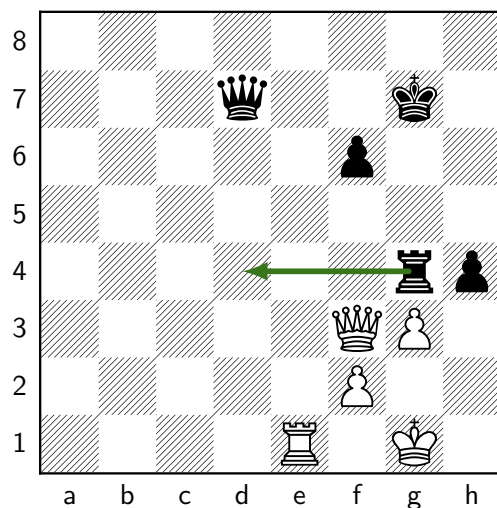
(a)



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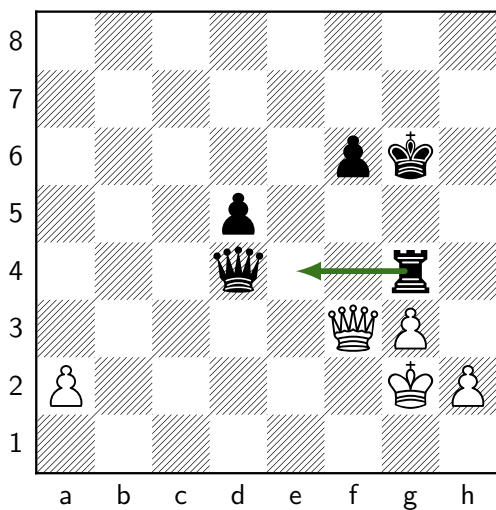


(c)

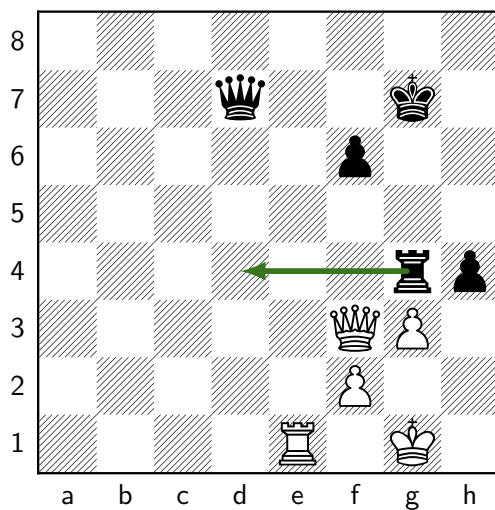


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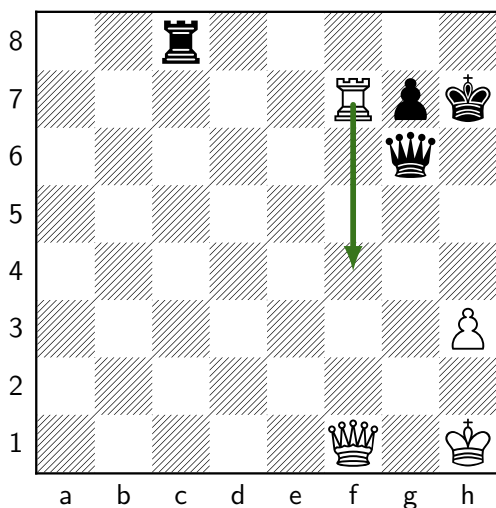
Figure 47: **Move 56, Black: Qd4.** (a) Our position; Melanie plays Qd4 (f4d4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3381` (ply 98) [structural distance $d=35$, QAP obj=15]: the master played Qe3; under μ our Qf5 plays the role of that game’s Qe5; cross-type roles: our Pf6 as their rd8; transferred move f4d4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3015` (ply 114) [structural distance $d=37$, QAP obj=47]: the master played Kf2; under μ our Kg3 plays the role of that game’s Kg1; cross-type roles: our pg3 as their bg7; transferred move g6h6. (d) Retrieved from `ccr1_top_p0000#3052` (ply 97) [structural distance $d=33$, QAP obj=40]: the master played Rd4; under μ our Rg5 plays the role of that game’s Rg5; cross-type roles: our pa2 as their Qd7, our Qf4 as their Ph4; transferred move g4h4. Synthesis: 2 retrieved analogs transferred to Qd4 (summed QAP objective 30, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



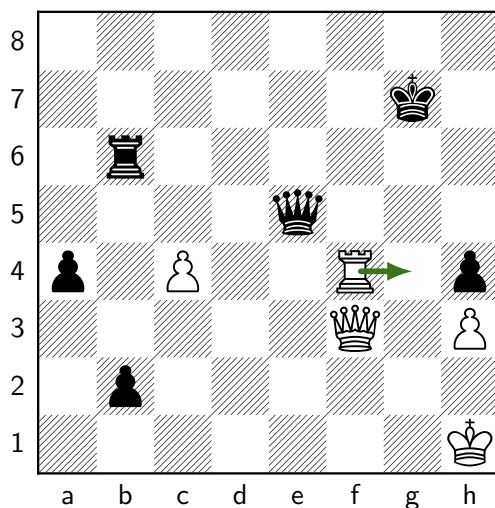
(a)



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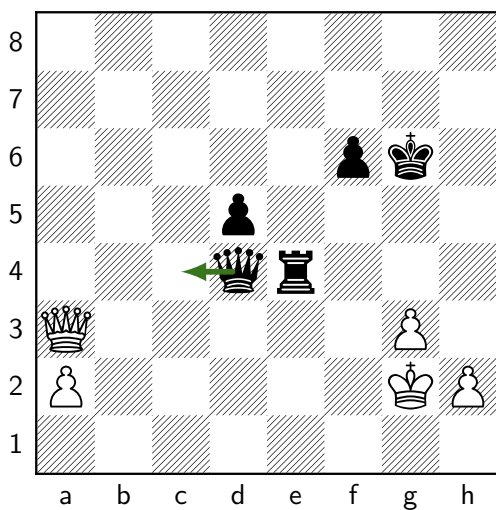


(c)

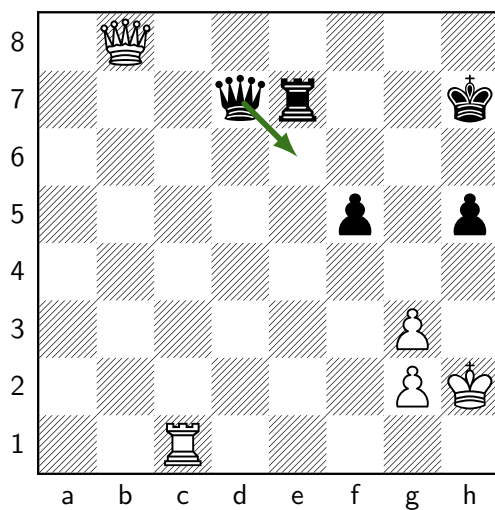


(d)

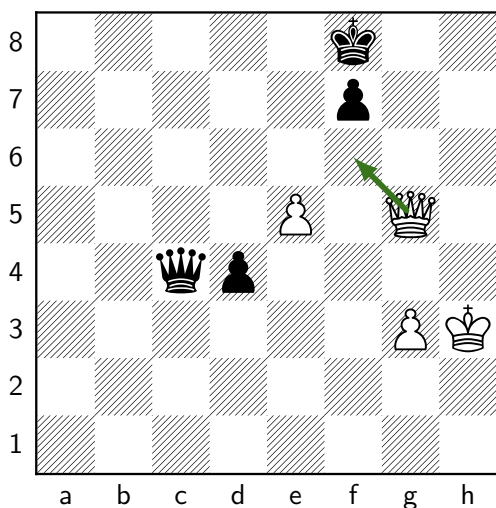
Figure 48: **Move 57, Black: Re4**. (a) Our position; Melanie plays **Re4 (g4e4)**. Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#3052` (ply 97) [structural distance $d=33$, QAP obj=88]: the master played **Rd4**; under μ our *Rg5* plays the role of that game’s *Rg5*; cross-type roles: our *pa2* as their *re1*; transferred move **g4e4 (endorses this move)**. (c) Retrieved from `ccrl_top_p0001#2374` (ply 98) [structural distance $d=48$, QAP obj=85]: the master played **Rf4**; under μ our *Rg5* plays the role of that game’s *Rf7*; cross-type roles: our *Pd5* as their *rc8*; transferred move **g4e4 (endorses this move)**. (d) Retrieved from `ccrl_top_p0002#3979` (ply 106) [structural distance $d=46$, QAP obj=29]: the master played **Rg4+**; under μ our *Rg5* plays the role of that game’s *Rf4*; cross-type roles: our *pa2* as their *rb6*; transferred move **g4e4 (endorses this move)**. Synthesis: 9 retrieved analogs transferred to **Re4** (summed QAP objective 329, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



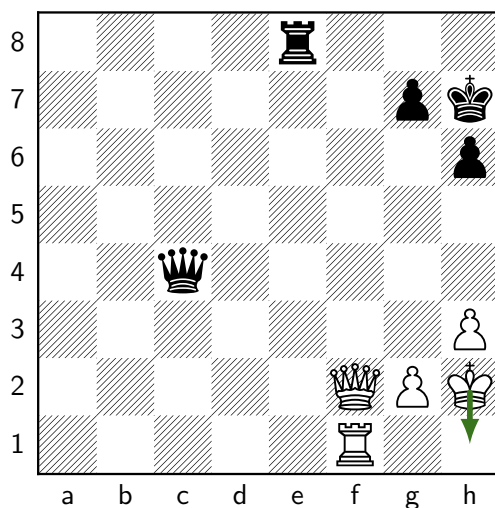
(a)



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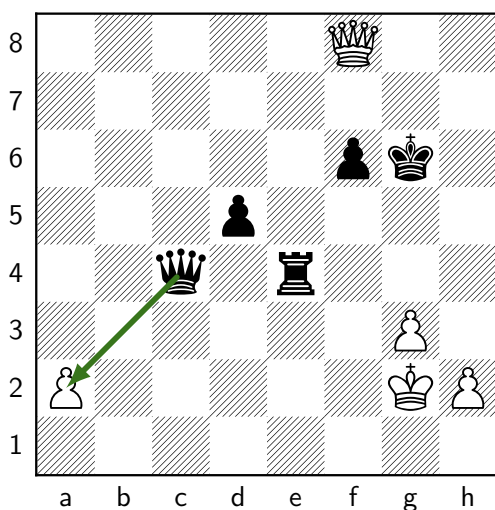


(c)

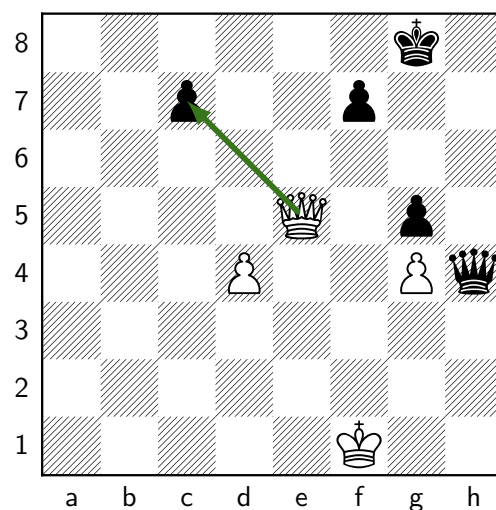


(d)

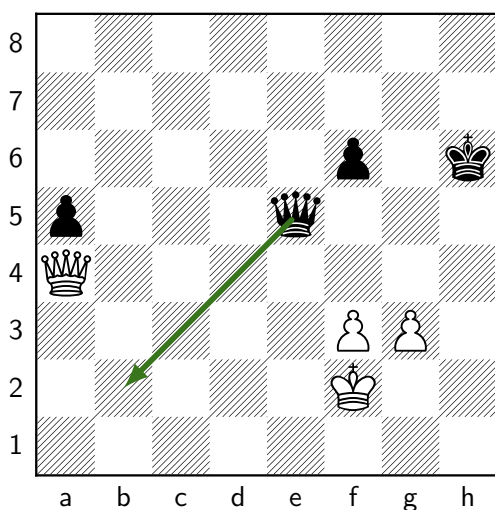
Figure 49: **Move 58, Black: Qc4**. (a) Our position; Melanie plays Qc4 (d4c4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2686` (ply 105) [structural distance $d=34$, QAP obj=1]: the master played Qe6; under μ our Qd5 plays the role of that game’s Qd2; cross-type roles: our Pd5 as their rc1; transferred move d4c4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1560` (ply 126) [structural distance $d=36$, QAP obj=1]: the master played Qf6; under μ our Qd5 plays the role of that game’s Qg5; cross-type roles: our Re4 as their Pg3, our ph2 as their qc4; transferred move d4c4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#556` (ply 102) [structural distance $d=36$, QAP obj=4]: the master played Kh1; under μ our Kg3 plays the role of that game’s Kh2; cross-type roles: our pg3 as their re8; transferred move g6f7. Synthesis: 2 retrieved analogs transferred to Qc4 (summed QAP objective 2, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



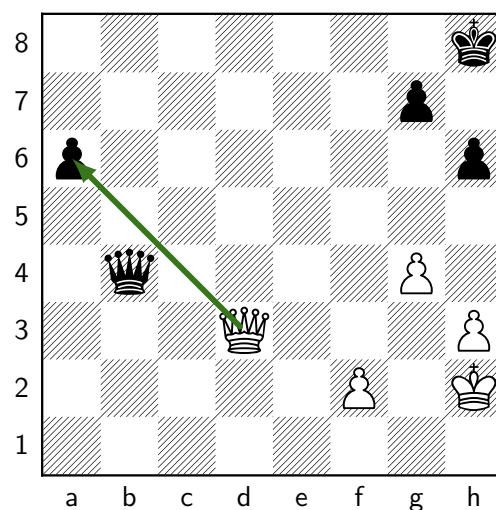
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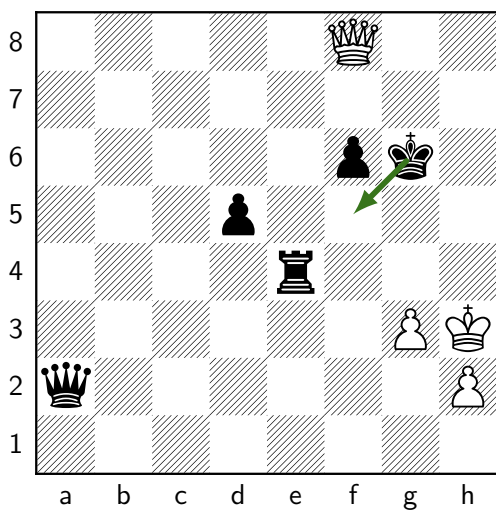


(c)

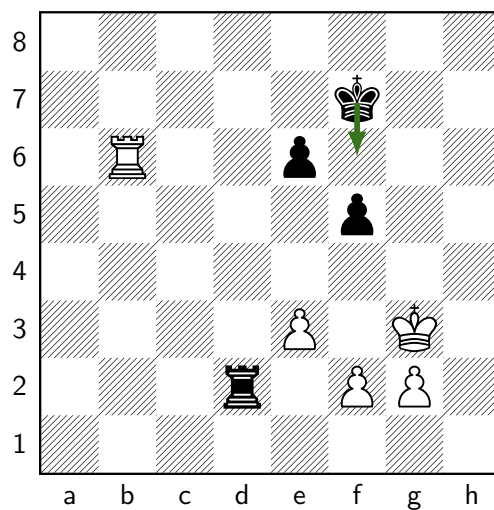


(d)

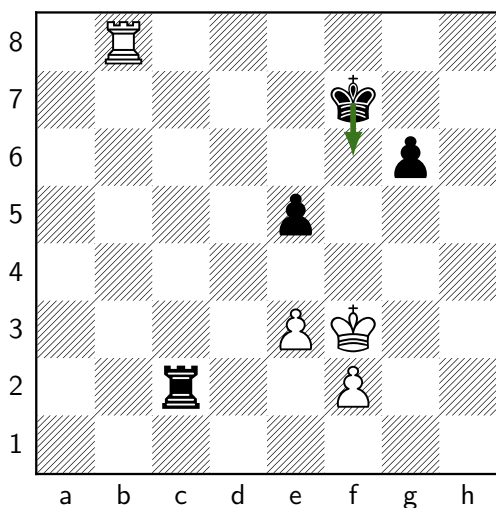
Figure 50: **Move 59, Black: Qxa2+.** (a) Our position; Melanie plays Qxa2+ (c4a2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#419` (ply 116) [structural distance $d=30$, QAP obj=26]: the master played Qxc7; under μ our Qc5 plays the role of that game’s Qe5; transferred move c4a2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2652` (ply 121) [structural distance $d=28$, QAP obj=21]: the master played Qb2+; under μ our Qc5 plays the role of that game’s Qe4; transferred move c4a2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3320` (ply 114) [structural distance $d=29$, QAP obj=20]: the master played Qxa6; under μ our Qc5 plays the role of that game’s Qd3; cross-type roles: our Re4 as their Ph3; transferred move c4a2 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Qxa2+ (summed QAP objective 82, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



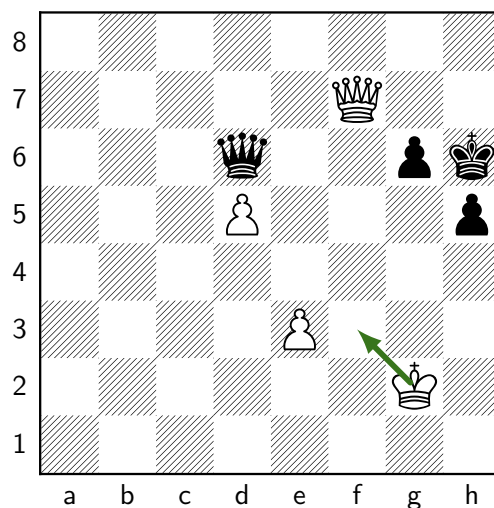
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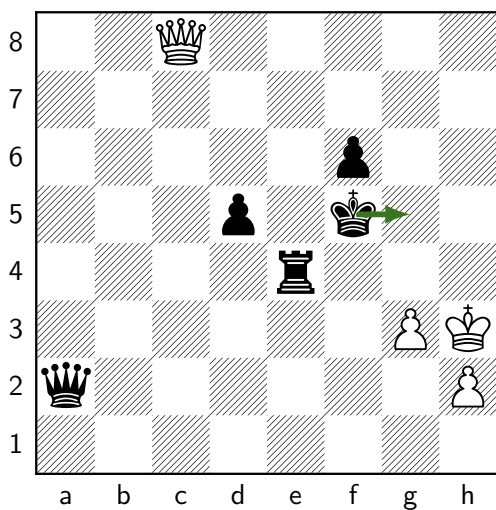


(c)

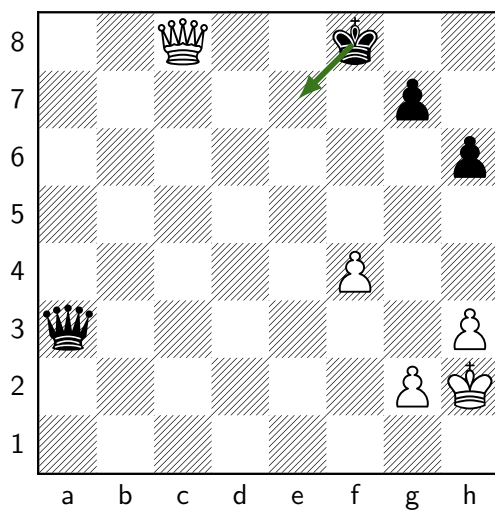


(d)

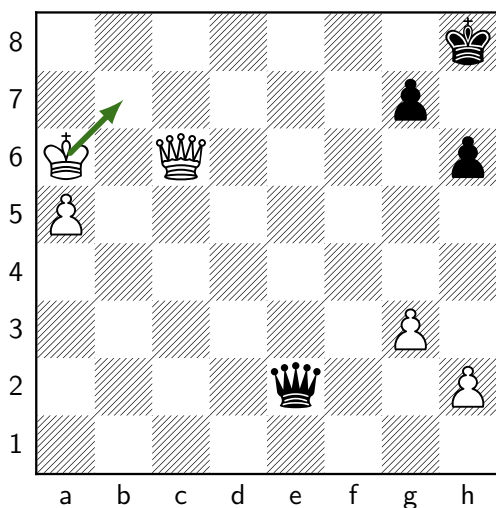
Figure 51: **Move 60, Black: Kf5.** (a) Our position; Melanie plays Kf5 (g6f5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1426` (ply 109) [structural distance $d=21$, QAP obj=24]: the master played Kf6; under μ our Kg3 plays the role of that game’s Kf2; cross-type roles: our Qa2 as their rb6, our Re4 as their pe3; transferred move g6f5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3048` (ply 123) [structural distance $d=22$, QAP obj=13]: the master played Kf6; under μ our Kg3 plays the role of that game’s Kf2; cross-type roles: our qf8 as their Rc2; transferred move g6f5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1637` (ply 122) [structural distance $d=21$, QAP obj=11]: the master played Kf3; under μ our Kg3 plays the role of that game’s Kg2; cross-type roles: our Re4 as their Pe3; transferred move g6f5 (**endorses this move**). Synthesis: 12 retrieved analogs transferred to Kf5 (summed QAP objective 84, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



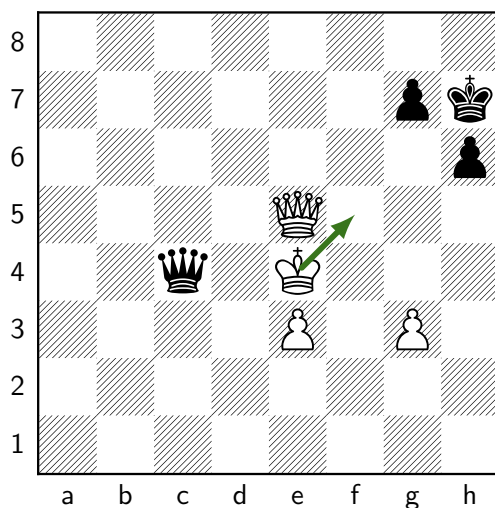
(a)



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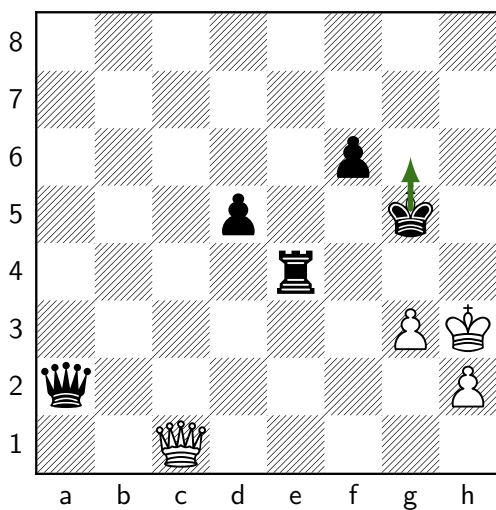


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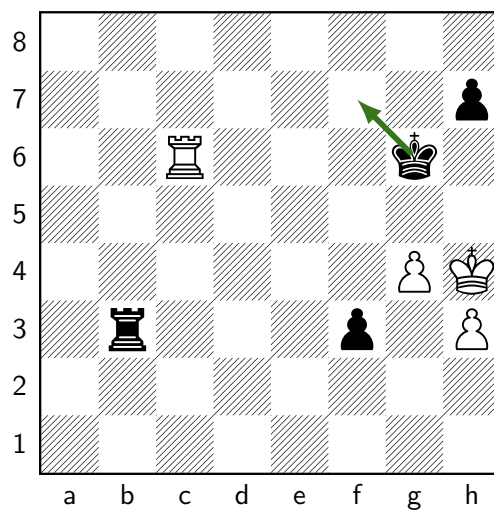


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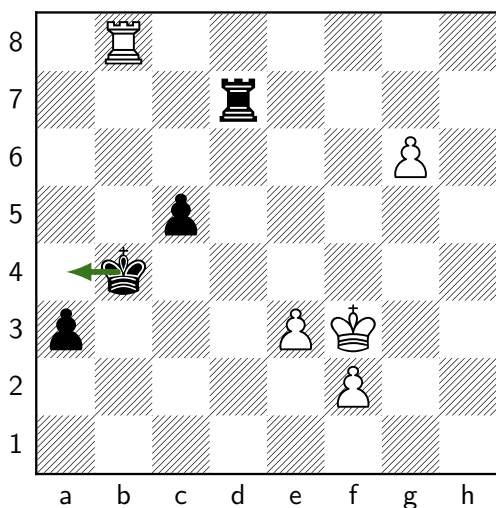
Figure 52: **Move 61, Black: Kg5**. (a) Our position; Melanie plays Kg5 (f5g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2607` (ply 111) [structural distance $d=28$, QAP obj=67]: the master played Ke7; under μ our Kf4 plays the role of that game’s Kf1; cross-type roles: our Re4 as their Ph6; transferred move f5g5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#923` (ply 136) [structural distance $d=29$, QAP obj=55]: the master played Kb7; under μ our Kf4 plays the role of that game’s Ka6; cross-type roles: our Re4 as their Pg3, our Pd5 as their Qc6; transferred move f5g5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#2748` (ply 122) [structural distance $d=26$, QAP obj=50]: the master played Kf5; under μ our Kf4 plays the role of that game’s Ke4; cross-type roles: our Re4 as their Qe5; transferred move f5g5 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to Kg5 (summed QAP objective 370, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



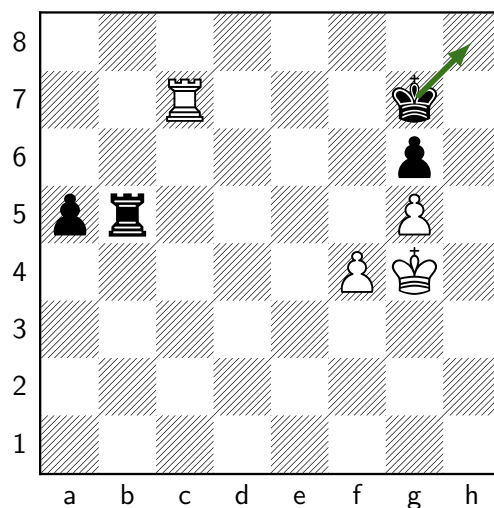
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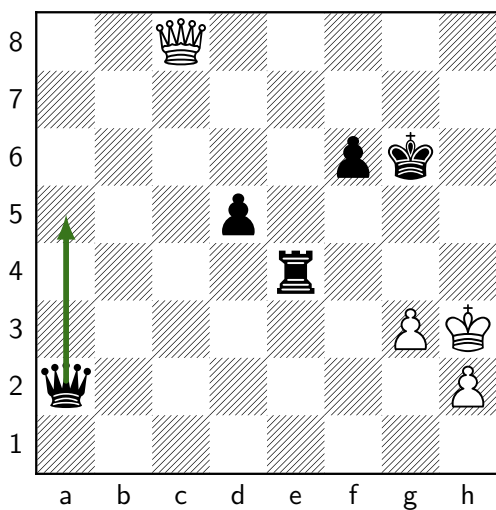


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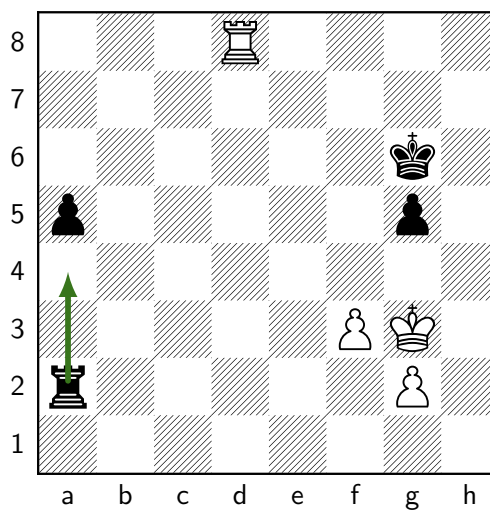


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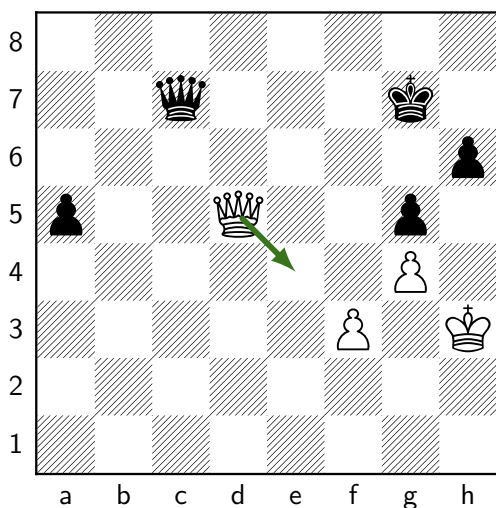
Figure 53: **Move 62, Black: Kg6**. (a) Our position; Melanie plays Kg6 (g5g6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3015` (ply 123) [structural distance $d=26$, QAP obj=73]: the master played Kf7; under μ our Kg4 plays the role of that game’s Kg3; cross-type roles: our qc1 as their rc6, our Pd5 as their Rb3; transferred move g5g6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2419` (ply 127) [structural distance $d=26$, QAP obj=69]: the master played Ka4; under μ our Kg4 plays the role of that game’s Kb5; cross-type roles: our qc1 as their rb8, our Qa2 as their Pa3; transferred move g5g6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3487` (ply 133) [structural distance $d=26$, QAP obj=69]: the master played Kh8; under μ our Kg4 plays the role of that game’s Kg2; cross-type roles: our qc1 as their rc7, our Qa2 as their Rb5; transferred move g5g6 (**endorses this move**). Synthesis: 37 retrieved analogs transferred to Kg6 (summed QAP objective 1724, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



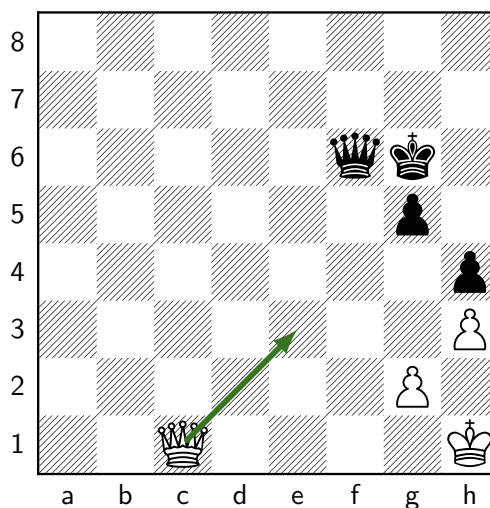
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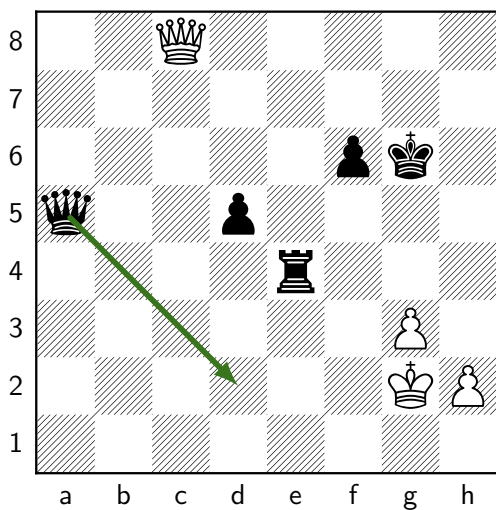


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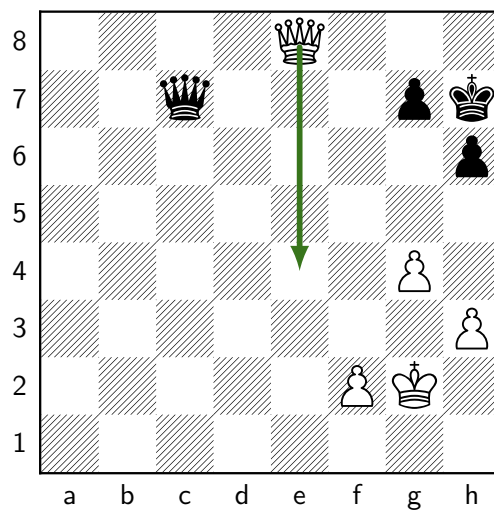


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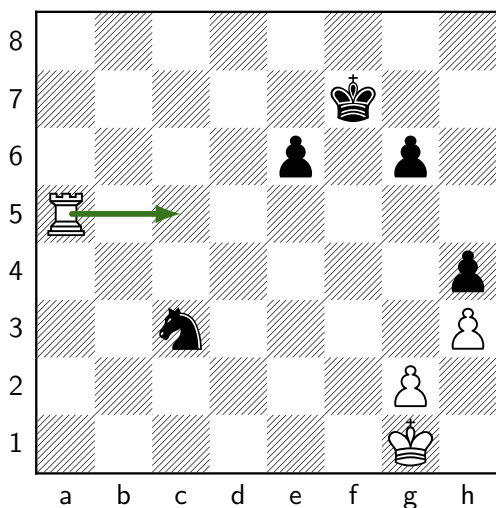
Figure 54: **Move 63, Black: Qa5**. (a) Our position; Melanie plays Qa5 (a2a5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3925` (ply 117) [structural distance $d=24$, QAP obj=21]: the master played Ra4; under μ our Qa7 plays the role of that game’s Ra7 (a **cross-type** correspondence); cross-type roles: our qc8 as their rd8, our Re4 as their Pg5; transferred move a2a5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1277` (ply 120) [structural distance $d=24$, QAP obj=14]: the master played Qe4; under μ our Qa7 plays the role of that game’s Qd5; cross-type roles: our Re4 as their Pg4; transferred move a2a5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3795` (ply 130) [structural distance $d=23$, QAP obj=13]: the master played Qe3; under μ our Qa7 plays the role of that game’s Qc1; cross-type roles: our Re4 as their Ph3; transferred move a2a5 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to Qa5 (summed QAP objective 98, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



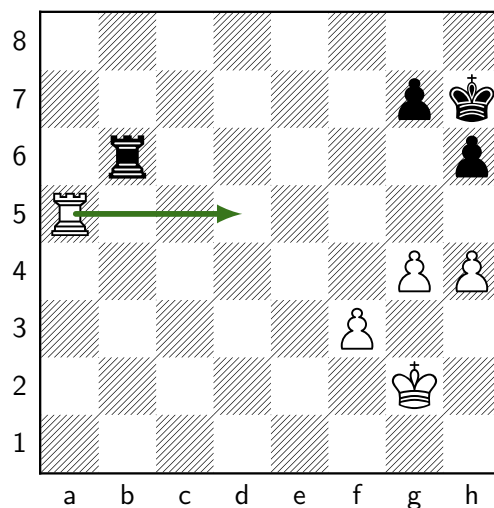
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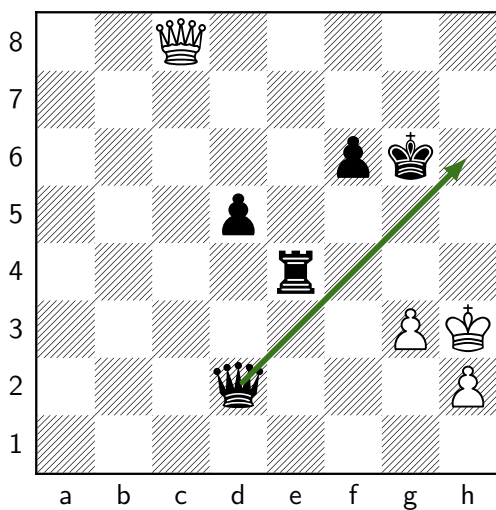


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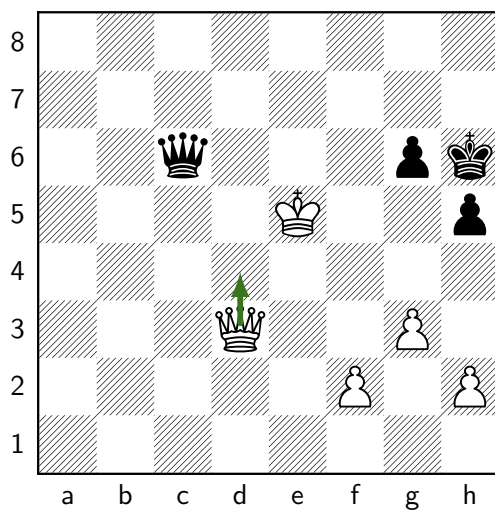


(d)

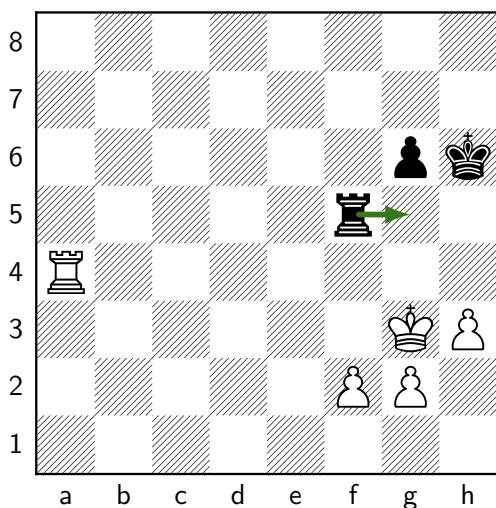
Figure 55: **Move 64, Black: Qd2+.** (a) Our position; Melanie plays Qd2+ (a5d2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top-p0000#3320` (ply 126) [structural distance $d=19$, QAP obj=4]: the master played Qe4+; under μ our Qa4 plays the role of that game's Qe8; cross-type roles: our Re4 as their Pg4; transferred move a5c5. (c) Retrieved from `ccr1_top-p0003#958` (ply 130) [structural distance $d=18$, QAP obj=4]: the master played Rc5; under μ our Re5 plays the role of that game's Ra5; cross-type roles: our Pd5 as their nc3, our qc8 as their Ph3; transferred move e4d4. (d) Retrieved from `ccr1_top-p0001#1420` (ply 122) [structural distance $d=18$, QAP obj=4]: the master played Rd5; under μ our Re5 plays the role of that game's Ra5; cross-type roles: our qc8 as their Pg4, our Qa5 as their Ph4; transferred move e4d4. Synthesis: no retrieved analog transferred to Qd2+ – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



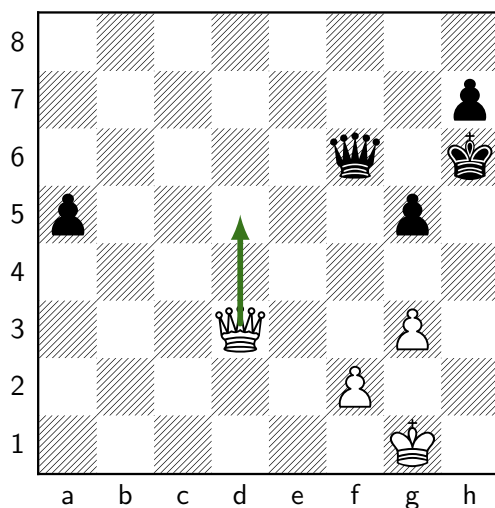
(a)



(b)

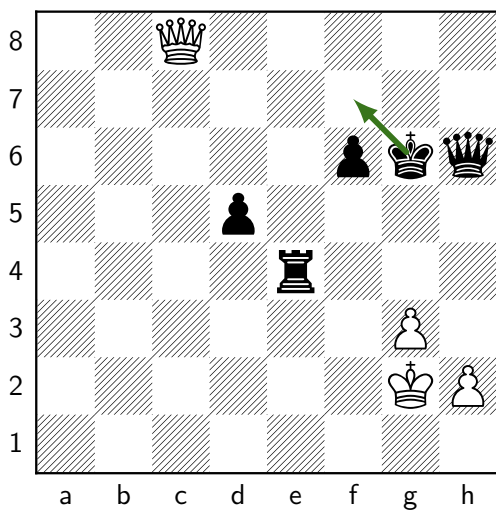


(c)

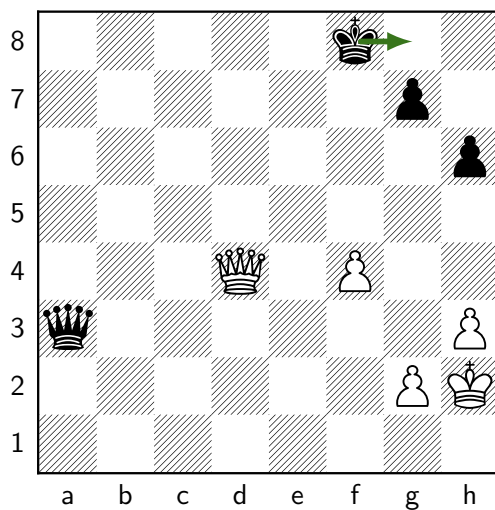


(d)

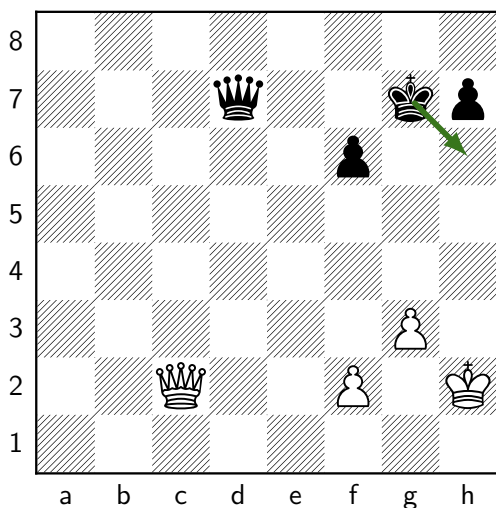
Figure 56: **Move 65, Black: Qh6+.** (a) Our position; Melanie plays Qh6+ (d2h6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1729` (ply 114) [structural distance $d=24$, QAP obj=17]: the master played Qd4; under μ our Qd7 plays the role of that game’s Qd3; cross-type roles: our Re4 as their Ph2; transferred move d2h6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1376` (ply 129) [structural distance $d=24$, QAP obj=16]: the master played Rg5+; under μ our Qd7 plays the role of that game’s Rf4 (a **cross-type** correspondence); cross-type roles: our qc8 as their ra4, our Re4 as their ph3; transferred move d2h6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1231` (ply 116) [structural distance $d=20$, QAP obj=33]: the master played Qd5; under μ our Qd7 plays the role of that game’s Qd3; cross-type roles: our qc8 as their pa5; transferred move d2d3. Synthesis: 2 retrieved analogs transferred to Qh6+ (summed QAP objective 33, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



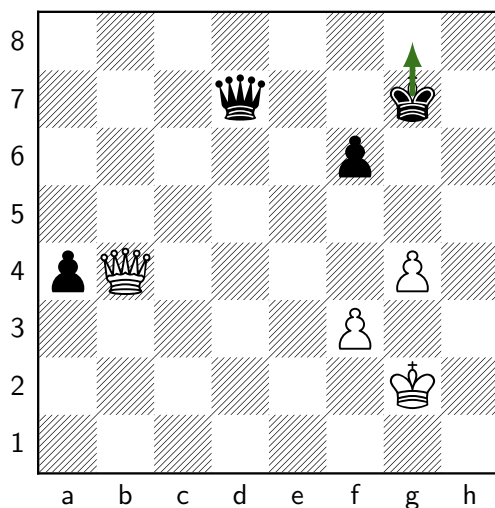
(a)



(b)

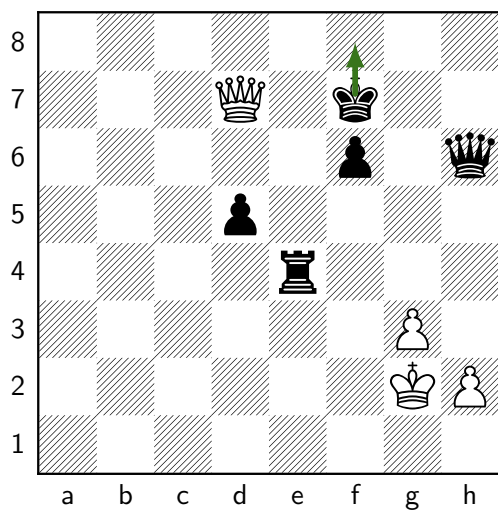


(c)

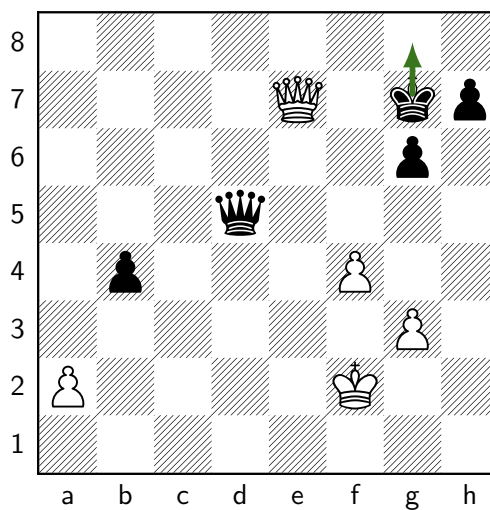


(d)

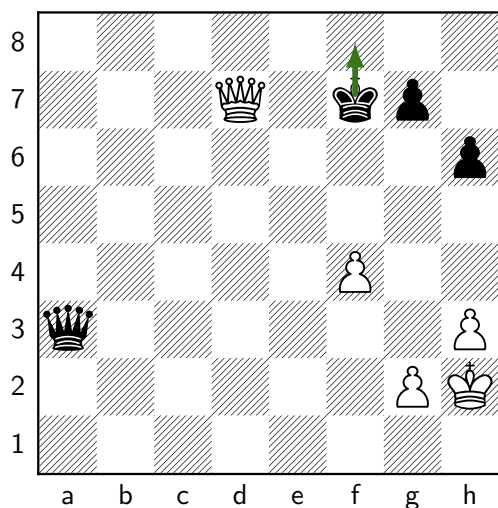
Figure 57: **Move 66, Black: Kf7.** (a) Our position; Melanie plays Kf7 (g6f7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2607` (ply 123) [structural distance $d=23$, QAP obj=15]: the master played Kg8; under μ our Kg3 plays the role of that game’s Kf1; cross-type roles: our Re4 as their pf4, our qc8 as their pg2; transferred move g6f7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#445` (ply 129) [structural distance $d=24$, QAP obj=15]: the master played Kh6; under μ our Kg3 plays the role of that game’s Kg2; cross-type roles: our Re4 as their pg3; transferred move g6f7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2652` (ply 143) [structural distance $d=24$, QAP obj=7]: the master played Kg8; under μ our Kg3 plays the role of that game’s Kg2; transferred move g6f7 (**endorses this move**). Synthesis: 18 retrieved analogs transferred to Kf7 (summed QAP objective 75, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



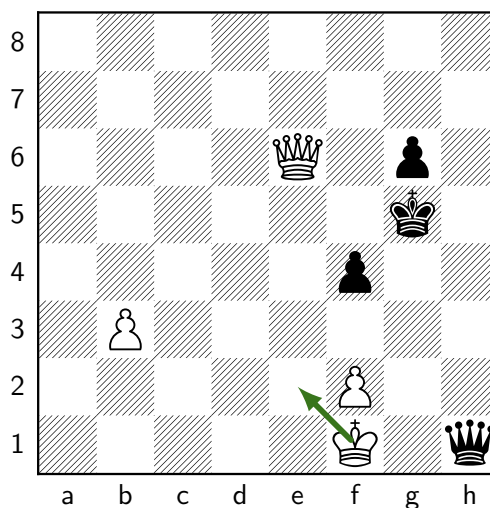
(a)



(b)

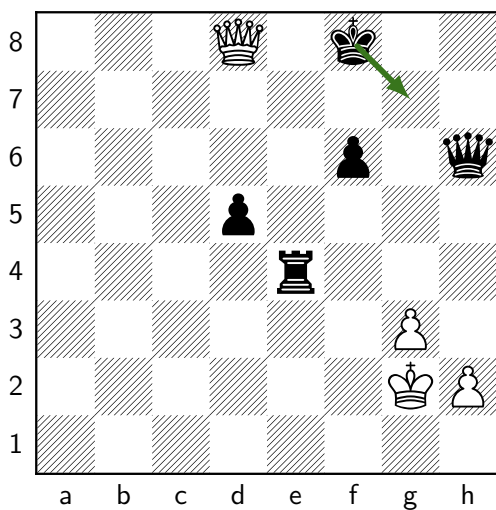


(c)

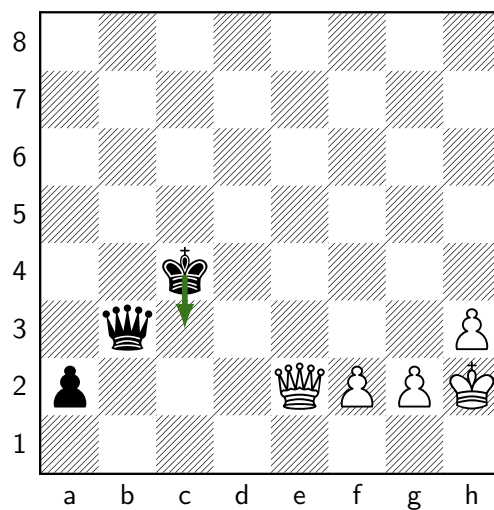


(d)

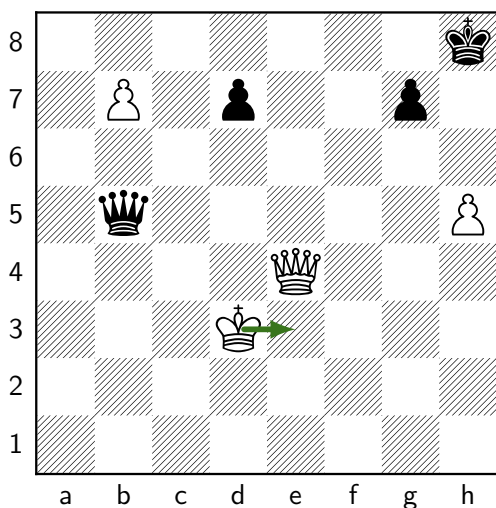
Figure 58: **Move 67, Black: Kf8**. (a) Our position; Melanie plays Kf8 (f7f8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1303` (ply 139) [structural distance $d=18$, QAP obj=72]: the master played Kg8; under μ our Kf2 plays the role of that game’s Kg2; cross-type roles: our Re4 as their Qd5, our Qh6 as their Pb4; transferred move f7f8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2607` (ply 121) [structural distance $d=18$, QAP obj=72]: the master played Kf8; under μ our Kf2 plays the role of that game’s Kf2; cross-type roles: our Re4 as their pf4; transferred move f7f8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2985` (ply 142) [structural distance $d=16$, QAP obj=69]: the master played Ke2; under μ our Kf2 plays the role of that game’s Kf1; cross-type roles: our Re4 as their Pf2; transferred move f7f8 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Kf8 (summed QAP objective 512, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



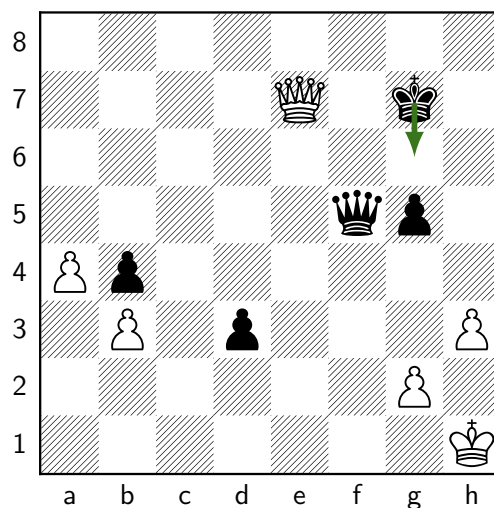
(a)



(b)

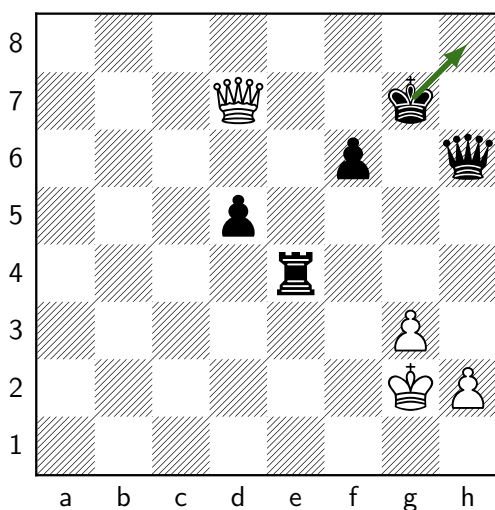


(c)

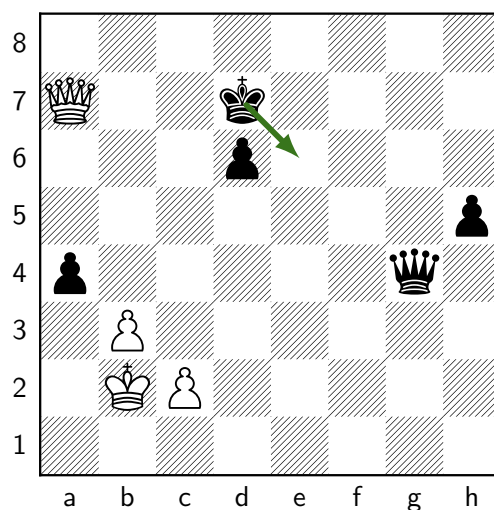


(d)

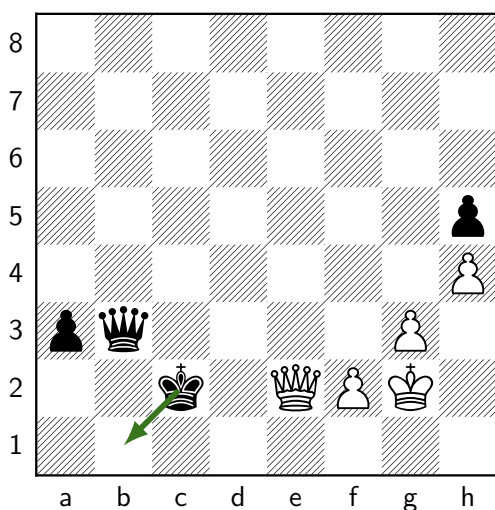
Figure 59: **Move 68, Black: Kg7 (f8g7).** (a) Our position; Melanie plays Kg7 (f8g7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1509` (ply 119) [structural distance $d=21$, QAP obj=69]: the master played Kc3; under μ our Kf1 plays the role of that game’s Kc5; cross-type roles: our Re4 as their ph3; transferred move f8g7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2475` (ply 140) [structural distance $d=21$, QAP obj=68]: the master played Ke3; under μ our Kf1 plays the role of that game’s Kd3; cross-type roles: our Re4 as their Ph5; transferred move f8g7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#2056` (ply 143) [structural distance $d=24$, QAP obj=67]: the master played Kg6; under μ our Kf1 plays the role of that game’s Kg2; cross-type roles: our Re4 as their Pb4; transferred move f8g7 (**endorses this move**). Synthesis: 16 retrieved analogs transferred to Kg7 (summed QAP objective 791, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



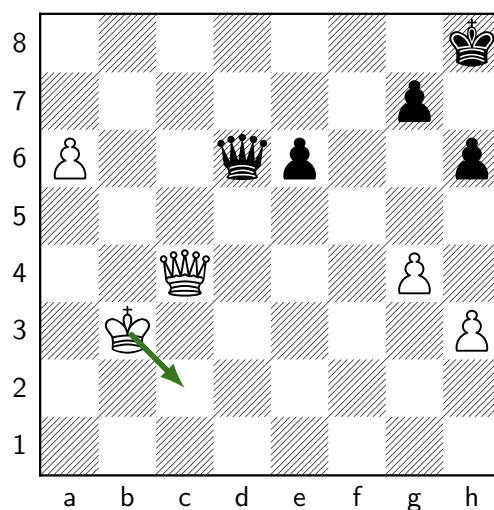
(a)



(b)

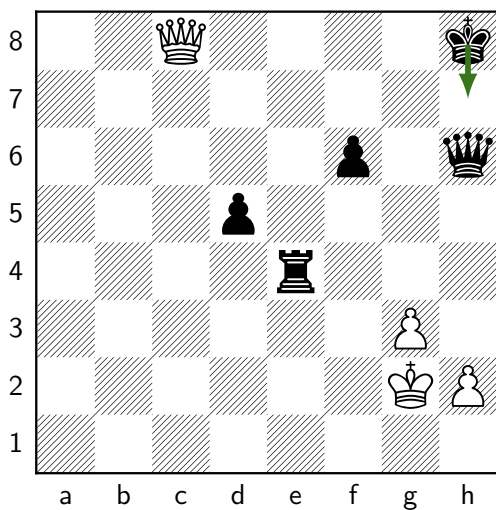


(c)

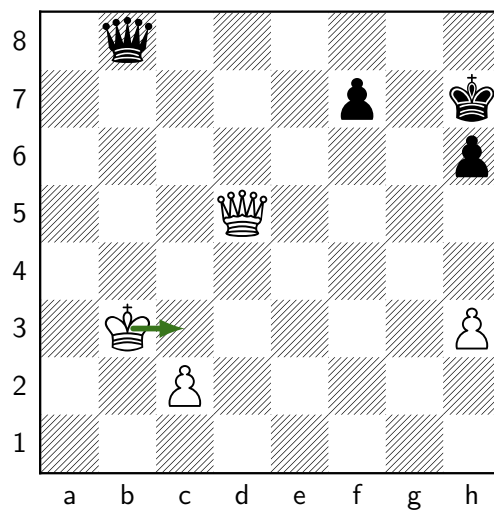


(d)

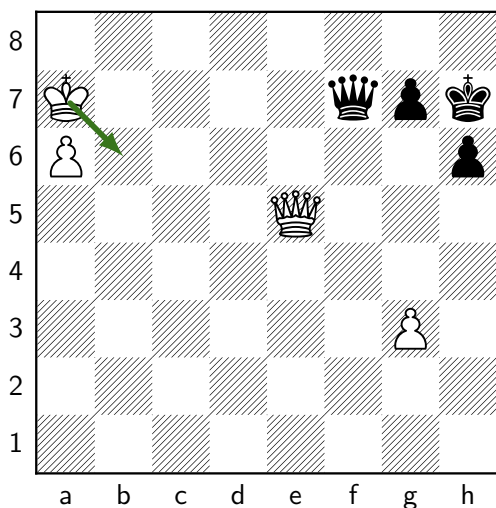
Figure 60: **Move 69, Black: Kh8**. (a) Our position; Melanie plays Kh8 (g7h8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2312` (ply 135) [structural distance $d=25$, QAP obj=68]: the master played Ke6; under μ our Kg2 plays the role of that game’s Kd2; cross-type roles: our Re4 as their pc2; transferred move g7h8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#173` (ply 133) [structural distance $d=21$, QAP obj=52]: the master played Kb1; under μ our Kg2 plays the role of that game’s Kc7; cross-type roles: our Re4 as their Ph5; transferred move g7h8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1716` (ply 126) [structural distance $d=24$, QAP obj=3]: the master played Kc2; under μ our Kg2 plays the role of that game’s Kb3; cross-type roles: our Re4 as their ph6; transferred move g7h8 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Kh8 (summed QAP objective 168, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



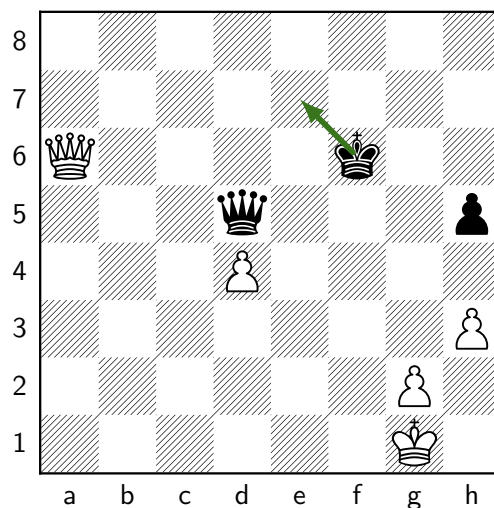
(a)



(b)



(c)



(d)

Figure 61: **Move 70, Black: Kh7.** (a) Our position; Melanie plays Kh7 (h8h7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1183` (ply 126) [structural distance $d=20$, QAP obj=70]: the master played Kc3; under μ our *Kh1* plays the role of that game’s *Kb3*; transferred move h8h7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#923` (ply 150) [structural distance $d=21$, QAP obj=68]: the master played Kb6; under μ our *Kh1* plays the role of that game’s *Ka7*; cross-type roles: our *Re4* as their *Qe5*, our *Qh6* as their *Pa6*; transferred move h8h7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#2871` (ply 129) [structural distance $d=22$, QAP obj=68]: the master played Ke7; under μ our *Kh1* plays the role of that game’s *Kf3*; transferred move h8h7 (**endorses this move**). Synthesis: 42 retrieved analogs transferred to Kh7 (summed QAP objective 1538, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

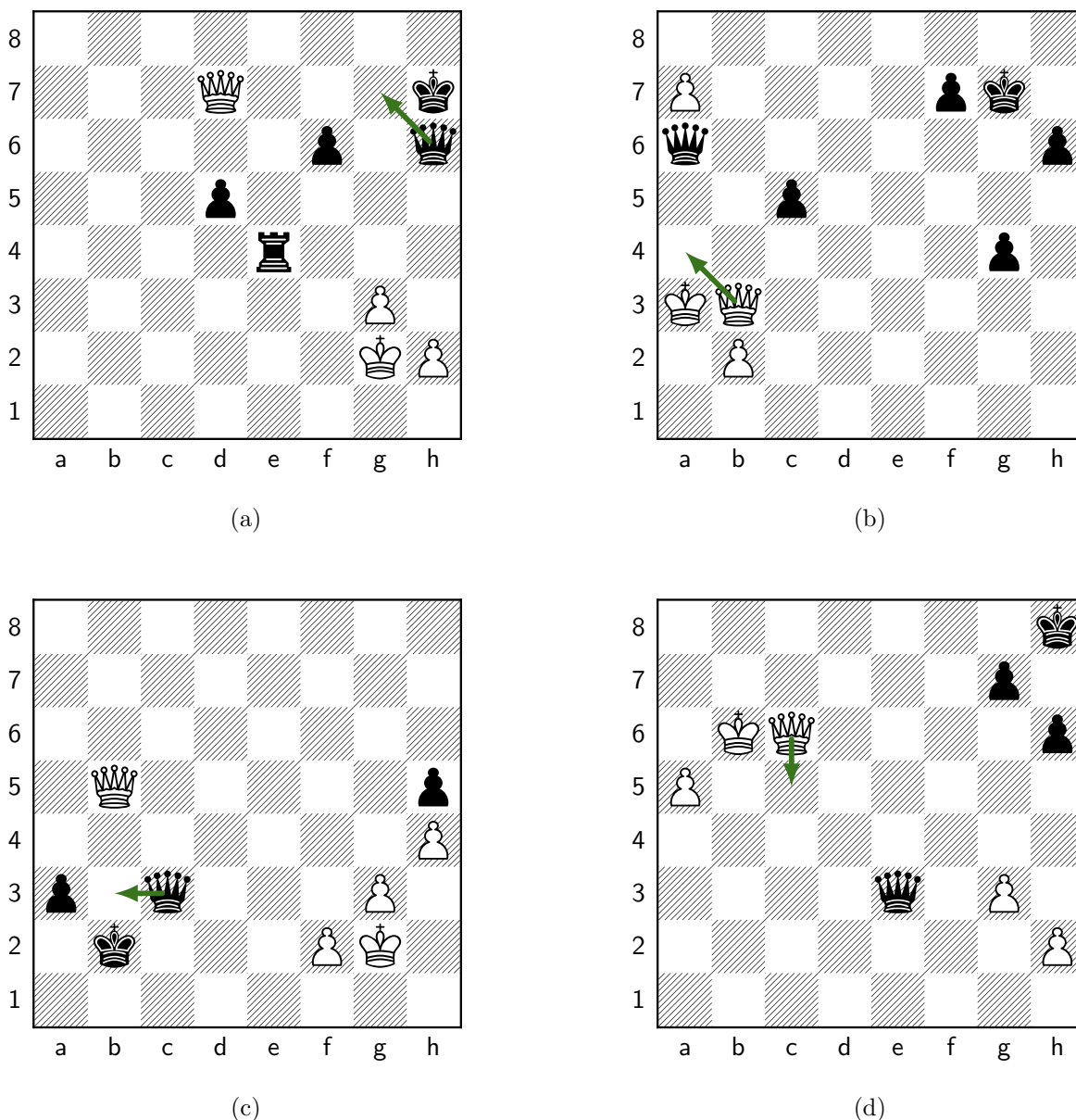
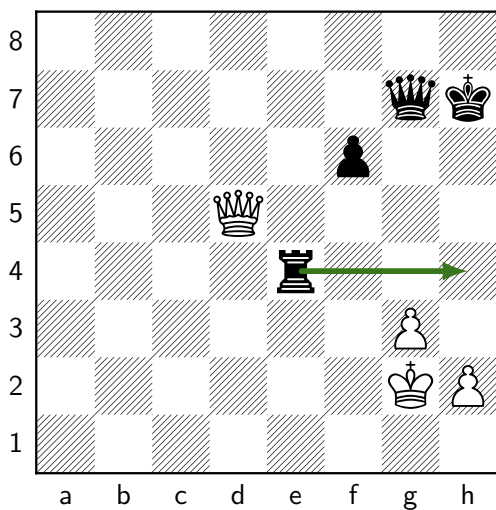
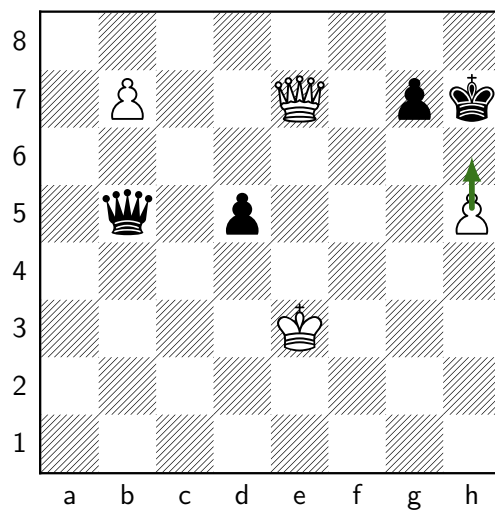


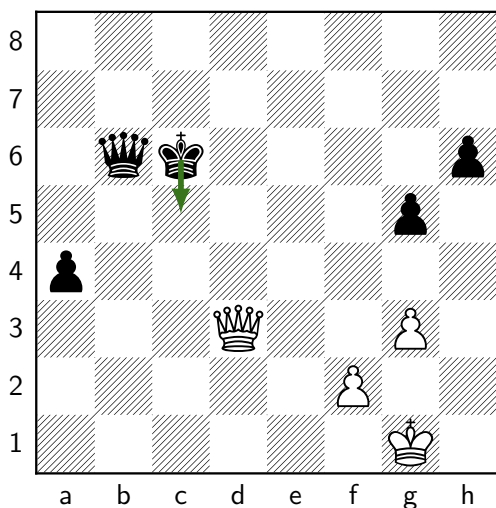
Figure 62: **Move 71, Black: Qg7**. (a) Our position; Melanie plays **Qg7** (**h6g7**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1168` (ply 130) [structural distance $d=24$, QAP obj=58]: the master played **Qa4**; under μ our *Qh3* plays the role of that game’s *Qb3*; cross-type roles: our *Re4* as their *Pb2*; transferred move **h6g7** (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#173` (ply 139) [structural distance $d=17$, QAP obj=58]: the master played **Qb3**; under μ our *Qh3* plays the role of that game’s *Qc6*; cross-type roles: our *Re4* as their *ph4*; transferred move **h6g7** (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#923` (ply 126) [structural distance $d=23$, QAP obj=57]: the master played **Qc5**; under μ our *Qh3* plays the role of that game’s *Qc6*; cross-type roles: our *Re4* as their *Pg3*; transferred move **h6g7** (**endorses this move**). Synthesis: 11 retrieved analogs transferred to **Qg7** (summed QAP objective 393, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



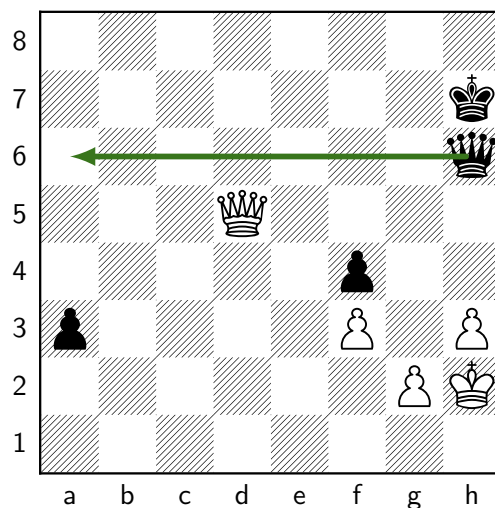
(a)



(b)

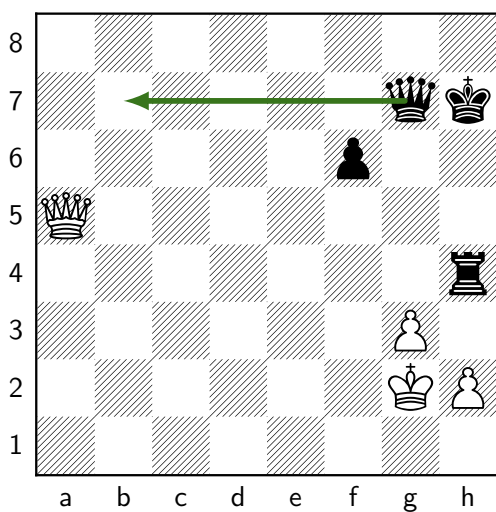


(c)

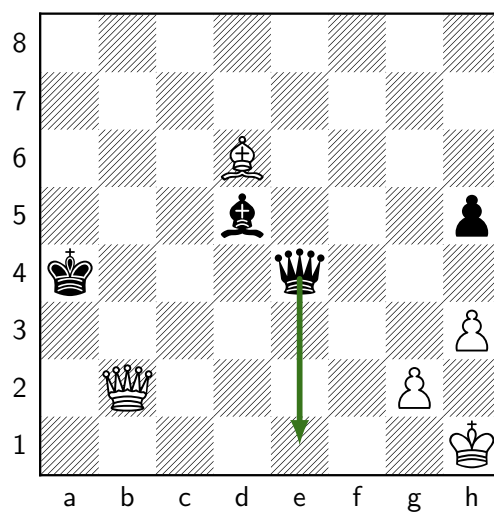


(d)

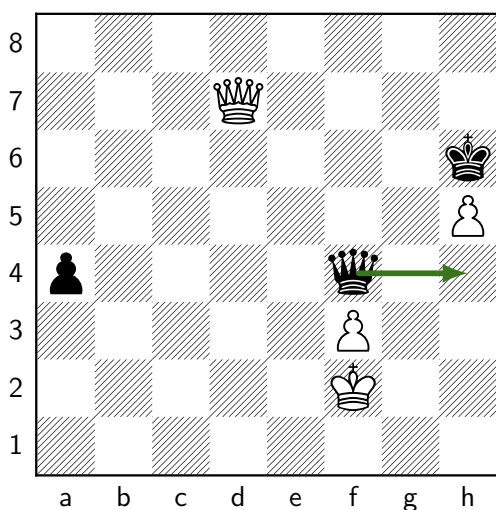
Figure 63: **Move 72, Black: Rh4.** (a) Our position; Melanie plays Rh4 (**e4h4**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2475` (ply 144) [structural distance $d=20$, QAP obj=17]: the master played **h6**; under μ our *Re5* plays the role of that game’s *Ph5* (a **cross-type** correspondence); transferred move **e4h4** (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1231` (ply 139) [structural distance $d=23$, QAP obj=31]: the master played **Kc5**; under μ our *Kh2* plays the role of that game’s *Kc3*; cross-type roles: our *Re4* as their *pg3*; transferred move **h7g6**. (d) Retrieved from `ccr1_top_p0001#1467` (ply 129) [structural distance $d=20$, QAP obj=28]: the master played **Qa6**; under μ our *Qg2* plays the role of that game’s *Qh3*; cross-type roles: our *Re4* as their *Pa3*; transferred move **g7a7**. Synthesis: 1 retrieved analog transferred to Rh4 (summed QAP objective 17, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



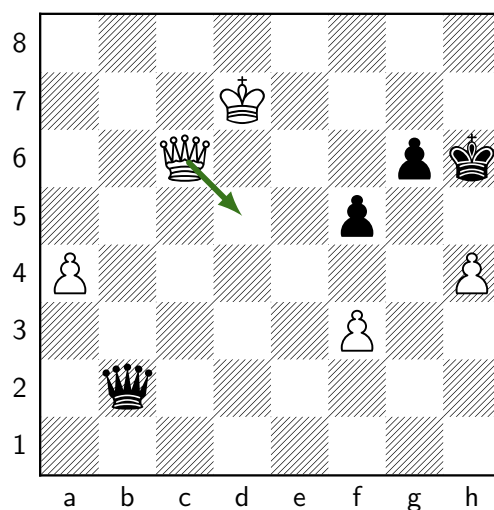
(a)



(b)

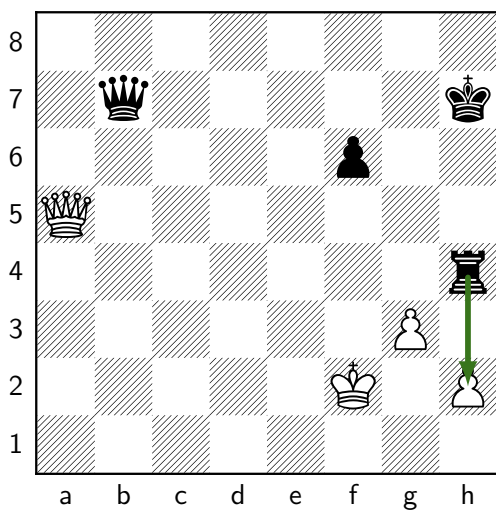


(c)

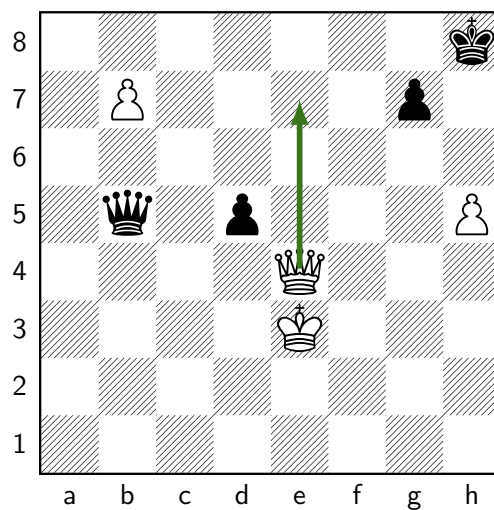


(d)

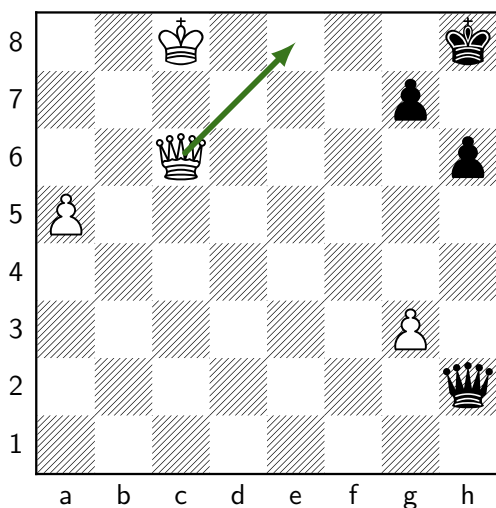
Figure 64: **Move 73, Black: Qb7+.** (a) Our position; Melanie plays Qb7+ (g7b7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2709` (ply 147) [structural distance $d=37$, QAP obj=17]: the master played Qe1+; under μ our Qg2 plays the role of that game's Qe5; cross-type roles: our Pf6 as their Bd5, our Rh4 as their Ph5; transferred move g7b7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1941` (ply 149) [structural distance $d=30$, QAP obj=11]: the master played Qh4+; under μ our Qg2 plays the role of that game's Qf5; transferred move g7b7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2930` (ply 130) [structural distance $d=35$, QAP obj=31]: the master played Qd5; under μ our Qg2 plays the role of that game's Qc6; cross-type roles: our Rh4 as their Ph4; transferred move g7e7. Synthesis: 2 retrieved analogs transferred to Qb7+ (summed QAP objective 28, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



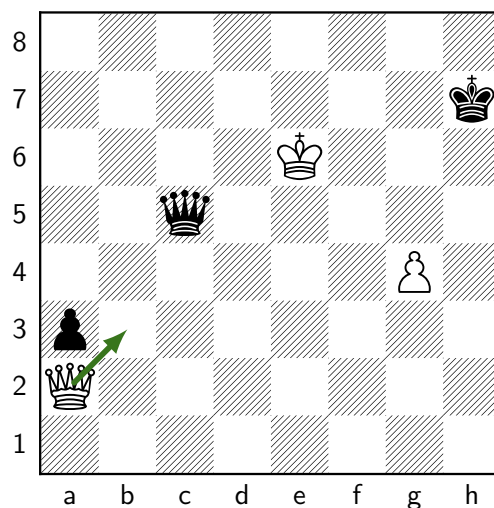
(a)



(b)

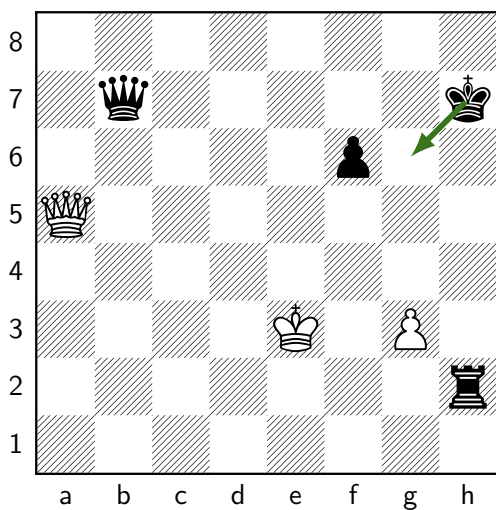


(c)

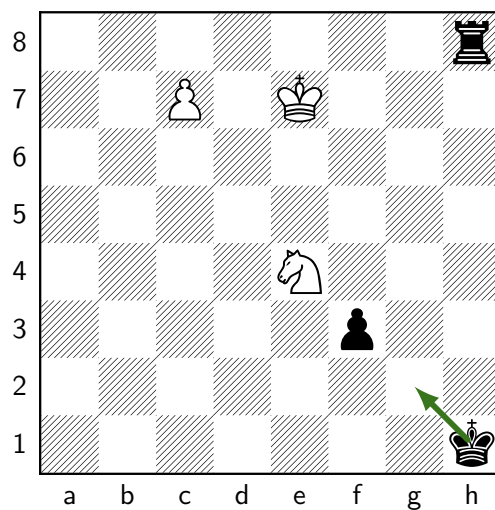


(d)

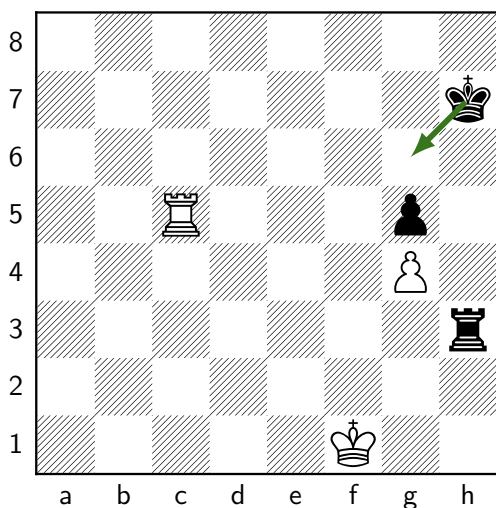
Figure 65: **Move 74, Black: Rxh2+.** (a) Our position; Melanie plays Rxh2+ (h4h2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2475` (ply 142) [structural distance $d=27$, QAP obj=13]: the master played Qe7; under μ our Rh5 plays the role of that game's Qe4 (a **cross-type** correspondence); cross-type roles: our ph2 as their qb5, our qa5 as their Ph5; transferred move h4h2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#923` (ply 140) [structural distance $d=21$, QAP obj=4]: the master played Qe8+; under μ our Rh5 plays the role of that game's Qc6 (a **cross-type** correspondence); cross-type roles: our Qb7 as their Pg3; transferred move h4h2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3123` (ply 156) [structural distance $d=26$, QAP obj=4]: the master played Qb3; under μ our Rh5 plays the role of that game's Qa2 (a **cross-type** correspondence); transferred move h4h2 (**endorses this move**). Synthesis: 11 retrieved analogs transferred to Rxh2+ (summed QAP objective 43, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



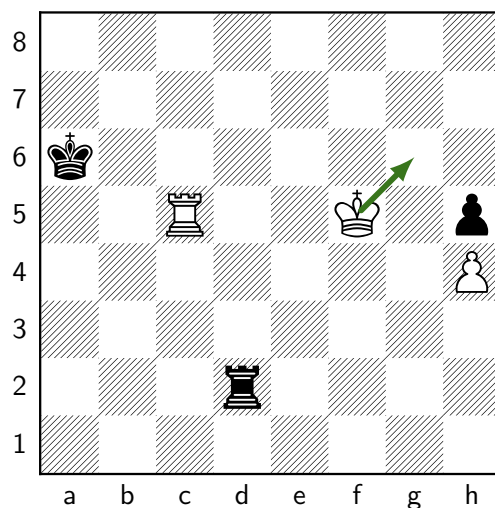
(a)



(b)

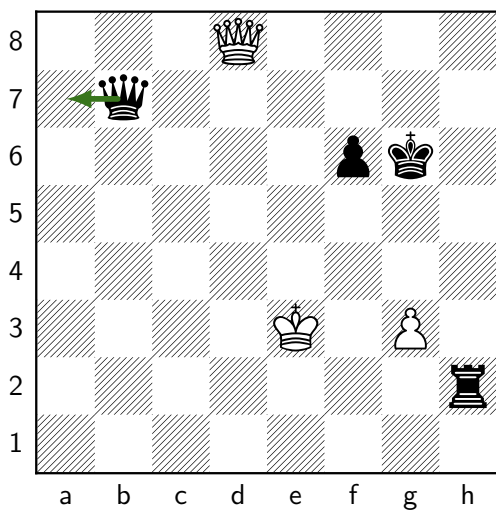


(c)

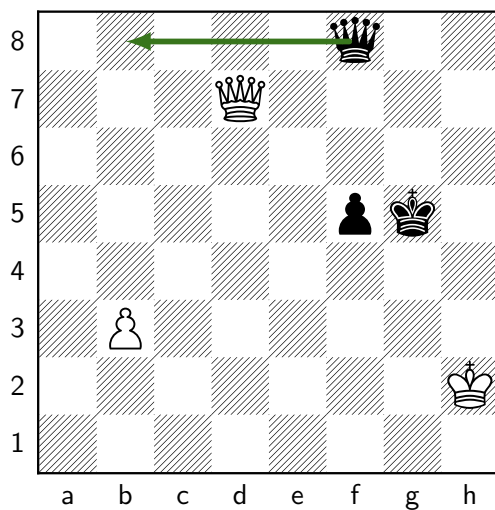


(d)

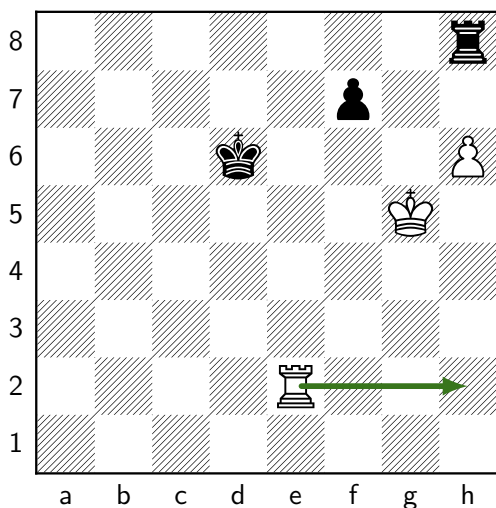
Figure 66: **Move 75, Black: Kg6**. (a) Our position; Melanie plays Kg6 (h7g6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0002#1382 (ply 137) [structural distance $d=13$, QAP obj=2]: the master played Kg2; under μ our Kh2 plays the role of that game’s Kh8; cross-type roles: our qa5 as their ne4, our Qb7 as their Pf3; transferred move h7g6 (**endorses this move**). (c) Retrieved from ccr1_top_p0002#566 (ply 159) [structural distance $d=16$, QAP obj=2]: the master played Kg6; under μ our Kh2 plays the role of that game’s Kh2; cross-type roles: our pg3 as their rc5, our Qb7 as their Pg5; transferred move h7g6 (**endorses this move**). (d) Retrieved from ccr1_top_p0001#3214 (ply 132) [structural distance $d=13$, QAP obj=2]: the master played Kg6; under μ our Kh2 plays the role of that game’s Kf5; cross-type roles: our pg3 as their rd2, our Qb7 as their Ph4; transferred move h7g6 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Kg6 (summed QAP objective 7, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



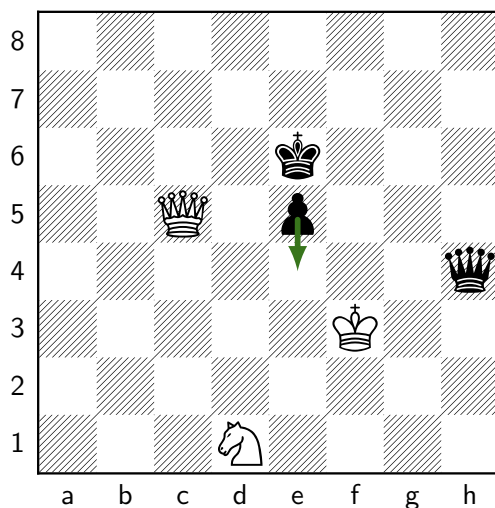
(a)



(b)

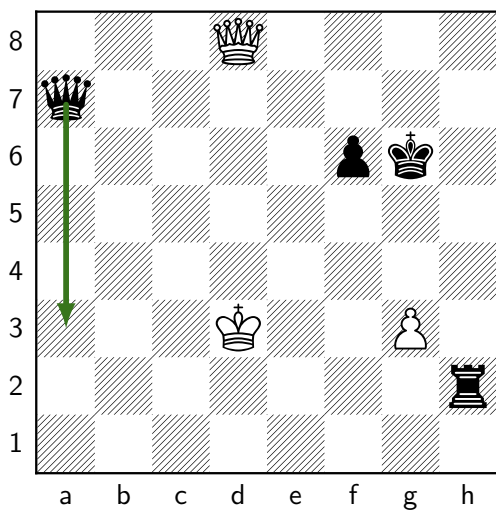


(c)

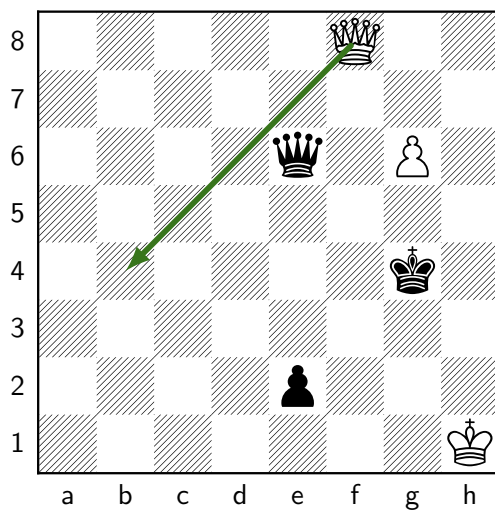


(d)

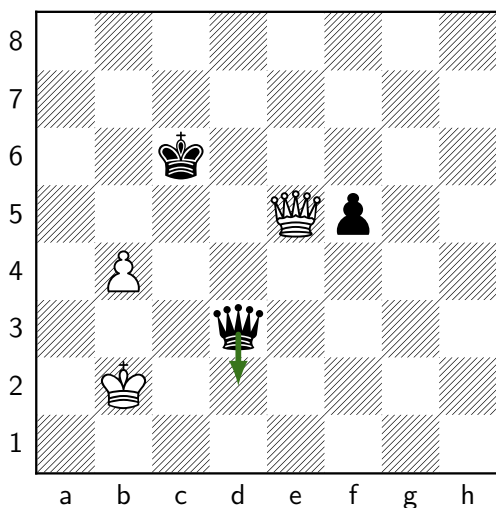
Figure 67: **Move 76, Black: Qa7+.** (a) Our position; Melanie plays Qa7+ (b7a7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top-p0001#1723` (ply 149) [structural distance $d=7$, QAP obj=22]: the master played Qb8+; under μ our Qb2 plays the role of that game’s Qf1; transferred move b7a7 (**endorses this move**). (c) Retrieved from `ccr1_top-p0003#2531` (ply 148) [structural distance $d=10$, QAP obj=21]: the master played Rh2; under μ our Rh7 plays the role of that game’s Re2; cross-type roles: our Qd8 as their rh8; transferred move h2h3. (d) Retrieved from `ccr1_top-p0002#2606` (ply 165) [structural distance $d=7$, QAP obj=21]: the master played e4+; under μ our Pf3 plays the role of that game’s Pe4; cross-type roles: our Rh2 as their Qh4, our pg3 as their nd1; transferred move f6f5. Synthesis: 3 retrieved analogs transferred to Qa7+ (summed QAP objective 66, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



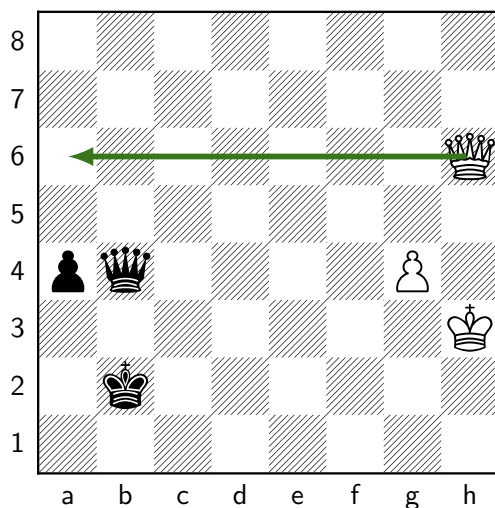
(a)



(b)

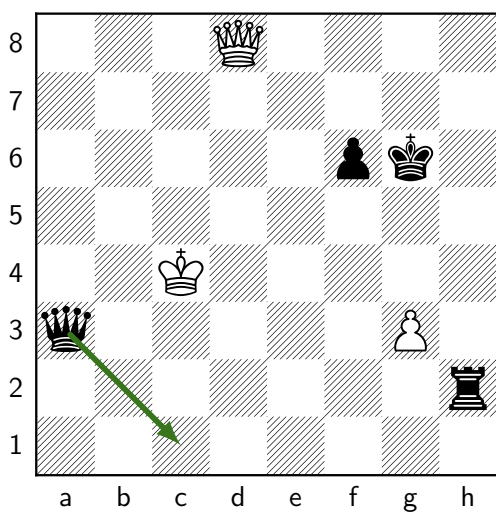


(c)

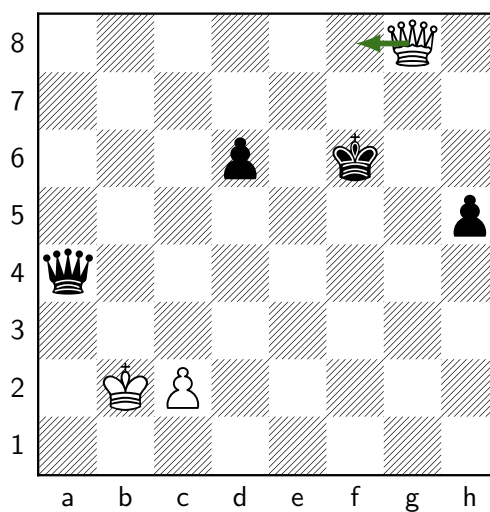


(d)

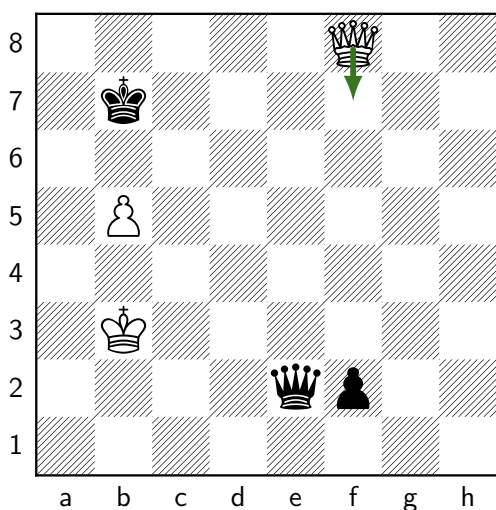
Figure 68: **Move 77, Black: Qa3+.** (a) Our position; Melanie plays Qa3+ (a7a3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#392` (ply 140) [structural distance $d=9$, QAP obj=21]: the master played Qb4+; under μ our Qa2 plays the role of that game's Qf8; transferred move a7a3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1751` (ply 139) [structural distance $d=10$, QAP obj=12]: the master played Qd2+; under μ our Qa2 plays the role of that game's Qd6; transferred move a7a3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1277` (ply 144) [structural distance $d=11$, QAP obj=12]: the master played Qa6; under μ our Qa2 plays the role of that game's Qh6; cross-type roles: our Rh2 as their pa4; transferred move a7a3 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Qa3+ (summed QAP objective 52, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



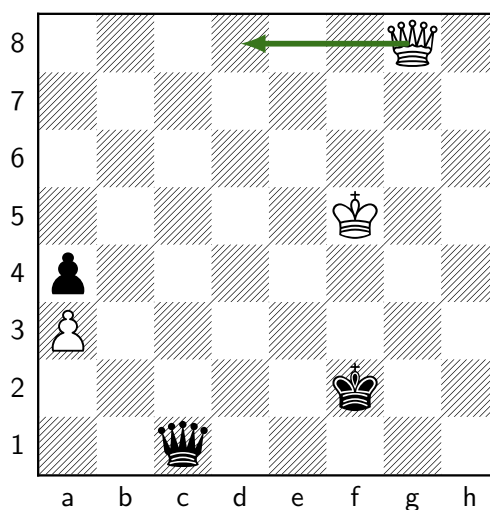
(a)



(b)

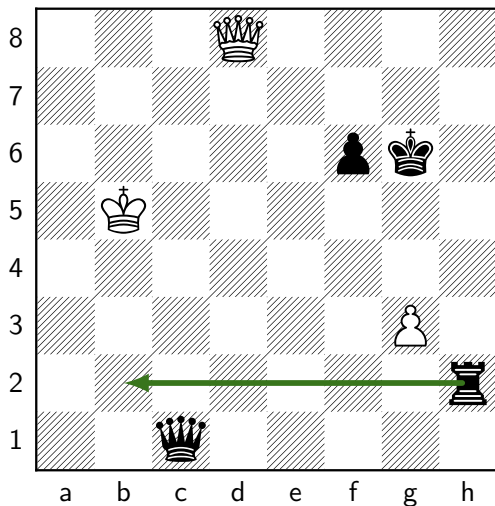


(c)

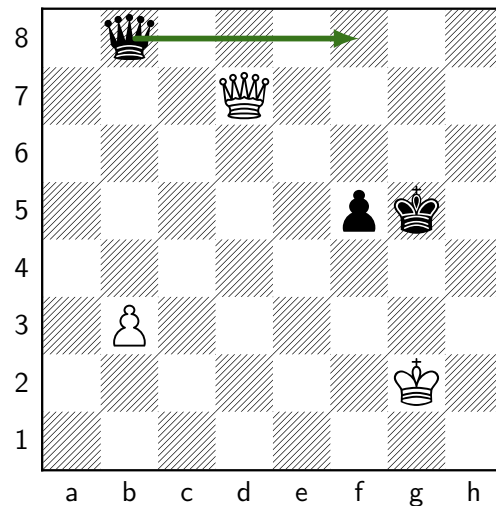


(d)

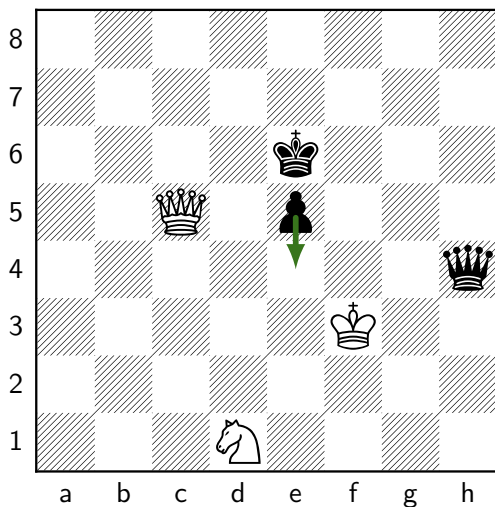
Figure 69: **Move 78, Black: Qc1+.** (a) Our position; Melanie plays Qc1+ (a3c1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top-p0003#2312` (ply 146) [structural distance $d=7$, QAP obj=21]: the master played Qf8+; under μ our Qa6 plays the role of that game’s Qg8; cross-type roles: our Rh2 as their pd6; transferred move a3c1 (**endorses this move**). (c) Retrieved from `ccr1_top-p0002#1751` (ply 168) [structural distance $d=10$, QAP obj=19]: the master played Qf7+; under μ our Qa6 plays the role of that game’s Qf8; transferred move a3c1 (**endorses this move**). (d) Retrieved from `ccr1_top-p0002#2753` (ply 138) [structural distance $d=8$, QAP obj=19]: the master played Qd8; under μ our Qa6 plays the role of that game’s Qg8; transferred move a3c1 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Qc1+ (summed QAP objective 73, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



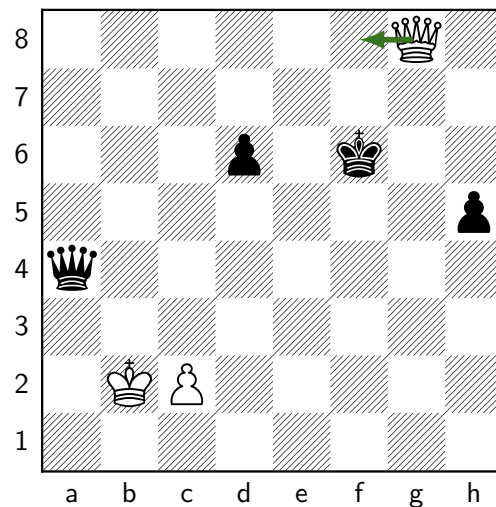
(a)



(b)

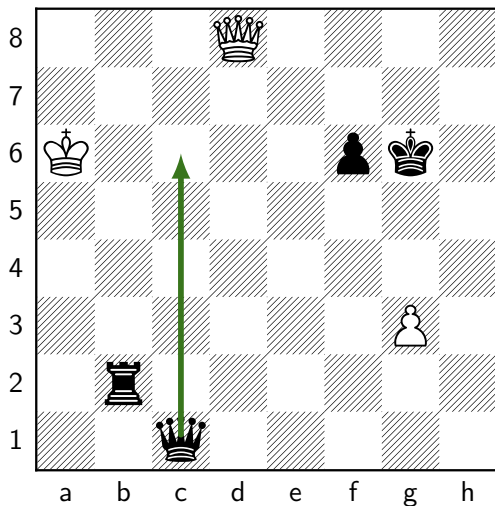


(c)

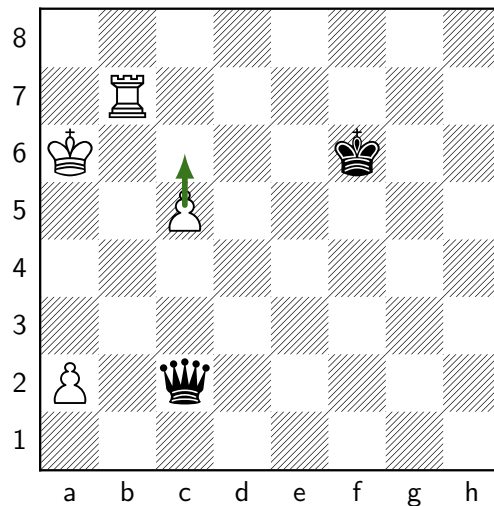


(d)

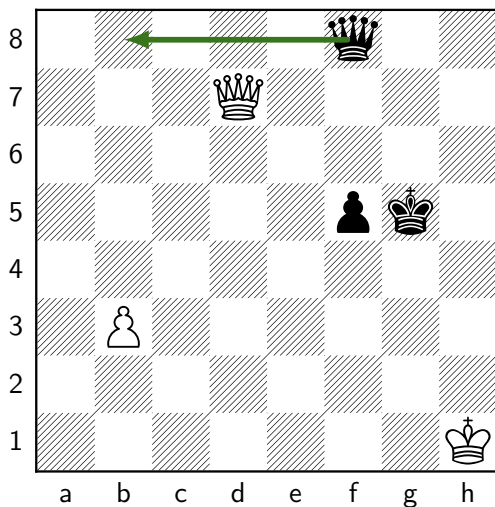
Figure 70: **Move 79, Black: Rb2+.** (a) Our position; Melanie plays Rb2+ (h2b2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top.p0001#1723` (ply 151) [structural distance $d=5$, QAP obj=22]: the master played Qf8; under μ our Qc8 plays the role of that game's Qb1; transferred move c1c8. (c) Retrieved from `ccr1_top.p0002#2606` (ply 165) [structural distance $d=7$, QAP obj=21]: the master played e4+; under μ our Pf3 plays the role of that game's Pe4; cross-type roles: our pg3 as their nd1; transferred move f6f5. (d) Retrieved from `ccr1_top.p0003#2312` (ply 146) [structural distance $d=4$, QAP obj=21]: the master played Qf8+; under μ our Qc8 plays the role of that game's Qg8; cross-type roles: our Rh2 as their ph5; transferred move c1e1. Synthesis: no retrieved analog transferred to Rb2+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



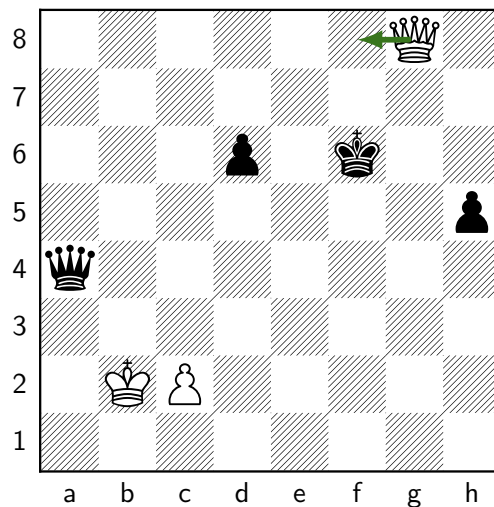
(a)



(b)



(c)



(d)

Figure 71: **Move 80, Black: Qc6+.** (a) Our position; Melanie plays Qc6+ (c1c6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3256` (ply 158) [structural distance $d=12$, QAP obj=26]: the master played c6; under μ our *Pf3* plays the role of that game’s *Pc5*; transferred move f6f5. (c) Retrieved from `ccr1_top_p0001#1723` (ply 153) [structural distance $d=7$, QAP obj=20]: the master played Qb8; under μ our *Qc8* plays the role of that game’s *Qf1*; transferred move c1c7. (d) Retrieved from `ccr1_top_p0003#2312` (ply 146) [structural distance $d=9$, QAP obj=20]: the master played Qf8+; under μ our *Rb7* plays the role of that game’s *Qg8* (a **cross-type** correspondence); cross-type roles: our *Qc1* as their *ph5*; transferred move b2a2. Synthesis: no retrieved analog transferred to Qc6+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.

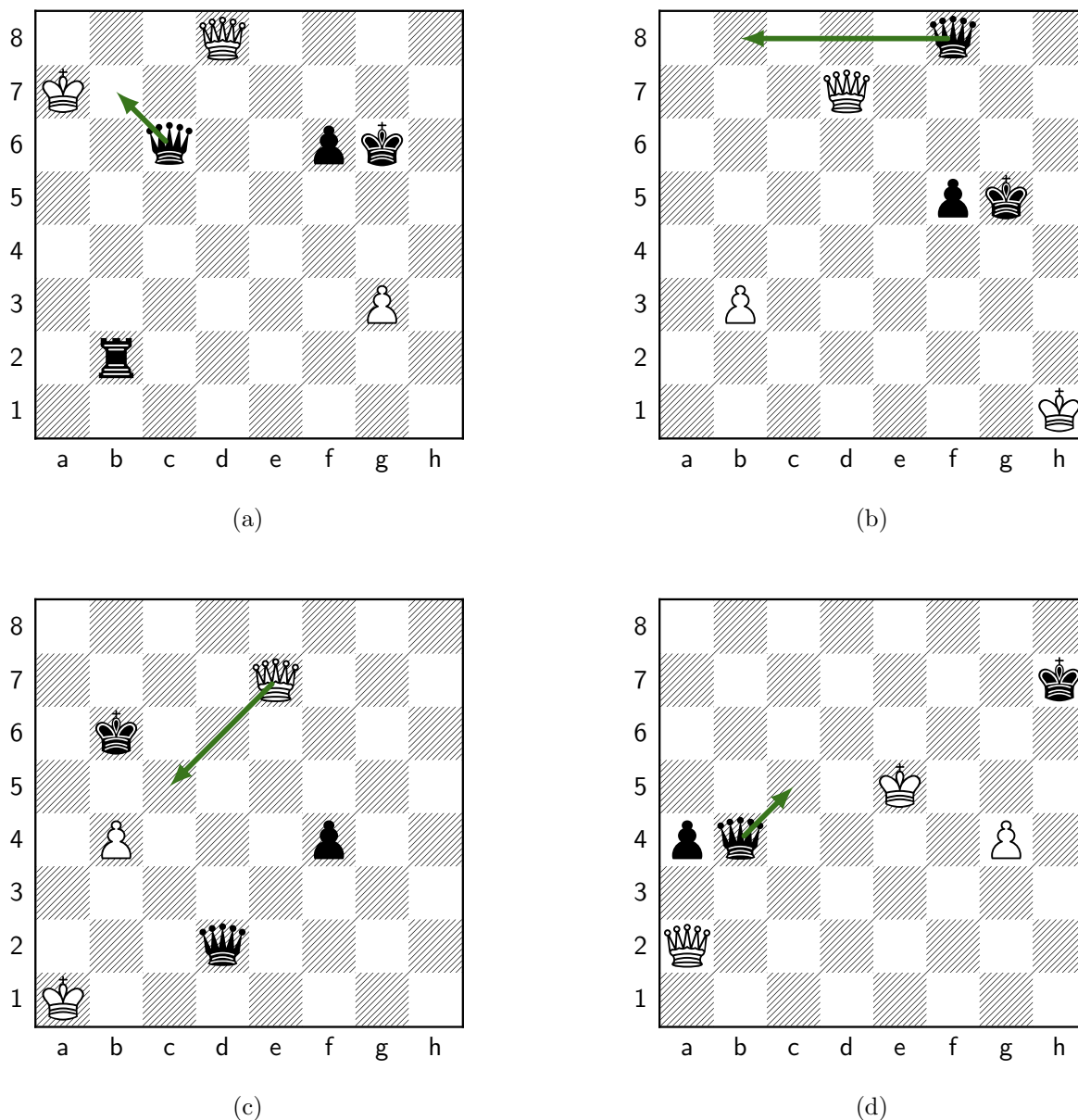


Figure 72: **Move 81, Black: Qb7#.** (a) Our position; Melanie plays Qb7# (c6b7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#1723` (ply 153) [structural distance $d=5$, QAP obj=24]: the master played Qb8; under μ our $Qc3$ plays the role of that game’s $Qf1$; transferred move c6c8. (c) Retrieved from `ccrl_top_p0002#1751` (ply 146) [structural distance $d=11$, QAP obj=21]: the master played Qc5+; under μ our $Qc3$ plays the role of that game’s $Qe7$; transferred move c6a6. (d) Retrieved from `ccrl_top_p0000#3123` (ply 153) [structural distance $d=10$, QAP obj=21]: the master played Qc5+; under μ our $Qc3$ plays the role of that game’s $Qb5$; transferred move c6c5. Synthesis: no retrieved analog transferred to Qb7# – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.